

ISO Host Interface Manual

BASE24[®]



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Preface

This manual describes the ISO Host Interface process, which is used to establish a communications link between BASE24 and a host.

The ISO Host Interface process uses an external message based on the International Organization for Standardization (ISO) 8583-1987 standard. The ANSI Host Interface process, which uses an external message based on the American National Standards Institute (ANSI) X9.2-1980 standard, is documented in the ***BASE24 ANSI Host Interface Manual***.

Audience

This manual is intended as a reference source for those persons responsible for the interface between BASE24 and a host. It provides a comprehensive description of all the functions performed by the ISO Host Interface process involving incoming and outgoing message traffic for both acquirer and issuer processors.

Additional Documentation

The BASE24 documentation set is arranged so that each BASE24 manual presents a topic or group of related topics in detail. When one BASE24 manual presents a topic that has already been covered in detail in another BASE24 manual, the topic is summarized and the reader is directed to the other manual for additional information. Information has been arranged in this manner to be more efficient for readers who do not need the additional detail and at the same time provide the source for readers who require the additional information. This manual contains references to the following BASE24 publications:

- The ***BASE24 ANSI Host Interface Manual*** discusses the ANSI Host Interface process. The ANSI Host Interface process allows communications with a host using the American National Standards Institute (ANSI) message format.

- The ***BASE24 Base Files Maintenance Manual*** documents BASE24 file maintenance screen fields, including those for the Host Configuration File (HCF), External Message File (EMF), Key File (KEYF), and Token File (TKN). These files are used in configuring the ISO Host Interface process.
- The ***BASE24-atm Transaction Processing Manual*** documents the complete set of BASE24-atm transactions.
- The ***BASE24-pos Transaction Processing Manual*** documents the complete set of BASE24-pos transactions.
- The ***BASE24-teller Transaction Processing Manual*** documents the complete set of BASE24-teller transactions.
- The ***BASE24 Remote Banking Transaction Processing Manual*** documents the complete set of BASE24 Remote Banking transactions.
- The ***BASE24-atm Files Maintenance Manual*** describes how to configure the BASE24-atm Terminal Data files (ATD). The ATD is used in configuring the ISO Host Interface process.
- The ***BASE24 External Message Manual*** documents the ISO external message format used by the ISO Host Interface process.
- The ***BASE24 Tokens Manual*** describes how to use the TKN to configure the tokens sent in the external message by the ISO Host Interface process.
- The ***BASE24 Transaction Security Manual*** describes in detail how to configure key management, message authentication, and message encryption.
- The ***BASE24 Logical Network Configuration File Manual*** documents the Logical Network Configuration File (LCONF). The LCONF is used in configuring the ISO Host Interface process.
- The ***BASE24 Text Command Reference Manual*** documents in detail the text commands that can be sent to the ISO Host Interface process.
- The ***BASE24 Event Message Reference Manual*** describes how to access event message documentation. The documentation includes causes and remedies for ISO Host Interface process event messages.
- The ***BASE24 Refresh and Extract Operators Manual*** describes how to perform batch file refreshes and extracts between BASE24 and its hosts.
- The ***BASE24 Remote Banking Standard Interface Support Manual*** documents the ISO external message format used by the BASE24-telebanking Integrated Authorization Server process to communicate with remote banking endpoint devices.

This manual also contains references to the following NET24-XPNET publications:

- The *NET24-XPNET Network Control Command Reference Manual* provides information about configuring, running, and maintaining an XPNET system that includes BASE24 applications.
- The *BASE24 Event Message Reference Manual* describes how to access online event message documentation and documents the file maintenance screen fields for the Log Message Configuration File (LMCF). The online documentation includes causes and remedies for event messages generated by Refresh and Super Extract processes. The LMCF can be used to modify the event messages generated by the Refresh or Super Extract process.

Software

This manual documents standard processing as of its publication date. Software that is not current and custom software modifications (CSMs) may result in processing that differs from the material presented in this manual. The customer is responsible for identifying and noting these changes.

Manual Summary

The following is a summary of the contents of this manual.

Section 1, “Introduction,” provides an introduction to the BASE24 ISO Host Interface process, identifies the files used by the ISO Host Interface process, and identifies the internal and external message types supported by the ISO Host Interface process.

Section 2, “Configuration and Control,” provides information about the configuration and control of the ISO Host Interface process.

Section 3, “Network Management and Operations,” provides an overview of the network management and operations for the ISO Host Interface process. It discusses monitoring ISO Host Interface process operation and the network management messages, including dynamic key management messages, that are used in communicating with the host.

Section 4, “Store-and-Forward Processing,” provides an overview of store-and-forward processing and store-and-forward message flows.

Section 5, “BASE24-atm Financial Message Handling,” provides an overview of BASE24-atm financial transaction processing and message flows.

Section 6, “BASE24-pos Financial Message Handling,” provides an overview of BASE24-pos financial transaction processing and message flows.

Section 7, “BASE24-telebanking Financial Message Handling,” provides an overview of BASE24-telebanking and BASE24-billpay financial transaction processing and message flows.

Section 8, “BASE24-teller Financial Message Handling,” provides an overview of BASE24-teller financial transaction processing and message flows.

Appendix A, “Processing Information,” contains ISO Host Interface processing information. It describes the steps that the ISO Host Interface process performs at startup and as a result of a warmboot request, and identifies the text commands that the ISO Host Interface process can process.

Publication Identification

Three entries appearing at the bottom of each page uniquely identify this BASE24 publication. The publication number (for example, BA-DH011-01 for the ***BASE24 ISO Host Interface Manual***) appears on every page to assist readers in identifying the manual from which a page of information was printed. The publication date (for example, Dec-2009 for December, 2009) indicates the issue of the manual. The software release information (for example, R6.0v10 for release 6.0, version 10) specifies the software that the manual describes. This information matches the document information on the copyright page of the manual.

1: Introduction

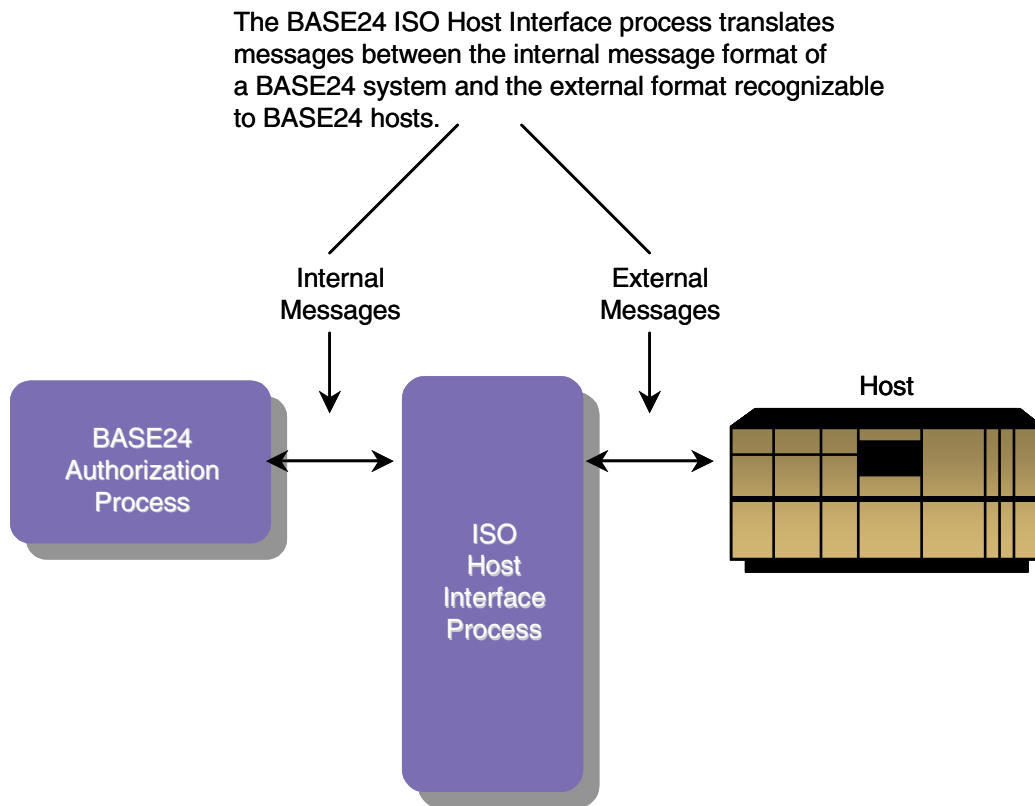
This section introduces the ISO Host Interface process and provides an overview of ISO Host Interface process functionality. It also identifies the BASE24 files used by the ISO Host Interface process, and the message types supported.

ISO Host Interface Process Overview

The ISO Host Interface process, shown in the illustration below, is responsible for the online message transfer between a BASE24 system and its hosts. This process performs the following functions:

- Controls message traffic to and from BASE24 hosts, enforcing an orderly message exchange.
- Translates messages from the internal message format used by BASE24 processes to an external message format recognizable to the host.
- Stores certain types of messages bound for the host for later transmittal when the host is unavailable.

ISO Host Interface Process



A BASE24 system supports communication with hosts using an external message format based on the International Organization for Standardization (ISO) 8583-1987 standard.

Hosts

A host is an external mainframe computer system with which a BASE24 system interacts—either online or by batch tape—to authorize transactions and update cardholder records.

The ISO Host Interface process is responsible for the online interaction between a BASE24 system and its hosts. The BASE24 Refresh and Super Extract processes are responsible for the batch tape interaction between the BASE24 system and its hosts. Refer to the ***BASE24 Refresh and Extract Operators Manual*** for more information on the Refresh and Super Extract processes.

Although the term *host* generally refers to an actual computer or system of computers, the term also refers to the institution or organization responsible for the host computer system and its processing.

Data Processing Centers

In a BASE24 system, hosts are referred to as *data processing centers* (DPCs). Generally, the terms *DPC* and *host* are synonymous. However, the term *host* implies either a host computer or the organization responsible for a host computer, and the term *DPC* always characterizes the host computer or computer network itself.

A DPC can be one mainframe computer communicating with a BASE24 system, or a DPC can represent several mainframe computers networked together at a host site. Likewise, one mainframe computer can have several applications that each function as a separate DPC. Regardless of the machine configuration, a DPC represents a single logical host. A host site can establish multiple DPCs; however, a BASE24 system considers each of these DPCs to be a separate host.

DPCs are assigned identification numbers and defined to a BASE24 system in the Host Configuration File (HCF). Each DPC must have a record in the HCF in order to be recognized by the BASE24 system. The HCF contains options at the DPC-level that allow hosts to specify individual processing for each DPC. The HCF also defines the stations used for communication between the BASE24 system and the DPC. Refer to the ***BASE24 Base Files Maintenance Manual*** for information on setting up the HCF.

Stations

Stations provide the means by which communication takes place between the BASE24 system and a DPC. All messages sent to a DPC are actually sent to the stations defined for the DPC. Likewise, all messages received from a DPC are actually received from stations.

Stations and their lines must be configured to the XPNET system using the ADD STATION and ADD LINE commands, respectively. Stations and lines are tied to their respective DPCs through the Host Configuration File (HCF). At least one station is required for each DPC, and a maximum of 32 stations are allowed per DPC. For information on configuring stations and lines to the XPNET system, refer to the *NET24-XPNET Network Control Command Reference Manual*.

Note: All protocol-level functions between a BASE24 system and the stations are actually handled by the XPNET process using the Tandem communications control software. The ISO Host Interface process sends and receives messages by way of the XPNET process. It is the responsibility of the XPNET process to maintain the communications links with the stations according to the protocols under which the stations are defined.

Issuer and Acquirer Hosts

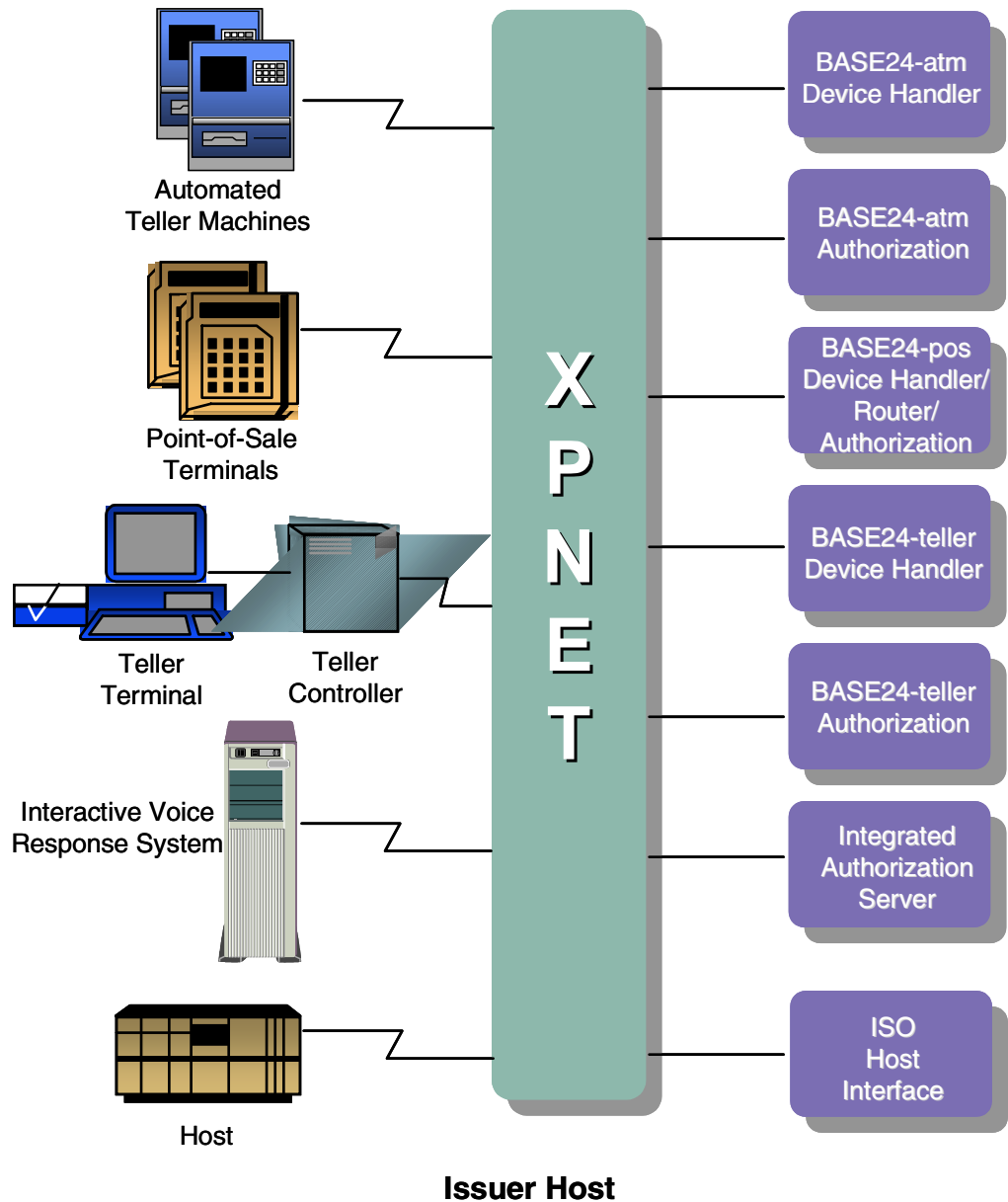
In the financial industry, the terms issuer and acquirer define the parties on opposite ends of the electronic message exchanges required to authorize and process transactions.

Issuers are card-issuing or account-owning institutions or their agents. These institutions or agents are responsible for authorizing cardholder transactions and keeping track of cardholder accounts. *Acquirers* are the parties initiating transactions on behalf of the card acceptors.

Issuer Hosts

Hosts acting on behalf of an issuer in a BASE24 message exchange are referred to as *issuer hosts*. Issuer hosts authorize or post transactions sent to them by a BASE24 system, as shown below.

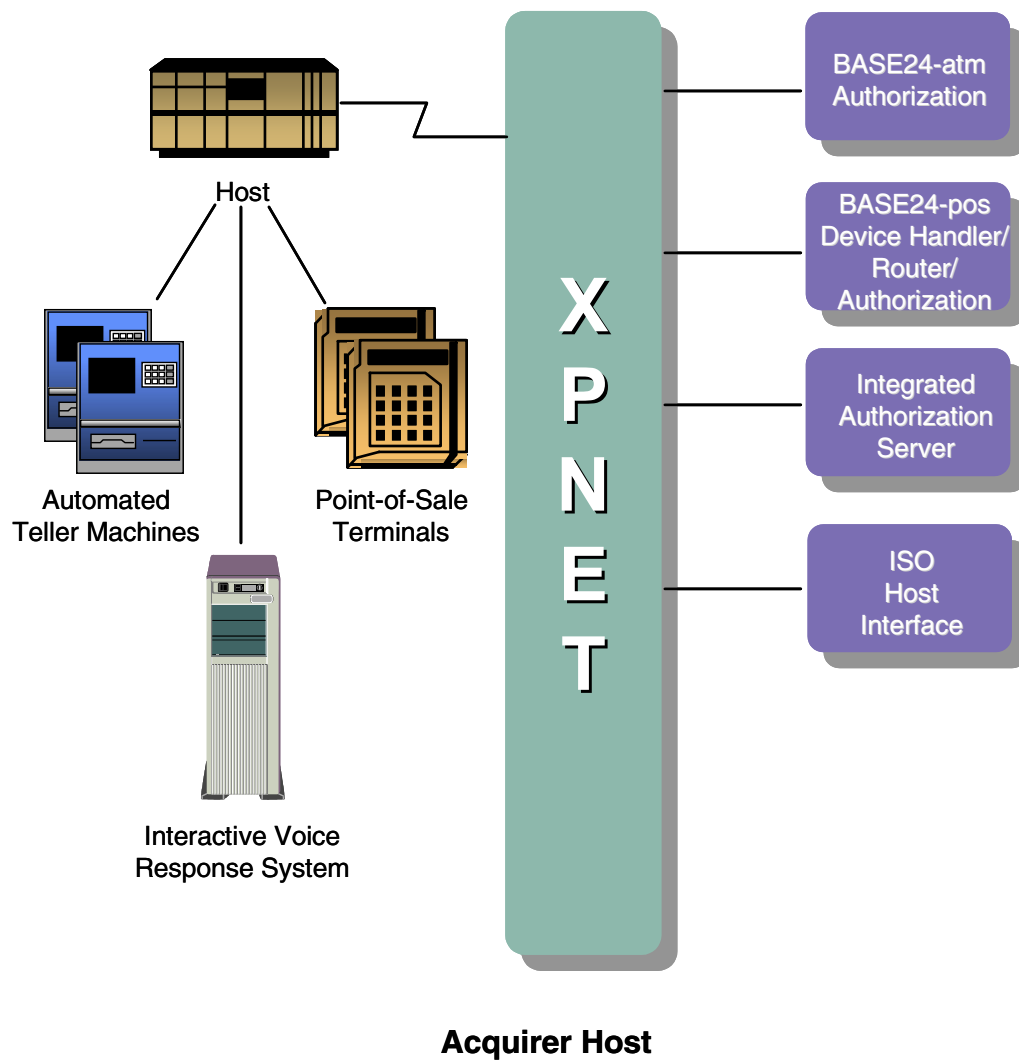
Note: Because of their general position in the transaction path, issuer hosts are also referred to as *back-end hosts*.



Acquirer Hosts

Hosts acting on behalf of an acquirer in a BASE24 message exchange are referred to as *acquirer hosts*. Acquirer hosts support ATM networks, POS networks, Teller terminals, or remote banking devices (e.g., Web servers, interactive voice response systems, or personal computers) that are external to BASE24 and forward transactions to the BASE24 system for authorization, as shown in the following diagram. Although Teller terminals are not depicted in this diagram, they are supported by acquirer hosts.

Note: Because of their general position in the transaction path, acquirer hosts are also referred to as *front-end hosts*.



Issuer and Acquirer Host Support

A BASE24 system can function in the role of an issuer, an acquirer, or both. When a BASE24 system sends financial transactions to an issuer host for authorization, the BASE24 system represents the transaction acquirer, even though the transaction may not have originated at a BASE24-controlled terminal or device. When financial transactions are sent to the BASE24 system for authorization from an acquirer host, the BASE24 system represents the card issuer or account owner, even though the BASE24 system may only forward the transaction to an issuer host for authorization.

Hosts can also function on behalf of both card issuers or account owners and transaction acquirers, as would be the case when an institution issues cards and supports its own network of ATM or POS devices or supports its own remote banking devices for its account owners. In order to allow for these different processing requirements, ACI provides two ISO Host Interface process modules: the Standard Issuer Host Interface module and the Acquirer Host Interface module.

The Standard Issuer Host Interface module supports communications with issuer hosts only. It can send transaction requests to a host and accept transaction responses from a host, but it cannot accept transaction requests from a host.

The Acquirer Host Interface module supports communications with both issuer and acquirer hosts. It can send transaction requests and responses to a host and accept transaction requests and responses from a host.

The set of transactions supported by each product can vary depending on whether the host is acting as an issuer or an acquirer. These transactions are listed below.

BASE24-atm

All BASE24-atm transactions are supported between BASE24-atm and an issuer host. All transactions, except for the following transactions, are supported between BASE24-atm and an acquirer host. Transaction codes are provided in parentheses. The complete set of BASE24-atm transactions is provided in the *BASE24-atm Transaction Processing Manual*.

Deposit with cash back (24)
Load value (41)
Payment to credit account (50)

BASE24-pos

All BASE24-pos transactions are supported between BASE24-pos and an issuer host or acquirer host, except for administrative transactions which are supported between BASE24-pos and an issuer host only. The complete set of BASE24-pos transactions is provided in the ***BASE24-pos Transaction Processing Manual***.

BASE24-teller

All BASE24-teller transactions are supported between BASE24-teller and an issuer host or acquirer host. The complete set of BASE24-teller transactions is provided in the ***BASE24-teller Transaction Processing Manual***.

BASE24-telebanking and BASE24-billpay

The following transactions are supported between BASE24-telebanking and BASE24-billpay and an issuer or acquirer host. Transaction type codes are provided in parentheses. The complete set of BASE24-telebanking and BASE24-billpay transactions is provided in the ***BASE24 Remote Banking Transaction Processing Manual***.

- Available funds inquiry (30)
- Check clearance inquiry (3A)
- Last count of debit/credit transactions inquiry (3B)
- Last count of account by source transactions inquiry (3C)
- Last count of check transactions inquiry (3D)
- Last count of debit transactions inquiry (3E)
- Last count of credit transactions inquiry (3F)
- Last count of account transfer transactions inquiry (3G)
- Customer vendor list inquiry (3H)
- Scheduled payments list inquiry (3J)
- Scheduled transfers list inquiry (3K)
- History inquiry – payments and transfers (3L)
- Customer account list inquiry (3N)
- Multiple account balance inquiry (3P)
- Transfer – immediate (40)
- PIN change (90)
- PIN verify (91)
- Schedule payment – immediate (9A)
- Schedule payment – future (9B)
- Schedule payment – recurring (9C)
- Schedule transfer – future (9D)

Schedule transfer – recurring (9E)
Scheduled payment delete (9F)
Scheduled transfer delete (9G)
Scheduled payment update (9H)
Scheduled transfer update (9J)
User defined transactions (Ux, where x is a value from 0–9)

The following transactions are supported between BASE24-telebanking and BASE24-billpay and an issuer host only:

Check stop payment add (A0)
Statement closing download inquiry (3Q)
Statement download inquiry (3R)

The BASE24 External Message

The ISO Host Interface process uses the external message to communicate with its hosts. The external message is based on the standard external message developed by the International Organization for Standardization (ISO). It is a variable-length, variable-content message that can be configured differently, based on the type of message sent. Specifications for this message can be found in the ***BASE24 External Message Manual***.

Store-and-Forward Function

A store-and-forward function is incorporated into the ISO Host Interface process to collect transaction messages during periods when a host is unavailable.

When a host is unavailable, messages required as notification to the host are stored in the Store-and-Forward File (SAF). These messages are then transmitted later when communications are restored. For more information on store-and-forward processing, refer to section 4.

Multiple Product Support

The same ISO Host Interface process can be set up to process host messages for multiple BASE24 products. If an institution purchases multiple BASE24 products, the appropriate product code is bound into the ISO Host Interface process to allow it to recognize and process transaction messages for each product.

Message Processing

Due to the variances in processing that must occur in handling messages for different products, the ISO Host Interface processes messages differently based on the message type and product ID for the message. For more information on how messages flow back and forth between the host and a BASE24 system, refer to the following sections, based on the type of message and BASE24 product:

- Network management message flows, including those for dynamic key management, are documented in section 3.
- Store-and-forward message flows are documented in section 4.
- BASE24-atm message flows are documented in section 5.
- BASE24-pos message flows are documented in section 6.
- BASE24-telebanking and BASE24-billpay message flows are documented in section 7.
- BASE24-teller message flows are documented in section 8.

File Usage

The ISO Host Interface process uses a number of files in processing. The majority of the files used by the ISO Host Interface process are control files. Control files contain configuration information and parameters that determine how the ISO Host Interface process operates. The files that contain configuration information for the ISO Host Interface process include the following:

- External Message File (EMF)
- Host Configuration File (HCF)
- Key File (KEYF)
- Key 6 File (KEY6)
- Log Message Configuration File (LMCF)
- Logical Network Configuration File (LCONF)
- Token File (TKN)

For more information on how these files are used to configure the ISO Host Interface process, refer to section 2.

In addition to the control files above, the ISO Host Interface process uses the Store-and-Forward File (SAF) to store information about individual transactions processed by the ISO Host Interface process. For more information on how the SAF is used by the ISO Host Interface process, refer to section 4.

Discretionary Data Suppression

Two data elements in the BASE24 external message contain consumer data identified by the Payment Card Industry (PCI) Data Security Standard (DSS) for the protection of sensitive consumer data (e.g., PIN-related data) in acquirer and consumer transactions. Track 1 Data (data element 45) and Track 2 Data (data element 35) contain the information encoded on the magnetic stripe of a card or on the chip used to originate the transaction.

BASE24-atm and BASE24-pos offer the option to suppress track 1 and track 2 discretionary data from the STM/PSTM response and reversal messages. Discretionary data is data other than the primary account number (PAN) and the card expiration date. Many interchanges may also suppress the track 1 and track 2 discretionary data. BASE24-atm and BASE24-pos ISO Host Interface customers may prefer to send the card details to the host in the same format in which they were received (i.e., just the PAN and expiration date). This can be accomplished by modifying the appropriate EMF settings for the impacted message types.

Refer to the *BASE24 External Message Manual* for more information.

Surcharging Support

The ISO Host Interface process supports the capability of surcharging. An institution can configure the BASE24-atm system to assess acquirer fees (surcharges) on transactions initiated at selected ATMs. Surcharges can be based on attributes such as terminal type (for example, the location of the ATM), card prefix or transaction routing, card type, and transaction type. In addition, surcharges can be based on a flat fee per transaction or on a percentage of the transaction amount. A negative flat fee (rebate) value is supported.

Surcharging is configured in the Surcharge File (SURF). Fields in the Card Prefix File (CPF) and the BASE24-atm Terminal Data files (ATD) link the SURF with a card prefix or terminal group. Refer to the ***BASE24 Base Files Maintenance Manual*** for more information about configuring the SURF and CPF for surcharging. Refer to the ***BASE24-atm Files Maintenance Manual*** for more information about configuring the ATD.

The Surcharge amount is carried in data element P-28 of the BIC ISO external message. The presence of data element P-28 in the BIC ISO external message is determined by the setting of this data element in the External Message File (EMF). In order for this data element to be included in the ISO Host external message, it must be defined as mandatory or conditional in the EMF. Refer to the ***BASE24 External Message Manual*** for more information about this data element.

In addition, the Surcharge Data token is appended to the message and passed along with any other tokens to the co-network. This token is used to store surcharge assessment information defined in the BASE24-atm Terminal Data files (ATD) and Surcharge File (SURF) and other information filled in at the entry point and by the Authorization process. Refer to the ***BASE24 Tokens Manual*** for more information about the Surcharge Data token.

ACI Commerce Gateway Support

The ACI Commerce Gateway product does not send the posting date in messages sent to the ISO Host Interface process. As a result, when the ACI Commerce Gateway product is used with the BASE24-pos product, a POS-SETL-NOTIFY*identifier* param must be configured in the Logical Network Configuration File (LCONF) for the ISO Host Interface process.

Each POS-SETL-NOTIFY*identifier* param contains the symbolic name of an ISO Host Interface process that the Main Settlement process must notify at the time specified in the POS-LN-CUTOVER param.

When the POS-SETL-NOTIFY*identifier* param is configured for the ISO Host Interface process, the Main Settlement process sends a 9502 cutover command to the ISO Host Interface process. When the ISO Host Interface process receives this command, it updates a global posting date value that it uses to set the posting date for a transaction when the posting date is not included in the transaction.

ACI Smart Chip Manager Support

ACI Smart Chip Manager is a card management system for multifunctional or multi-application smart cards. The ACI Smart Chip Manager application supports issuing multi-application cards and manages the life cycles of these cards and the life cycles of the card's applications. BASE24 can be configured as a smart card application optionally managed by ACI Smart Chip Manager; therefore, when updates are made to the BASE24 Cardholder Authorization File (CAF), the applicable updates are forwarded to the ACI Smart Chip Manager product for updating the ACI Smart Chip Manager database.

When the CAF is altered using file maintenance screens or the BASE24-from host maintenance product, BASE24 can optionally send a notification message to the ACI Smart Chip Manager. The 0300 ISO message format is used for update messages between BASE24 and ACI Smart Chip Manager.

If the ACI Smart Chip Manager functionality is being used, BASE24-from host maintenance forwards applicable CAF add, update, delete, and increment messages to a Host Interface process that supports BASE24-from host maintenance and applicable connections to ACI Smart Chip Manager. If a station and line are configured to ACI Smart Chip Manager, messages can be sent online. If a station or line is not configured or if the online connection is unavailable, these messages are written to the Store-and-Forward File (SAF). The SAF can be extracted and sent to ACI Smart Chip Manager in a SAF extract tape or file format.

Configuring the Host Interface

When the ACI Smart Chip Manager functionality is being used, an ISO Host Interface must be configured to communicate with the ACI Smart Chip Manager application. The ISO Host Interface process must include the BASE24-from host maintenance ISO Host Interface module. If no lines or stations are configured or if the online connection is unavailable to the ACI Smart Chip Manager application, the ISO Host Interface logs the CAF add, update, and delete messages to the SAF.

The DPC number of the ACI Smart Chip Manager process and the name of the ISO Host Interface process are used to route messages to the correct ISO Host Interface process that communicates with the ACI Smart Chip Manager application and reads the Host Configuration File (HCF). The From Host Maintenance process reads the MONAD-NOTIFY-PROCESS param from the LCONF to obtain the ACI Smart Chip Manager DPC number and ISO Host

Interface process name. The format is XXXXYYYYYYYYYYYYYYYYYY, where XXXX is the DPC number and YYYYYYYYYYYYYYYYYY is the ISO Host Interface process name.

EMF records should be added for outbound BASE24-from host maintenance 0300 messages and for inbound 0310 BASE24-from host maintenance messages.

Enhanced Reversal Routing Support

Enhanced reversal routing is required when multiple Interchange Interface processes are used to communicate with a single interchange, because each process writes to its own Interchange Log File (ILF). When an Interchange Interface process receives a reversal, it attempts to retrieve the original transaction data from its ILF. If the original transaction was processed by another Interchange Interface process, the process cannot retrieve the original transaction data from its ILF and, therefore, drops the reversal.

Enhanced reversal routing ensures that the reversal transaction is sent to the same Interchange Interface process that processed the original transaction. Use this functionality in conjunction with the POS-RVSL-ROUTE Logical Network Configuration (LCONF) param and the enhanced reversal routing enhancement in the BASE24-pos Router module. When this param is set to a value of 1, the Router module saves the symbolic name of the Interchange Interface process that authorized the original transaction. The module then sends a subsequent reversal transaction to the symbolic name saved from the original transaction response. For more information about the POS-RVSL-ROUTE param, refer to the ***BASE24 Logical Network Configuration File Manual***.

Otherwise, when this param is set to a value of 0, the Router module uses normal reversal routing logic (i.e., using the Authorization Selection Table (AST), SPROUTE, or Institution Definition File (IDF) routing methods, followed by additional star routing, which is also known as RTBL character substitution). Refer to the ***BASE24-pos Transaction Processing Manual*** for more information about RTBL character substitution.

The ISO Host Interface process returns the Enhanced Reversal Routing token in the 0210 message that is sent to the host. This token contains the saved symbolic name of the Interchange Interface process that authorized the original transaction. The host must be modified to accept this token and store its contents when it is received in a 0210 message. The host must also be modified to add this token and its contents to the 0420 reversal message before attempting to reverse the transaction. Refer to the ***BASE24 Tokens Manual*** for more information about the Enhanced Reversal Routing token.

Supported Message Types

The ISO Host Interface process supports the external message types shown below for transactions sent between the BASE24 system and a host. The following table includes external message types for BASE24-atm (ATM), BASE24-from host maintenance (FHM), BASE24-pos (POS), BASE24-telebanking/BASE24-billpay (TB), and BASE24-teller (TLR).

Type	Description	BASE24 Product				
		ATM	FHM	POS	TB	TLR
0100	Authorization Request			✓	✓	
0110	Authorization Request Response			✓	✓	
0120	Authorization Advice			✓	✓	
0121	Authorization Advice Repeat			✓	✓	
0130	Authorization Advice Response			✓	✓	
0200	Financial Transaction Request	✓		✓	✓	✓
0205	Statement Print Request	✓				
0210	Financial Transaction Request Response	✓		✓	✓	✓
0215	Statement Print Response	✓				
0220	Financial Transaction Advice	✓		✓	✓	✓
0221	Financial Transaction Advice Repeat	✓		✓	✓	✓
0230	Financial Transaction Advice Response	✓		✓	✓	✓
0300	File Inquiry/Update Request					✓
0300	File Update Request		✓			
0301	File Update Request Repeat		✓			

Type	Description	BASE24 Product				
		ATM	FHM	POS	TB	TLR
0310	File Inquiry/Update Response					✓
0310	File Update Response		✓			
0320	File Inquiry/Update Advice					✓
0321	File Inquiry/Update Advice Repeat					✓
0330	File Inquiry/Update Advice Response					✓
0402	Card Issuer Reversal Request			✓		
0412	Card Issuer Reversal Request Response			✓		
0420	Acquirer Reversal Advice	✓		✓	✓	✓
0421	Acquirer Reversal Advice Repeat	✓		✓	✓	✓
0430	Reversal Advice Response	✓		✓	✓	✓
0500	Acquirer Reconciliation Request			✓		
0510	Acquirer Reconciliation Request Response			✓		
0520	Acquirer Reconciliation Advice			✓		
0521	Acquirer Reconciliation Advice Repeat			✓		
0530	Acquirer Reconciliation Advice Response			✓		
0600	Administrative Request					✓
0610	Administrative Request Response					✓
0620	Administrative Advice					✓
0621	Administrative Advice Repeat					✓

Type	Description	BASE24 Product				
		ATM	FHM	POS	TB	TLR
0630	Administrative Advice Response					✓
0800	Network Management Request	✓		✓	✓	✓
0810	Network Management Request Response	✓		✓	✓	✓

Enhanced EMV Response Code Mapping

The ISO Host Interface process maps internal BASE24-atm and BASE24-pos response codes to and from ISO-based response codes carried in the BASE24 external message. Two types of mapping are supported: standard mapping and enhanced EMV mapping.

The difference between standard mapping and enhanced EMV mapping is how EMV-related response codes are handled. Using standard mapping, EMV-related response codes are mapped to more general response codes which can obscure the reason an EMV transaction is declined. Using enhanced EMV mapping, however, unique ISO response codes are used that allow EMV-related response codes to retain their EMV characteristics. For example, an outbound BASE24-atm EMV response code of 084 would map to 05 under standard mapping and U4 under enhanced EMV mapping as shown in the table below. Where the former does not retain the specific reason for the response (due to an ATC check failure), the latter does.

BASE24-atm	Standard Mapping	Enhanced EMV Mapping
084 ATC check failure	05 Do not honor	U4 ATC check failure

For a complete list of BASE24-atm and BASE24-pos response code mappings, refer to the ***BASE24 External Message Manual***.

The type of mapping used by the ISO Host Interface process is controlled by the ENHANCED-EMV-RC-MAPPING param in the LCONF. Param values control the type of mapping as follows:

Param Value	Response Code Mapping	
	BASE24-atm	BASE24-pos
0	Standard Mapping	Standard Mapping
1	Enhanced EMV Mapping	Standard Mapping
2	Standard Mapping	Enhanced EMV Mapping
3	Enhanced EMV Mapping	Enhanced EMV Mapping

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2: Configuration and Control

This section describes the configuration and control of the ISO Host Interface process and includes the following topics:

- Descriptions of files used by the ISO Host Interface process
- Process configuration information
- Descriptions of institution- and DPC-level controls
- A description of how stations are defined and used by the ISO Host Interface process
- Descriptions of the timers used by the ISO Host Interface process and the processing performed when each timer expires

LCONF Assigns and Params

The ISO Host Interface process uses a number of Logical Network Configuration File (LCONF) assigns and params in its processing. These assigns and params are identified below. For more information on configuring the ISO Host Interface process using these assigns and params, refer to the ***BASE24 Logical Network Configuration File Manual***.

Assigns

The following assigns identify the names of files that contain ISO Host Interface process configuration and control information, or that the ISO Host Interface process uses during processing. The ISO Host Interface process reads these assigns regardless of the products supported.

EMF — Identifies the name of the External Message File (EMF).

HCF — Identifies the name of the Host Configuration File (HCF).

KEY6 — Identifies the name of the Key 6 File (KEY6).

KEYF — Identifies the name of the Key File (KEYF).

LMCF — Identifies the name of the Log Message Configuration File (LMCF).

SAF — Identifies the name of the Store-and-Forward File (SAF).

SECURE-DEV1

SECURE-DEV2 — Identify the names of the security modules the ISO Host Interface process uses if BASE24 PIN management, message authentication, message encryption, or card verification is performed using a security module. If BASE24 PIN management, message authentication, message encryption, or card verification is performed using a security module, at least one of these assigns is required. For more information on configuring these assigns, refer to the ***BASE24 Transaction Security Manual***.

TKN — Identifies the name of the Token File (TKN).

Params

The following params contain configuration information for the ISO Host Interface process. Unless otherwise noted, the ISO Host Interface process reads these params regardless of the products supported.

ACCT-NUM-TYP — Specifies whether account numbers are carried in the system in decimal or binary format. This param is only read if BASE24-atm or BASE24-pos support is bound in to the ISO Host Interface process.

ADJ-ADVICE-RC — Specifies for outgoing BASE24-pos 0220 adjustment messages whether the transaction response code or reversal code is to be included in the P-39 (Response Code) data element of the external message. Typically, only a response code or reversal code (one or the other) is present and meaningful for a transaction. For example, in a normal response message, the response code is present and meaningful, and in an approved adjustment/reversal request/advice, the reversal code is present and meaningful. However, in the case of declined adjustment advices, both a reversal code (the reason for the adjustment) and a response code (the reason for the decline) are present in the PSTM. Since declined adjustments are supported by BASE24-pos and can be sent to a host, hosts are provided the option of including the response code or reversal code in the P-39 external message data element for 0220 adjustment messages. Valid values are as follows:

- 1 = Reversal code (default)
- 2 = Reversal code for approved advices; response codes for declined advices
- 3 = Response code

ATM-LN-CUTOVER — Specifies the logical network cutover time for the BASE24-atm product. This param is only read if BASE24-atm support is bound in to the ISO Host Interface process.

CICS-ID-OFFSET — The message offset at which the ISO Host Interface process starts checking for control messages when it receives a message from an IBM CICS host. If the message value starting at the indicated offset is DFH or DFS, the message is a control message and is ignored. This param is used for DPCs using the SNA/CICS protocol.

DES-TRIPLE-SINGLE — Specifies the actions for the ISO Host Interface process to take when the interface must support the Triple Data Encryption Algorithm (T-DEA) without using the Security Services process.

ENHANCED-EMV-RC-MAPPING — A code indicating whether enhanced EMV response code mapping is used. Certain BASE24-atm and BASE24-pos response codes are used to identify EMV-specific failure conditions. Without enhanced EMV response code mapping, these response codes are mapped to a generic ISO response code (e.g., 05—do not honor), and the acquiring system often cannot identify the reason for a declined EMV transaction. With enhanced EMV response code mapping, unique ISO response codes are added to identify EMV-specific failure conditions. Valid values are as follows:

- 0 = Do not perform enhanced EMV response code mapping. (This is the default value.)
- 1 = Perform enhanced EMV response code mapping for ATM transactions only.
- 2 = Perform enhanced EMV response code mapping for POS transactions only.
- 3 = Perform enhanced EMV response code mapping for ATM and POS transactions.

HOST-SELECTED-ACCT — For host-acquired BASE24-atm transactions, this param indicates whether the transaction is prescreened when account numbers are already present in the 0200 request message sent to the BASE24 system.

LOGICAL-NET — Specifies the name of the logical network of which the ISO Host Interface process is a part.

MAIL-HISO-GEN-TRACE-NUM — A flag that indicates whether the ISO Host Interface process is to generate a mail trace number and insert it into the headers of messages it processes or allow the host to generate and insert the trace number.

MAIL-SEND-RESTRICTED-BY-FIID — A code that indicates whether the ISO Host Interface process is restricted from sending mail messages to terminals owned by another FIID.

MULT-LN-PIN-KEY — A value containing PIN encryption key information used to reencrypt PINs on transactions passed between logical networks in a BASE24 system with multiple logical networks.

PRE-VERIFIED-PIN-PAD-CHAR — Specifies the value to be repeated in a PIN block to identify the PIN as pre-verified. The default value is F, but it is possible (although highly unlikely) for an actual encrypted PIN to result in a PIN block value of all Fs. If used, this value should be set to a non-hexadecimal character (i.e., a character other than 0-9 or A-F) because the use of any repeated hexadecimal character could have the same issue as the F character. If this value is set, the host must also be modified to recognize the new pre-verified PIN block character.

BASE24-pos Note: BASE24-pos transactions may or may not contain a PIN, and authorizing systems must be able to differentiate between transactions with a pre-verified PIN and transactions that do not contain a PIN (i.e., that were not entered with a PIN). Therefore, for incoming and outgoing BASE24-pos transactions, a PIN block containing this LCONF pad character is always used to indicate a PIN has been pre-verified. The absence of the P-52 (Personal Identification Number Data) data element in the external message is used to indicate the transaction did not contain a PIN (for incoming BASE24-pos messages, a space-filled P-52 data element is also acceptable to indicate the transaction did not contain a PIN).

BASE24-atm and BASE24-teller Note: BASE24-atm and BASE24-teller transactions always require a PIN. Therefore, for these product transactions, a pre-verified PIN can be accurately identified by the absence of the P-52 (Personal Identification Number Data) data element in the external message. For outgoing messages, BASE24-atm and BASE24-teller always identify a pre-verified PIN by omitting the P-52 data element. For incoming messages, the host has several options—BASE24-atm and BASE24-teller assumes a PIN has been pre-verified if the P-52 data element is omitted, contains spaces, or is filled by the pad character specified in this param.

POS-LN-CUTOVER — Specifies the logical network cutover time for the BASE24-pos product. This param is only read if BASE24-pos support is bound in to the ISO Host Interface process.

POS-RVSL-ROUTE — Specifies whether the BASE24-pos Router module and acquirer HISO use normal or enhanced routing when sending reversal transactions to Interchange Interface processes acting as transaction authorizers (issuers).

PSEUDO-3DES-CBC-MAC-TYPE — Specifies whether to use the Pseudo 3DES Cipher Block Chaining method of MACing instead of the default 3DES Cipher Block Chaining method used by the ISO Host Interface process.

Note: This param is not used if the BASE24 Transaction Security Services (TSS) process is being used.

RACAL-ANSI-METHOD — Specifies whether to use the ANSI X9.17 Encryption Method for key exchanges instead of the default Variant Encryption Method used by the ISO Host Interface process. The ANSI X9.17 Encryption Method can be used with a Thales (formerly Racal) Hardware security module.

Note: This param is not used if the BASE24 Transaction Security Services (TSS) process is being used.

REVERSE-BAL-INQ — Specifies whether balance inquiry transactions can be reversed. This param is only read if BASE24-atm or BASE24-pos support is bound in to the ISO Host Interface process.

RSM-TERM-MASTER-KEY — Specifies an encrypted terminal master key used during hardware message authentication by Thales (formerly Racal) security devices.

Note: This param is not used if the BASE24 Transaction Security Services (TSS) process is being used.

SAF-NOTIFY-INTERVAL — Specifies the interval, in number of messages written to the SAF, at which the ISO Host Interface process calculates how full the SAF is.

SAF-NOTIFY-PERCENTAGE — Specifies the percentage of file full at which the ISO Host Interface process begins generating event messages indicating that the SAF is becoming full.

SAF-PRIOR-TO-SEND — A Y (yes) or N (no) code indicating whether or not the Host Interface process places transactions in the Store-and-Forward File (SAF) prior to sending them to the host.

SECURE-DEV-CONF-CMD100 — Specifies the command text sent to Atalla Network Security Processors (NSPs) when the BASE24 ISO Host Interface process is initialized or warmbooted. For more information on configuring this param, refer to the *BASE24 Transaction Security Manual*.

SECURE-DEV-CONF-TXT — Specifies the configuration text sent to Atalla NSPs when the BASE24 ISO Host Interface process is initialized or warmbooted. For more information on configuring this param, refer to the *BASE24 Transaction Security Manual*.

SECURE-DEV-FAIL-NOTIFY — Specifies the interval, in number of transactions, at which the ISO Host Interface process logs messages once a security module is determined to be inoperative. For more information on configuring this param, refer to the *BASE24 Transaction Security Manual*.

SECURE-DEV-MONITOR-THRESHOLD — Specifies the interval, in number of transactions performed by the security module defined in the SECURE-DEV1 assign, at which the security module defined in the SECURE-DEV2 assign is checked. For more information on configuring this param, refer to the *BASE24 Transaction Security Manual*.

SECURE-DEV-TIM-LMT — Specifies the number of seconds the ISO Host Interface process waits for a response from the security module. For more information on configuring this param, refer to the *BASE24 Transaction Security Manual*.

SECURE-DEV-TYP — Specifies the type of security modules named in the SECURE-DEV1 and SECURE-DEV2 assigns. For more information on configuring this param, refer to the *BASE24 Transaction Security Manual*.

SEND-ETX — A flag indicating whether an interchange interface process sends the end-of-text (ETX) control character.

TLR-LN-CUTOVER — Specifies the logical network cutover time for the BASE24-teller product. This param is only read if BASE24-teller support is bound in to the ISO Host Interface process.

Configuring a DPC

Each DPC with which the BASE24 system communicates requires a record in the Host Configuration File (HCF). DPCs are assigned their DPC numbers and are tied to the ISO Host Interface processes with which they communicate in the HCF.

A number of processing options are controlled at the DPC level, which enable institutions to set up DPCs with individual processing requirements. The majority of processing options available at the DPC level are stored in the HCF. Other options are specified in the KEYF or KEY6. The processing options controlled at the DPC level are described below.

Specifying Timer Lengths

The ISO Host Interface process uses internal timers to control host message traffic. The length of each of these timers is specified in the HCF or KEYF. Some timer intervals are specified once at the DPC level, while others are specified at the DPC and product level (for example, you can specify one time limit for inbound BASE24-atm messages and a different time limit for inbound BASE24-pos messages).

The table below identifies each timer used by the ISO Host Interface process, along with the level at which it is defined and the file in which it is defined. Each of these timers is discussed later in this section.

Timer	DPC Level	DPC/ Product Level
Network management message timer	HCF	
Extended network management message timer	HCF	
Store-and-forward message timer	HCF	
Store-and-forward pacing timer	Set by command	
Wait-for-traffic timer	HCF	
Performance timer	HCF	
Outbound request timer		HCF

Timer	DPC Level	DPC/ Product Level
Inbound request timer		HCF
Completion acknowledgment timer		HCF
PIN key exchange timer	KEYF or KEY6	
MAC key exchange timer	KEYF or KEY6	
Clear old key timer	KEYF or KEY6	

Interspersing Store-and-Forward Messages

Institutions can specify how the host receives store-and-forward messages. Store-and-forward messages can be interspersed with request messages, or all store-and-forward messages can be sent to the host prior to any further requests.

Interspersing store-and-forward messages allows other transactions to be processed during store-and-forward activity, while receiving all store-and-forward messages prior to further requests may cause requests to be processed offline or be declined because of an inability to send them to the host.

Institutions use the STORE AND FORWARD field on HCF screen 1 to specify the method of store-and-forward processing.

Specifying Single-Threaded and Multithreaded Processing

DPC stations can be *single-threaded* or *multithreaded*. Single-threaded means that only one message of any kind can be outstanding to a station at a time. In this case, the ISO Host Interface process system must receive a response or the message must time out before another message can be sent to the station. Multithreaded means that one or more messages of any kind can be outstanding to a single station at one time. In this case, a response to one message is not required before another message can be sent to the station.

Single-threading and multithreading is allowed for both issuer and acquirer hosts, although the ISO Host Interface process only controls the threading for issuer hosts.

Institutions use the DPC TYPE field on HCF screen 1 to specify the type of threading the ISO Host Interface process uses. Because this setting is at the DPC level, all stations associated with that DPC are either single-threaded or multithreaded.

Note: Acquirer hosts can use single-threading or multithreading without regard for the DPC TYPE field setting.

Sending Text-Level Acknowledgments to a DPC

Acquirer institutions must specify whether the host expects text-level acknowledgments from the BASE24 system. If text-level acknowledgments are required, the ISO Host Interface process returns a 0130 message for each 0120 or 0121 message; a 0230 message for each 0220 or 0221 message; a 0430 message for each 0420 or 0421 message; and a 0530 message for each 0520 or 0521 message.

Institutions use the ACK TO DPC field on HCF screen 1 to indicate whether text-level acknowledgments are required to be sent.

Expecting Text-Level Acknowledgments from a DPC

Issuer institutions must indicate whether the ISO Host Interface process should expect to receive text-level acknowledgments. If text-level acknowledgments are to be sent, the ISO Host Interface process expects a 0130 message for each 0120 or 0121 message; a 0230 message for each 0220 or 0221 message; a 0330 message for each 0320 or 0321 message; a 0430 message for each 0420 message; a 0530 for each 0520 or 0521 message; and a 0630 for each 0620 or 0621 message.

Institutions use the ACK FROM DPC field on HCF screen 1 to indicate whether hosts send text-level acknowledgments to the BASE24 system.

Note: The ISO Host Interface process always requires acknowledgments for its store-and-forward messages. If text-level acknowledgments are not sent, the ISO Host Interface process requests a *logical acknowledgment* from the XPNET process. A logical acknowledgment notifies the ISO Host Interface process that the message was sent successfully to the host.

If message authentication is performed, the setting for the ACK FROM DPC field should be carefully considered. Some message verification errors can be retried (for example, if the host's security device is down at the time it receives the message to be authenticated). If this field indicates that text-level acknowledgments are not required and the message was a store-and-forward message, the original message will not be available to resend to the host. This is because the message would have been deleted when the XPNET process returned a logical acknowledgment that the message was sent successfully.

Specifying Maximum Timeouts

Institutions must indicate the number of consecutive timeouts of non-network management and non-store-and-forward messages that can occur at a station before the station is marked down. For example, if three consecutive timeouts are specified, the ISO Host Interface process marks the station down upon the third consecutive timeout. However, if only two consecutive timeouts have occurred, the station is not marked down.

Institutions use the MAX TIMEOUTS field on HCF screen 1 to control when a station is marked down because of timeouts.

Enabling Network Management

Institutions specify whether the ISO Host Interface process is allowed to initiate network management messages (that is, 0800 messages for logon, logoff, or echo-test). This option can limit the ability of the ISO Host Interface process to log on and solicit communications with a host.

Institutions use the NMM ENABLED field on HCF screen 1 to indicate whether the ISO Host Interface process can initiate network management messages.

Note: This option only controls whether the ISO Host Interface process can initiate the network management actions; the ISO Host Interface process always responds to the network management messages it receives.

Also, this option does not control whether the ISO Host Interface process initiates network management messages used for dynamic key management (that is, 0800 messages for new key, change key, verify key, or repeat key functions).

Enabling Dynamic Key Management

Institutions specify whether the ISO Host Interface process is allowed to initiate network management messages for dynamic key management (DKM). This includes 0800 messages for new key, change key, verify key, or repeat key functions. This option limits the ability of the ISO Host Interface process to perform or solicit key management requests.

Network management messages for dynamic key management are controlled by the timers and thresholds on KEYF screen 3 or KEY6 screen 4. If all of the timers and thresholds on KEYF screen 3 are set to the value 0, the ISO Host Interface process does not initiate network management messages for DKM. If any of the timers or thresholds contains a nonzero value, the ISO Host Interface process initiates a key management message when the timer expires or when the threshold is passed.

Specifying Maximum Outstanding Store-and-Forward Messages

The BASE24 system allows institutions to control the maximum number of store-and-forward messages that can be outstanding to a host before the BASE24 system stops sending additional messages. Outstanding messages are those messages that have not timed out, but for which the required responses have not been received (for example, a 0230 message for a 0220 message).

This option allows the ISO Host Interface process to initiate multithreaded store-and-forward processing. For more information on issues to consider when establishing multithreaded store-and-forward processing, refer to section 4.

Once the number of outstanding store-and-forward messages reaches this limit, the ISO Host Interface process does not send any additional store-and-forward messages until the number of outstanding messages has decreased.

Institutions use the MAX OUTSTANDING SAFS field on HCF screen 1 to specify the maximum number of outstanding store-and-forward messages to a host.

Specifying Maximum Store-and-Forward Retries

Institutions set the maximum number of times the ISO Host Interface process attempts to send a store-and-forward message to a host before writing that message in an event message to the network logging facility and deleting it from the SAF.

This option can be used to stop processing a message when, due to the number of timeouts on a store-and-forward message, it becomes apparent that the host cannot process the message. Institutions control this option with the MAX SAF RETRY field on HCF screen 1.

Specifying Maximum Outbound Messages

The BASE24 system allows institutions to specify the maximum number of request messages (that is, 0100, 0200, 0300, 0402, 0500, and 0600 messages) or advice messages (that is, 0120, 0220, 0320, 0420, 0520, and 0620 messages) that can be outstanding to an issuer host before the BASE24 system stops sending additional messages. Outstanding messages are those messages that have not timed out, but for which the required responses have not been received (for example, a 0210 message for a 0200 message or a 0230 message for a 0220 message).

This option allows the ISO Host Interface process to prevent processing when it becomes apparent that the host cannot process a message within the time limits required because of the number of messages already outstanding to the host.

Once the number of outstanding messages reaches this limit, the ISO Host Interface process marks any new request messages destined for the host as failed and returns them to their originating processes. The ISO Host Interface process places any new advice messages destined for the host in the SAF. At this point, the host does not receive additional request or advice messages until the messages in the queue are either responded to or have timed out.

Institutions use the OUTBOUND field on HCF screen 1 to control the maximum number of outstanding request and advice messages to a host.

Specifying Maximum Inbound Requests

The BASE24 system also provides a parameter for limiting the number of outstanding 0100 and 0200 messages to BASE24 Authorization processes. This parameter controls requests sent from acquirer hosts. Once the number of

outstanding 0100 and 0200 messages reaches this limit, the ISO Host Interface process declines any new 0100 and 0200 messages coming from the host and returns them as 0110 and 0210 messages.

The maximum number of outstanding requests to an Authorization process is controlled by the value in the INBOUND field on HCF screen 1.

Note: This feature affects only 0100 and 0200 messages. Other message types are not included in the count of outstanding messages, nor is their processing affected in any way.

Checking for Time Discrepancies Between the BASE24 System and the Host

Institutions can specify whether the ISO Host Interface process should check incoming 0800 messages for time discrepancies. If the messages are checked and the difference between the time the message was sent and the time the message was received is greater than five minutes, a warning event message is logged.

Institutions use the TIME DISCREPANCY CHECK field on screen 1 of the HCF to select the option to check for time discrepancies.

Converting Messages from ASCII to EBCDIC

The ISO Host Interface process can send messages to a DPC in ASCII or EBCDIC format. Institutions select the character format to be used by specifying a value of A (ASCII) or E (EBCDIC) in the CHARACTER FORMAT field on screen 1 of the HCF.

The ISO Host Interface process creates messages in ASCII format. If the CHARACTER FORMAT field is set to E, the ISO Host Interface process converts the messages to EBCDIC prior to sending them to the host. Similarly, if this field is set to E, responses from the DPC are expected in EBCDIC and converted by the ISO Host Interface process to ASCII. This option applies to all messages (i.e., transaction messages and network management messages).

For outbound messages, when message authentication and full message encryption are also being performed, creation of the MAC occurs first, followed by the ASCII-to-EBCDIC conversion, followed by the message encryption. For inbound messages, that order is reversed.

Where the SNA protocol is in use between BASE24 and a host, the ASCII-to-EBCDIC conversion can be performed at the controller level. However, if an institution is using full message encryption, they should not have the ASCII-to-EBCDIC conversion performed at the controller level, but rather use this ISO Host Interface option—because the conversion must take place prior to encrypting the message.

Tracking and Reporting DPC Performance Statistics

During processing, the ISO Host Interface process tracks the following performance statistics for each DPC.

Outbound Statistics	Inbound Statistics
Number of transaction messages sent to the DPC	Number of transaction messages sent from the DPC.
Number and percentage of transaction messages sent to the DPC that timed out.	Number and percentage of transaction messages sent from the DPC that timed out at the ISO Host Interface process before a BASE24 response was returned.
Average response time of the transactions messages sent to the DPC (in seconds, to the hundredth).	Average response time of the transactions messages sent from the DPC (in seconds, to the hundredth). Response time is the amount of time it takes for the BASE24 response to return to the ISO Host Interface process.
Number of transaction messages that could not be sent to the DPC (due to problems such as format errors, stations down, etc.).	Number of transaction messages sent from the DPC that could not be sent to a BASE24 authorization process (due to problems such as format errors, message errors, etc.).

Note: Transaction message counts include requests, advices, and reversals—essentially all messages except network management and store-and-forward messages.

Performance statistics are tracked by interval on a rolling basis and are available from the ISO Host Interface process for the most current four intervals. The length of the tracking interval is controlled by a performance timer which triggers the end of one interval and the beginning of the next. The timer length is specified, in minutes, in the PERFORMANCE PERIOD field on HCF screen 1.

Performance Statistics Reporting

Performance statistics are reported in a series of event messages generated by the ISO Host Interface process. These event messages can be generated operationally by entering one of the following commands from a network control facility.

- PERFORMANCE – the event messages generated by this command include, by tracking interval, all of the performance statistics listed in the table above for the DPC specified in the command.
- PERFSTAT– the event messages generated by this command include, by tracking interval, only the outbound performance statistics listed in the table above for the DPC specified in the command. Inbound performance statistics are not included. However, additional event messages are generated to provide the following information for the DPC:
 - ❑ Whether the host link is marked up or down.
 - ❑ Whether or not the host is logged on.
 - ❑ The number of host stations defined for the DPC and how many of the stations are up.
 - ❑ Number of store-and-forward records that need to sent to the DPC and how many are in progress.

For information concerning the entry of the PERFORMANCE and PERFSTAT command, refer to the *BASE24 Text Command Reference Manual*.

Automated Performance Statistics Reporting

If the ENHANCED STATUS field on screen 1 of the HCF is set to Y, the ISO Host Interface process will automatically generate the PERFSTAT command event messages each time the performance timer expires.

Identifying Data Prefix Characters

The BASE24 products allow institutions to define certain characters they want included in front of the messages they receive from the BASE24 system. An institution can specify up to nine characters to precede messages to their host system. Each time the ISO Host Interface process creates an external message to a DPC, it checks the HCF for data prefix characters. If the HCF contains data prefix characters for the DPC, the ISO Host Interface process places these characters at the front of the message.

Data prefix characters are placed in the HCF using the hexadecimal equivalents for the ASCII versions of the characters to be sent. For example, suppose a host required the following three characters in front of the message:

ESC 1 C
(escape)

To place these three values in an HCF record, you would have to enter hexadecimal equivalents (six characters) for the ASCII representations of these values, which would be:

1B 31 43

These values would be stored in the HCF as six characters (1B3143) and would be converted back to the three ASCII characters at the time the message is created.

Because a BASE24 system does not support transparent data communications, protocol characters must not be used as data prefix characters. Some characters to avoid are as follows:

HEX	ASCII DISPLAY	HEX	ASCII DISPLAY
01	SOH	02	STX
03	ETX	04	EOT
05	ENQ	06	ACK
10	DLE	15	NAK
16	SYN	17	ETB
1F	ITB		

Note: Data prefix characters are converted from ASCII to EBCDIC along with the rest of the message if the CHARACTER FORMAT field on screen 1 of the HCF is set to E.

Identifying Authorization Processes

ISO Host Interface processes handling requests from acquirer networks determine the Authorization processes to which to route inbound transaction requests using the AUTH PROCESS field on HCF screen 5 for BASE24-atm, HCF screen 7 for BASE24-pos, or HCF screen 10 for BASE24-teller. ISO Host Interface processes handling requests from acquirer networks determine the Integrated Authorization Server process to which to route inbound BASE24 Remote Banking transaction requests using the AUTH PROCESS field on HCF screen 22.

Setting Up DPC Stations

Stations serve as the links between the BASE24 system and a host. All messages sent to a DPC are actually sent to the stations defined for that DPC. Likewise, all messages received from a DPC are actually received from the stations defined to that DPC.

Up to 32 stations can be defined for each DPC, although a recommended maximum is 10 to 12. The Network Environment File (NEF) contains the attribute configuration for all lines, stations, processes, and devices in each XPNET node. You must first configure a station as a host station in the XPNET node. The XPNET process uses the information in the NEF to route messages received from the station to the ISO Host Interface process.

The symbolic name of the station, as defined in the NEF, is placed in the HCF to fully define the station to the ISO Host Interface process. The ISO Host Interface process uses the information in the HCF only; it does not read the NEF.

The ISO Host Interface process reads the HCF at initialization and builds a table of the stations defined for each DPC.

Station Types

Hosts can define three types of stations for use by the ISO Host Interface process: send/receive stations, receive-only/no output stations, and statement print send/receive stations. These station types are described below.

Station types are defined using the TYPE fields on HCF screen 2.

Send/Receive Stations

Send/receive stations allow for both incoming and outgoing messages. The ISO Host Interface process can send messages to and receive messages from the host using this type of station.

Receive-Only/No Output Stations

Receive-only/no output stations only allow for messages coming in to the BASE24 system. The ISO Host Interface process never uses receive-only/no output stations for sending messages to the host (this includes network management messages).

Receive-only/no output stations are used to avoid line contention when using a point-to-point or SNA protocol. In this case, communications are received on one station and sent from a different station.

Statement Print Send/Receive Stations

Statement print send/receive stations are special use stations that allow for incoming and outgoing statement print transaction messages only. Statement print send/receive stations can only be used in conjunction with BASE24-atm Self-Service Banking (SSB), and are designed to segregate statement print message traffic from other transactions to a host, which may be desirable due the size and number of messages involved in each statement print transaction.

If a statement print send/receive station is defined, the ISO Host Interface process will use only this type of station for sending SSB statement print transaction messages to the host (subject to the alternate routing option described below). The ISO Host Interface process will not use this type of station for sending any other type of financial messages to the host.

For the purposes of network management, statement print send/receive stations are treated the same as send/receive stations, with the exception that they are not used for dynamic key management.

Caution: Although statement print send/receive stations can only be used in conjunction with BASE24-atm SSB, it is actually possible to define this type of station in the HCF without BASE24-atm SSB. In this case, the ISO Host Interface process would start and manage the station normally; however, since it would never receive SSB statement print transactions, it would never actually use the station.

Alternate Routing for Statement Print Send/Receive Stations

If a transaction is a statement print transaction and the BASE24-atm SSB Base Application is installed, the ISO Host Interface process will attempt to send the message to the host by way of a statement print send/receive station if one has been defined.

If a statement print send/receive station is unavailable for any reason, the ISO Host Interface process can take one of two actions: deny the transaction with a response code of 72 (destination not available), or send it to the host using another available send/receive station. Users specify how this situation is to be handled using the STATEMENT PRINT ALTERNATE ROUTE field on HCF screen 20.

Setting Up the HCF Station Table

Stations are numbered from left to right, top to bottom on the HCF screen 2. This numbering scheme is of particular importance to hosts with multiple stations on their lines.

When selecting a station to which to send a message, the ISO Host Interface process goes through its station table from start to finish until it finds a station available to receive the message.

If lines have multiple stations on them, the stations from each line should be alternated when setting up the HCF station table, and not numbered sequentially. Alternating the stations balances the transaction load so that if one station is chosen from one line, the next station chosen is on a different line.

Selecting Stations for Sending Messages

To select the station to which to send a message, the ISO Host Interface process begins with the first station in the HCF station table for that DPC and evaluates each station based on the following criteria:

- It must be a send/receive station and the station must be marked up. The ISO Host Interface process never sends messages to receive-only/no output stations.
Note: If a transaction is a statement print transaction and the BASE24-atm SSB Base Application is installed, the ISO Host Interface process searches for a statement print send/receive station if one has been defined.
 - ❑ If one has been defined and it is marked up, the ISO Host Interface process will use that station.
 - ❑ If one has been defined but it is not marked up, the ISO Host Interface process will either deny the transaction or use another send/receive station as determined by the value in the STATEMENT PRINT ALTERNATE ROUTE field on HCF screen 20.
 - ❑ If one has not been defined, the ISO Host Interface process will use a send/receive station.
- If the DPC is single-threaded, the station cannot be busy; that is, it must not have a message pending.

- If the DPC is multithreaded, the station must have the least number of messages pending in relation to other stations.

The ISO Host Interface process sends the message to the first station that meets the above criteria. To send the next message, the ISO Host Interface process performs station selection in the same manner, starting with the next station in the table. For example, if the last message was sent to the second station in the table, the ISO Host Interface process would begin station selection with the third station.

Configuring Timers

The ISO Host Interface process uses internal timers to control host message traffic. The length of each of these timers is specified in the HCF.

Network Management Message Timer

A network management message timer is set each time an 0800 message is sent to a DPC station. The network management message timer limits the amount of time that the ISO Host Interface process waits for a response after transmitting an 0800 message. The number of seconds to use for this timer is taken from the NETWORK MANAGEMENT field on HCF screen 1.

The actions taken when a network management message timer expires depend on the type of message for which the timer was set. If the 0800 message was a logon, logoff, or echo-test message, the ISO Host Interface process performs the following steps:

1. Marks the station down.
2. Verifies that the DPC still has a station available for sending messages. If no send/receive stations are left up for the DPC, the ISO Host Interface process marks the DPC down.
3. Sets an extended network management message timer for the station.

If the 0800 message was a DKM message, the ISO Host Interface process performs the following steps:

1. Adds one to the count of consecutive timeouts accumulated for the station.
2. Checks the number of consecutive timeouts accumulated for the station against the maximum allowed by performing the following steps. The maximum allowed is specified in the MAX TIMEOUTS field on HCF screen 1.
 - a. If the current number of consecutive timeouts is less than the maximum allowed, processing continues with step 3.
 - b. If the current number of consecutive timeouts is equal to the maximum allowed, the ISO Host Interface process performs the following steps:
 - 1) Marks the station down.

- 2) Verifies that the DPC still has a station available for sending messages. If no send/receive stations are left up for the DPC, the ISO Host Interface process marks the DPC down.
3. Sets an extended network management message timer for the station.
4. If the message that timed out was a new key or a change key message, sends an event message to the network logging facility indicating that the message expired and that the message will automatically be retried.

Extended Network Management Message Timer

An extended network management message timer is set each time a DPC station is marked down and each time a new key or change key message times out. The extended network management message timer specifies the following intervals:

- The amount of time that the ISO Host Interface process waits after a station is marked down before attempting to reestablish communications with the station.
- The amount of time that the ISO Host Interface process waits after a new key or change key message times out before attempting to resend the message.

The number of seconds to use for this timer is taken from the EXTENDED NETWORK field on HCF screen 1.

If an extended network management message timer that was set when a station was marked down expires before any messages are received from the station, the ISO Host Interface process performs the following steps:

1. Sends an echo-test message to the station.
2. Sets a network management message timer for the echo-test message, as follows:

When an extended network management message timer that was set after a new key or change key message timed out expires, the ISO Host Interface process determines whether there is a station available to which to send the message as follows:

- If a station is available, the ISO Host Interface process sends the new key message or change key message to that station and sets a network management message timer to wait for the response to the new key or change key message.

- If no stations are available, the ISO Host Interface process sets another extended network management message timer. When this timer expires, the ISO Host Interface process attempts to send the message again.

Store-and-Forward Message Timer

A store-and-forward message timer is set each time a store-and-forward message is sent to a DPC. The store-and-forward message timer limits the amount of time that the ISO Host Interface process waits for an acknowledgment after transmitting a store-and-forward message. The number of seconds to use for this timer is taken from the STORE AND FORWARD field on HCF screen 1.

The ISO Host Interface process only expects text-level acknowledgments to store-and-forward messages if the ACK FROM DPC field on HCF screen 1 contains the value Y. If text-level acknowledgments are not required from the DPC, the ISO Host Interface process expects a logical acknowledgment from the XPNET process instead. In this case, the XPNET process returns a logical acknowledgment once the message has been sent successfully to the DPC. When logical acknowledgments are used, if the host rejects the message for any reason, the message will not be resent because it will have already been deleted from the SAF.

If this timer expires before an appropriate acknowledgment is received in response, the ISO Host Interface process performs the following steps:

1. Reads the message from the SAF.
2. Adds one to the SAF retries count for that record. The SAF retries count indicates the number of times the SAF record has been sent to the host. A separate count is kept for each SAF record.
3. Checks the SAF retries count in the record against the maximum specified in the MAX SAF RETRIES field on HCF screen 1, as follows:
 - a. If the SAF retries count is equal to the maximum allowed, the ISO Host Interface process performs the following steps:
 - 1) Writes the SAF record in an event message to the network logging facility.
 - 2) Deletes the record from the SAF.
 - b. If the SAF retries count in the record is less than the maximum allowed, the ISO Host Interface process continues with step 4.

4. Marks the SAF message to be resent.
5. Updates the message type, if necessary.

If the message type is 0120, 0220, 0320, 0420, 0520, or 0620, the ISO Host Interface process reads the corresponding record in the SAF and updates the message type in place to 0121, 0221, 0321, 0421, 0521, or 0621 respectively. Message types 0402 and 0412 are not changed.

Store-and-Forward (SAF) Pacing Timer

Store-and-Forward (SAF) pacing timers are only set if SAF pacing is enabled for a DPC. The SAF pacing timer controls the amount of time (in hundredths of seconds) that the ISO Host Interface process waits between sending store-and-forward messages to the DPC.

If SAF pacing is enabled, a SAF pacing timer is first set when SAF pacing is enabled and then subsequently each time the current SAF pacing timer expires. As long as SAF pacing is still enabled, a SAF pacing timer will be set regardless of whether there are any SAF records in the file. If there are SAF records to send, one record gets sent upon expiration of this timer.

Use of this timer, as well as its length, is controlled operationally through the entry of the SAFPACING command from the network control facility. For information concerning the entry of the SAFPACING command, refer to the ***BASE24 Text Command Reference Manual***. For more information on SAF pacing, refer to section 4

Wait-for-Traffic Timer

A wait-for-traffic timer is set each time a station is marked up and each time a previously set wait-for-traffic timer expires. The wait-for-traffic timer specifies the amount of time that the ISO Host Interface process waits for traffic on the line before sending an echo-test message. The number of seconds to use for this timer is taken from the WAIT FOR TRAFFIC field on HCF screen 1.

The ISO Host Interface process sets wait-for-traffic timers for its send/receive stations only. Wait-for-traffic timers are not set for receive-only/no output stations.

When this timer expires, the ISO Host Interface process determines whether there has been any message traffic to or from the station since the timer was set. Depending on whether there has been traffic on the line, the ISO Host Interface process performs the following steps:

1. If there has been message traffic, the ISO Host Interface process sets another wait-for-traffic timer.
2. If there has not been message traffic, the ISO Host Interface process performs the following steps:
 - a. Sends an echo-test message to the station.
 - b. Sets a network management message timer for the echo-test message.

Performance Timer

A performance timer is set for each DPC at the start of the new performance statistics accumulation period for that DPC. The performance timer specifies the interval the ISO Host Interface process uses for gathering performance statistics. The number of minutes to use for this timer is taken from the PERFORMANCE PERIOD field on HCF screen 1.

Performance statistics are tracked on an interval basis and are available for the most current four intervals (interval zero through interval three, with interval zero being the most current). The performance timer triggers the end of one interval and the beginning of the next.

When a performance timer expires, the ISO Host Interface process performs the following steps:

1. Reports the DPC's performance statistics and information if configured to do so.

If the ENHANCED STATUS field on screen 1 of the HCF is set to Y, the ISO Host Interface process automatically generates event messages containing the DPC's performance information. Otherwise, the ISO Host Interface process skips directly to the next step.

2. Moves the performance statistics from interval two to interval three.
3. Moves the performance statistics from interval one to interval two.
4. Moves the performance statistics from interval zero to interval one.

5. Zeros out the performance statistics in interval zero to begin accumulating statistics again.
6. Sets a performance timer for the DPC.

Note: Performance statistics can be reported by entering a PERFORMANCE or PERFSTAT command from the network control facility. For information about the entry of the PERFORMANCE and PERFSTAT command, refer to the ***BASE24 Text Command Reference Manual***.

Outbound Request Timer

An outbound request timer is set each time a 0100, 0200, 0205, 0300, or 0600 message is sent to a DPC. The outbound request timer limits the amount of time that the ISO Host Interface process waits for a response from the host after transmitting a request message. The number of seconds to use for this timer is taken from the OUTBOUND field on the product-specific HCF screen.

If this timer expires before a response is received to the message, the ISO Host Interface process performs the following steps:

1. Performs product-specific processing.
 - a. If the message is a BASE24-atm message, the ISO Host Interface process checks the value of the EXTRN-RQST-TIMEOUT param in the LCONF and performs the following steps:
 - 1) If the param is set to a value of 0, the process performs the following:
 - a) Retrieves the original message from memory.

The ISO Host Interface process retains a copy of the internal message for each message it sends to the host until it is notified that the message was received.
 - b) Sets the RTE.STAT field in the Standard Internal Message (STM) to a value of 11, to indicate the destination was not available.

This identifies the 0200 message as a failure.

- c) Adjusts the RQST.COMPL-REQ field in the STM, if necessary.
 - If the RQST.COMPL-REQ field in the STM contains a value of 0 (completions not required) or 1 (approved completions required), the ISO Host Interface process sets the RQST.COMPL-REQ field in the STM to a value of 2 to indicate that an advice is required if the request is approved offline.
 - If the RQST.COMPL-REQ field in the STM contains a value of 4 (completions required), 5 (completions required for approved BASE24-authorized transactions), or 6 (completions required for all BASE24-authorized transactions - approved and denied), the ISO Host Interface process does not change the value.
 - d) Returns the 0200 message (STM) to the process indicated in the ORIG-PRO-NAME field of the STM.
- 2) If the param is set to a value of 1, the process performs the following:
- a) Retrieves the original message from memory and generates a 0420 reversal message, which it sends to the host. The Originator Code field in the external message header is set to a value of 4 (Host Interface process) and the Response Code (P-39) data element is set to a value of 1 (timeout).
 - b) Retrieves the original message from memory.
 - c) Sets the RTE.STAT field in the Standard Internal Message (STM) to a value of 11, to indicate the destination was not available.

This identifies the 0200 message as a failure.
 - d) Adjusts the RQST.COMPL-REQ field in the STM, if necessary.
 - If the RQST.COMPL-REQ field in the STM contains a value of 0 (completions not required) or 1 (approved completions required), the ISO Host Interface process sets the RQST.COMPL-REQ field in the STM to a value of 2 to indicate that an advice is required if the request is approved offline.

- If the RQST.COMPL-REQ field in the STM contains a value of 4 (completions required), 5 (completions required for approved BASE24-authorized transactions), or 6 (completions required for all BASE24-authorized transactions - approved and denied), the ISO Host Interface process does not change the value.
- e) Returns the 0200 message (STM) to the process indicated in the ORIG-PRO-NAME field of the STM.

Note: The outbound request timer is also used for 0205 messages. If this timer expires before a 0215 message is received from the host, the ISO Host Interface process creates a 0215 message with a response code of 72, declining the transaction.

- b. If the message is a BASE24-pos message, the ISO Host Interface process checks the value of the EXTRN-RQST-TIMEOUT param in the LCONF and performs the following steps:

- 1) If the param is set to a value of 0, the process performs the following:
- a) Retrieves the 0200 message from memory (all 0100 messages are handled as 0200 messages internally).

The ISO Host Interface process retains a copy of the POS Standard Internal Message (PSTM) for each 0100 and 0200 message it sends to the host until it is notified that the message was received.

- b) Determines whether an alternate routing destination has been configured. If so, sets the ORIG-PRO-NAME field of the PSTM to that destination process name, and continues processing with part e of this step. If there is no alternate destination, ensures that the current authorizer is set to AUTHH.
- c) Sets the RTE-STAT field in the PSTM to a value of 11, indicating the destination was not available.

This identifies the 0200 message as failed.

- d) Adjusts the RQST.COMPL-REQ field in the PSTM, if necessary.
 - If the RQST.COMPL-REQ field in the PSTM contains a value of 0 (completions not required) or 1 (approved completions required), the ISO Host Interface process sets the RQST.COMPL-REQ field in the PSTM to a value of 2 to indicate that an advice is required if the request is approved offline.
 - If the RQST.COMPL-REQ field in the PSTM contains a value of 4 (completions required), 5 (completions required for approved BASE24-authorized transactions), or 6 (completions required for all BASE24-authorized transactions - approved and denied), the ISO Host Interface process does not change the value.
 - e) Returns the 0200 message (PSTM) to the process indicated in the ORIG-PRO-NAME field of the PSTM.
- 2) If the param is set to a value of 1, the process performs the following:
- a) Retrieves the original message from memory and generates a 0420 reversal message, which it sends to the host. The Originator Code field in the external message header is set to a value of 4 (Host Interface process) and the Response Code (P-39) data element is set to a value of 1 (timeout).
 - b) Retrieves the 0200 message from memory (all 0100 messages are handled as 0200 messages internally).
 - c) Determines whether an alternate routing destination has been configured. If so, sets the ORIG-PRO-NAME field of the PSTM to that destination process name, and continues processing with part f of this step. If there is no alternate destination, ensures that the current authorizer is set to AUTHH.
 - d) Sets the RTE-STAT field in the PSTM to a value of 11, indicating the destination was not available.

This identifies the 0200 message as failed.

- e) Adjusts the RQST.COMPL-REQ field in the PSTM, if necessary.
 - If the RQST.COMPL-REQ field in the PSTM contains a value of 0 (completions not required) or 1 (approved completions required), the ISO Host Interface process sets the RQST.COMPL-REQ field in the PSTM to a value of 2 to indicate that an advice is required if the request is approved offline.
 - If the RQST.COMPL-REQ field in the PSTM contains a value of 4 (completions required), 5 (completions required for approved BASE24-authorized transactions), or 6 (completions required for all BASE24-authorized transactions - approved and denied), the ISO Host Interface process does not change the value.
 - f) Returns the 0200 message (PSTM) to the process indicated in the ORIG-PRO-NAME field of the PSTM.
- c. If the message is a BASE24-telebanking message, the ISO Host Interface process checks the value of the EXTRN-RQST-TIMEOUT param in the LCONF and performs the following steps:
- 1) If the param is set to a value of 0, the process performs the following:
 - a) Retrieves the original message from memory.

The ISO Host Interface process retains a copy of the internal message for each message it sends to the host until it is notified that the message was received.
 - b) Sets the FAILURE-FLG field of the Telebanking Standard Internal Message (BSTM) to a value of 1 to indicate a failure.

This identifies the 0100 or 0200 message as a failed message.
 - c) Sends the BSTM, including the internal transaction data (ITD), to the process identified in the SRC-SYM-NAM field of the BSTM.
 - 2) If the param is set to a value of 1, the process performs the following:
 - a) Retrieves the original message from memory and generates a 0420 reversal message, which it sends to the host. The Originator Code field in the external message header is set to a value of 4 (Host Interface process) and the Response Code (P-39) data element is set to indicate that the request timed out.

- b) Retrieves the original message from memory.
 - c) Sets the FAILURE-FLG field of the Telebanking Standard Internal Message (BSTM) to a value of 1 to indicate a failure.
This identifies the 0100 or 0200 message as a failed message.
 - d) Sends the BSTM, including the internal transaction data (ITD), to the process identified in the SRC-SYM-NAM field of the BSTM.
- d. If the message is a BASE24-teller message, the ISO Host Interface process checks the value of the EXTRN-RQST-TIMEOUT param in the LCONF and performs the following steps:
- 1) If the param is set to a value of 0, the process performs the following:
 - a) Retrieves the original message from memory.
The ISO Host Interface process retains a copy of the internal message for each message it sends to the host until it is notified that the message was received.
 - b) Sets the SYS.ORIG.RTE-STAT field in the Teller Standard Internal Message Header (TSTMH) to a value of 11, to indicate the destination was not available.
This identifies the 0200, 0300, or 0600 message as a failure.
 - c) Sends the Teller Standard Internal Message (TSTM) to the process identified as the originator of the message in the system message header.
 - 2) If the param is set to a value of 1, the process performs the following:
 - a) Retrieves the original message from memory and generates a 0420 reversal message, which it sends to the host. The Originator Code field in the external message header is set to a value of 4 (Host Interface process) and the Response Code (P-39) data element is set to indicate that the host is down.
 - b) Retrieves the original message from memory.
 - c) Sets the SYS.ORIG.RTE-STAT field in the Teller Standard Internal Message Header (TSTMH) to a value of 11, to indicate the destination was not available.
This identifies the 0200, 0300, or 0600 message as a failure.

- d) Sends the Teller Standard Internal Message (TSTM) to the process identified as the originator of the message in the system message header.
2. Adds one to the count of consecutive timeouts accumulated for the station.
3. Adds one to the number of request timeouts for the DPC.

The ISO Host Interface process tracks this information as part of its DPC performance statistics.
4. Checks the current number of consecutive timeouts accumulated for the station against the maximum allowed by the value in the MAX TIMEOUTS field on HCF screen 1.
 - a. If the current number of consecutive timeouts is less than the maximum allowed, no further action is taken.
 - b. If the current number of consecutive timeouts is equal to the maximum allowed, the ISO Host Interface process performs the following steps:
 - 1) Marks the station down.
 - 2) Verifies that the DPC still has a station available for sending messages. If no send/receive stations are left up for the DPC, the ISO Host Interface process marks the DPC down.
 - 3) Sets an extended network management message timer for the station.

Inbound Request Timer

An inbound request timer is set each time a 0100 or 0200 message is sent to an authorization process, which includes the BASE24-atm Authorization process, BASE24-pos Authorization module, and BASE24-telebanking Integrated Authorization Server process. The inbound request timer limits the amount of time that the ISO Host Interface process waits for a response from a BASE24 authorization process or module after transmitting a request message. The number of seconds to use for this timer is taken from the INBOUND field on the product-specific HCF screen.

If an inbound request timer expires before a response message is received, the ISO Host Interface process performs the following steps:

1. Retrieves the 0100 or 0200 message from memory.
 The ISO Host Interface process retains a copy of the external message for each 0100 or 0200 message it sends to an authorization process until a response is received (BASE24-pos handles all 0100 messages as 0200 messages internally).
2. Formats a 0110 or 0210 external message to send to the host using the information from the request. The ISO Host Interface process performs the following steps:
 - a. Sets the Responder Code field in the external message header to a value of 4, indicating that the request is being responded to by the ISO Host Interface process.
 - b. Sets the Response Code data element (P-39) in the external message to a value of 91 (issuer inoperative).
 - c. Sets the Card Issuer FIID and Logical Network data element (P-61) in the external message to zeros.
3. Sends the external message to the host.
4. Adds one to the number of request timeouts for the BASE24 system.

Completion Acknowledgment Timer

A completion acknowledgment timer is set each time a 0120, 0220, 0320, 0402, 0412, 0420, 0500, 0520, or 0620 message is sent to a DPC. The completion acknowledgment timer limits the amount of time that the ISO Host Interface process waits for an acknowledgment from the host after transmitting an advice message. The number of seconds to use for this timer is taken from the COMPLETION ACK field on the product-specific HCF screen.

Note: If the message is a BASE24-teller message and the RQST.ADVC-RESP-REQ field in the TSTMH is set to a value of 1, the number of seconds to use for this timer is taken from the OUTBOUND field on HCF screen 10. If the message is a BASE24-pos 0500 message, the number of seconds to use for this timer is taken from the OUTBOUND field on HCF screen 7.

The ISO Host Interface process only expects text-level acknowledgments to completion messages if the ACK FROM DPC field on HCF screen 1 contains the value Y. If text-level acknowledgments are not required from the DPC, the ISO

Host Interface process expects an acknowledgment from the XPNET process instead. In this case, the XPNET process returns an acknowledgment once the message has been sent successfully to the DPC.

If a completion acknowledgment timer expires before an appropriate acknowledgment is received, the ISO Host Interface process performs the following steps.

Note: If the message is a BASE24-teller message and the RQST.ADVC-RESP-REQ field in the TSTMH is set to a value of 1, the ISO Host Interface process changes the message type from 0220, 0320, 0420, or 0620 to 0230, 0330, 0430, or 0630, respectively; declines the transaction with the response code F07 (host is unavailable); and sends the message to the transaction originator.

1. Retrieves a copy of the external message from memory.

The ISO Host Interface process retains a copy of the external message for each message it sends to the host until it is notified that the message was received.
2. Changes the message type from 0120, 0220, 0320, 0420, 0500, 0520, or 0620 to 0121, 0221, 0321, 0421, 0520, 0521, or 0621 respectively. Message types 0402 and 0412 are not changed.
3. Places the external message in the SAF.
4. Adds one to the count of consecutive timeouts accumulated for the station.
5. Adds one to the number of request timeouts for the DPC.
6. Checks the current number of consecutive timeouts accumulated for the station against the maximum specified in the MAX TIMEOUTS field on HCF screen 1, as follows:
 - a. If the current number of consecutive timeouts is less than the maximum allowed, the ISO Host Interface process takes no further action.
 - b. If the current number of consecutive timeouts is equal to the maximum allowed, the ISO Host Interface process performs the following steps:
 - 1) Marks the station down.
 - 2) Verifies that the DPC still has a station available for sending messages. If no send/receive stations are left up for the DPC, the ISO Host Interface process marks the DPC down.
 - 3) Sets an extended network management message timer for the station.

Key Exchange Timer

Key exchange timers are used to automatically initiate PIN, MAC, and message encryption key exchanges. If the BASE24 system is configured for dynamic key management, one key exchange timer is set for each unique PIN, MAC, and message encryption key used by the ISO Host Interface process. Key exchange timers are set or reset at the following times:

- When the ISO Host Interface process is initialized or warmbooted
- When the key for which the timer has been set is exchanged for any reason

The ISO Host Interface process sets and uses key exchange timers for PIN keys when the PIN KEY TIMER VALUE field on KEYF screen 3 or KEY6 screen 4 is set to a nonzero value. PIN key exchange timers limit the amount of time that the BASE24 system uses a particular PIN key to encrypt and decrypt PINs sent between the BASE24 system and the host. The amount of time to use for these timers is taken from the PIN KEY TIMER INTERVAL and PIN KEY TIMER VALUE fields on KEYF screen 3 or KEY6 screen 4.

The ISO Host Interface process sets and uses key exchange timers for MAC keys when the MAC KEY TIMER VALUE field on KEYF screen 3 or KEY6 screen 4 is set to a nonzero value. MAC key exchange timers limit the amount of time that the BASE24 system uses a particular MAC key to authenticate messages sent between the BASE24 system and the host. The amount of time to use for these timers is taken from the MAC KEY TIMER INTERVAL and MAC KEY TIMER VALUE fields on KEYF screen 3 or KEY6 screen 4.

The ISO Host Interface process sets and uses key exchange timers for message encryption keys when the MSG KEY TIMER VALUE field on KEYF screen 4 is set to a nonzero value. Message encryption key exchange timers limit the amount of time that the BASE24 system uses a particular message encryption key to encrypt and decrypt messages sent between the BASE24 system and the host. The amount of time to use for these timers is taken from the MSG KEY TIMER INTERVAL and MSG KEY TIMER VALUE fields on KEYF screen 4.

When a key exchange timer expires, the ISO Host Interface process performs the following steps:

1. Determines the key that needs to be exchanged.
2. Changes the status of the DPC to indicate that a key exchange is pending.

3. Determines the type of message to send based on the value in the KEY PROCESSING TYPE field on KEYF screen 3 or KEY6 screen 4 and the key direction as follows:
 - If the KEY PROCESSING TYPE field contains the value N (none—the interface does not generate or receive keys), then no further steps are taken.
 - If the KEY PROCESSING TYPE field contains the value M (main—the interface is responsible for all key generation), the message type is a new key message.
 - If the KEY PROCESSING TYPE field contains the value S (secondary—the interface can request key generation but does not generate keys itself), the message type is a change key message.
 - If the KEY PROCESSING TYPE field contains the value C (co-network—the interface generates outbound keys but can only request inbound keys) and the key direction is outbound, the message type is a new key message.
 - If the KEY PROCESSING TYPE field contains the value C (co-network—the interface generates inbound keys, but can only request outbound keys) and the key direction is inbound, the message type is a change key message.
4. Determines whether a key exchange is already in progress for the specific key. If a key exchange is already in progress for the specific key, no further steps are taken. If a key exchange is not in progress for the specific key, processing continues with step 5.
5. Determines whether a station is available to which to send the message. Depending on whether a station is available, the ISO Host Interface process performs the following steps:
 - a. If no stations are available, the ISO Host Interface process sets an extended network management message timer. When this timer expires, the ISO Host Interface process attempts to send the message again. No further processing is performed at this time.
 - b. If a station is available, the ISO Host Interface process performs the following steps according to the type of message:
 - 1) If the message is a new key message, the ISO Host Interface process sends the message to the selected station. The ISO Host Interface process sets a network management message timer to wait for the response. The ISO Host Interface process also sets a flag to indicate that a key exchange is in progress and deletes the clear old key timer for the key being exchanged (if one is still set).

- 2) If the message is a change key message, the ISO Host Interface process sends the message to the selected station. The ISO Host Interface process sets a network management message timer to wait for the response.
6. Adds one to the count of messages outstanding to the station.

Note: Upon receiving a response from the DPC, the ISO Host Interface process clears the flag indicating a key exchange is pending and sets a new key exchange timer.

Clear Old Key Timer

The clear old key timer limits the amount of time the BASE24 system attempts to use an old key to verify a PIN or MAC or decrypt a message after a key exchange has taken place. The number of seconds to use for this timer is set in the CLEAR OLD KEY TIMER VALUE field on KEYF screen 3 or KEY6 screen 4.

The system maintains two versions (current and old) of its various PIN, MAC, and message encryption keys to allow for a transition period following a key change. When processing a transaction, the BASE24 system uses the new key for PIN processing, message authentication, or message decryption. If the security device cannot verify the PIN or the MAC or decrypt the message because of a key error, the transaction is tried again using the old key, if one exists. This timer controls how long the old key exists. When the timer expires, the old key is cleared.

A clear old key timer is set under the following circumstances:

- When the ISO Host Interface process receives a duplicate new key message from the host. A duplicate new key message carries a key that was already received in a previous new key message and stored in the KEYF or KEY6.
- When the BASE24 system approves a new key message from the host.
- When the BASE24 system receives an approved new key response message from the host (that is, the BASE24 system sent the host a new key, and the host has responded that the new key has been implemented successfully).

When a clear old key timer expires, the ISO Host Interface process performs the following steps.

Note: KEYF or KEY6 information is read into memory by the ISO Host Interface process at initialization, and the ISO Host Interface process performs the following steps on the KEYF or KEY6 information in memory and in the KEYF or KEY6.

1. Determines the type of key for which the timer was set.
2. Determines the key to be cleared (i.e., the old key).
3. Sets the selected key value and its check digits in memory to ASCII zeros.
4. Sets the selected key value and its check digits in the KEYF or KEY6 to binary zeros.

Stale Transaction Time

Issuer hosts can specify the maximum length of time that the XPNET process can take to send a message before returning the message to the ISO Host Interface process as *stale*.

This option is particularly useful for request messages (that is, 0100, 0200, 0300, and 0600 messages) because it allows the ISO Host Interface process to prevent processing when it becomes apparent that the host will not have time to respond to a message before the request times out.

If a message stays queued by the XPNET process for this length of time, the XPNET process returns the message to the ISO Host Interface process as a failed message rather than send it to the host. The ISO Host Interface process then returns the message to its originating process as a failed message.

This length of time is derived using the value in the QUEUE SUBTRACT field on the product-specific HCF screens. There is one QUEUE SUBTRACT field each for BASE24-atm, BASE24-pos, BASE24-telebanking, and BASE24-teller. The value in the QUEUE SUBTRACT field is subtracted from the corresponding timer value of each message to determine the maximum length of time to allow it to be queued.

For 0100, 0200, 0300, and 0600 messages, the value in the QUEUE SUBTRACT field is subtracted from the value in the OUTBOUND field on the product-specific HCF screen to arrive at the maximum queuing time. For 0120, 0220, 0320, 0402, 0420, 0520, and 0620 messages, the value in the QUEUE SUBTRACT field is subtracted from the value in the COMPLETION ACK field on the product-specific HCF screen to arrive at the maximum queuing time.

Configuring External Messages

The ISO Host Interface process has a standard set of message structures, or defaults, that it uses for creating and interpreting external messages. These message defaults are hard-coded in the ISO Host Interface process. The ***BASE24 External Message Manual*** documents the ISO external message format.

If the default message structures need to be modified, records can be created in the External Message File (EMF) specifying the new message structures. The BASE24 system allows incoming and outgoing messages to be configured individually by an institution, depending on the information the host chooses to send and receive. Refer to the ***BASE24 Base Files Maintenance Manual*** for more information on the EMF.

Data Elements

Data elements can be defined in the EMF as mandatory, conditional, or not required.

- If a data element is defined as mandatory, it is always sent in outgoing external messages and must be present in incoming external messages.
- If a data element is defined as conditional, it is sent in outgoing messages under certain conditions. If the element contains data and the data is valid, the ISO Host Interface process sends the element in its outgoing external messages. If the element contains blanks or zeros, it is not sent. Likewise, the data element may or may not be present in incoming messages.
- If a data element is defined as not required, it is not sent in outgoing external messages nor is it expected in incoming external messages.

Message Type

A code in the MSG TYP field on the EMF screens identifies the external message type defined by the EMF record. Any external message type allowed by the ISO Host Interface process is valid in this field, except for message types for repeat messages. A repeat message contains the same information as the original message. The value entered in this field can only be used with certain values entered in the PROD # field and the IN-OUT-IND field. A table showing the valid relationships for external messages to and from a host is provided below.

PROD #	PRODUCT	MSG TYP	IN-OUT-IND
00	Base	0800, 0810	I, O, B
		9000	O
01	ATM	0200, 0205, 0210, 0215, 0220, 0230, 0420, 0430	I, O, B
02	POS	0100, 0110, 0120, 0130, 0200, 0210, 0220, 0230, 0402, 0412, 0420, 0430	I, O, B
		0510, 0530	I
		0500, 0520	O
03	Teller	0200, 0210, 0220, 0230, 0300, 0310, 0320, 0330, 0420, 0430, 0600, 0610, 0620, 0630	I, O, B
08	FHM	0300	I, O
		0310	I, O
11	Mail	0620, 0630	B
14	Telebanking	0110, 0130, 0210, 0230, 0430	I
		0100, 0120, 0200, 0220, 0420	O

Note: Host Interface process message types in the 9000 range are used to denote rejects. For example, a message type of 0200 would be changed to 9200 if it were rejected and a message type of 0420 would be changed to a 9420 if it were rejected. Setting up a record in the EMF for a message type of 9000 is not to control message structure, but rather to allow for assigning a single IMS/CICS transaction code equivalent to all reject messages returned to the host. For further information on assigning an IMS/CICS transaction code equivalent to a 9000 message, refer to the description of EMF screen 3 in the ***BASE24 Base Files Maintenance Manual***.

Changing the Defaults

When the default values for a particular message type are not appropriate, a separate EMF record can be created for each message type. To modify the default values for any message type, the following steps must be performed. These steps assume that you are logged on to the BASE24 screen access system.

Warning: Special care must be used when adding or updating an EMF record. If IMS or CICS transaction code equivalents are used, access to EMF screen 3 is required in order to perform file maintenance operations (in addition to any other EMF screen access that may be defined). EMF screen 3 is used to assign IMS or CICS transaction code equivalents to BASE24 transaction codes and can consist of up to five pages. The last page containing transaction codes must be displayed when the function key is pressed to add or update the record or transaction codes are lost. For example, if the function key is pressed while screen 1 or screen 2 is displayed, all transaction codes on screen 3 are not included in the new or updated record. If the function key is pressed while one of the screen 3 pages is displayed, but it is not the last page containing transaction codes, the transaction codes on the remaining screen 3 pages are not included in the new or updated record.

1. Display EMF screen 1 by entering EMF in the FILE DESTINATION field at the bottom of the CRT access screen. From a menu screen, press the **F1** key. From a file screen, press the **F16** key.
2. Read the EMF record to be changed by completing the field information shown below and pressing the **F2** key. If no record is found, the ISO Host Interface process is using EMF default values. The default values can be displayed by pressing the **F7** key.
 - a. Set the value in the INTERF TYP field to HOST.
 - b. Set the value in the DPC # field to the DPC number assigned to the host in the HCF.
 - c. Set the value in the PROCESS NAM field to the symbolic name of the ISO Host Interface process.

- d. Set the PROD #, MSG TYP, and IN-OUT-IND fields using any combination of values from each row of the table below:

PROD #	Product	MSG TYP	IN-OUT-IND
00	Base	0800, 0810	I, O, B
01	ATM	0200, 0205, 0210, 0215, 0220, 0230, 0420, 0430	I, O, B
02	POS	0100, 0110, 0120, 0130, 0200, 0210, 0220, 0230, 0402, 0412, 0420, 0430	I, O, B
		0510, 0530	I
		0500, 0520	O
03	Teller	0200, 0210, 0220, 0230, 0300, 0310, 0320, 0330, 0420, 0430, 0600, 0610, 0620, 0630	I, O, B
08	FHM	0300	I, O
		0310	I, O
11	Mail	0620, 0630	B
14	Telebanking	0110, 0130, 0210, 0230, 0430	I
		0100, 0120, 0200, 0220, 0420	O

- Review the values for the message type displayed. If you need to change the value for a specific data element, enter an M (mandatory), a C (conditional), or a blank (not required) in the field directly to the right of the appropriate data element number.
- Display EMF screen 2 by pressing the **F9** key. EMF screen 2 allows you to specify individual data elements to be used during message authentication.
- Review the values for the message type displayed. If you need to change the value for a specific data element, enter a Y (include in authentication) or a blank (do not include in authentication) in the field directly to the right of the appropriate data element number.

6. Display EMF screen 3 by pressing the **F9** key. EMF screen 3 allows you to assign CICS or IMS equivalents to BASE24 transaction codes.
7. Determine whether any BASE24 transaction codes require IMS or CICS transaction code equivalents, and perform the following steps:
 - a. If no IMS or CICS transaction codes are required, or if all of the IMS or CICS transaction codes are displayed on the current page of screen 3, press the **F3** key to add the record. Continue with step 8.
 - b. If additional IMS or CICS transaction codes are required, add the IMS or CICS equivalents by making the appropriate entries in the TRAN, IMS TRAN, and L fields.

If you need to add more than 30 transaction codes, go to the next screen 3 page by pressing the **F8** key, and add the necessary IMS or CICS equivalents. You must go to a new screen 3 page each time you add 30 transaction codes.

When you have finished adding the necessary IMS or CICS transaction code equivalents, add the record to the EMF by pressing the **F3** key.

Note: The last screen 3 page containing entries must be displayed when you add or update the record.

8. Return to EMF screen 1 by pressing the **F9** key. Repeat steps 2 through 7 for any remaining message types to be added to the EMF.
9. Reinitialize the ISO Host Interface process in order to read the new values.

Configuring Tokens to be Sent in the External Message

A *token* is a collection of related data required to perform a specific function. As transactions are processed and additional information is required, BASE24 processes add tokens to the internal message. The Token File (TKN) is used to configure the following token processing:

- The tokens that get logged to the transaction log files. The transaction log files affected are the BASE24-atm Transaction Log File (TLF), ITS Transaction Log File (ITLF), POS Transaction Log File (PTLF), Teller Transaction Log File (TTLF), and Interchange Log Files (ILFs).
- The tokens that get extracted from the transaction log files by the Super Extract process. Extracting allows the token information to be processed by the host.

- The tokens that get sent to the host with each transaction in the ISO-based external message. The ANSI-based external message does not support message tokens.

For more information on configuring the tokens that get logged, extracted, and sent in the external message, refer to the ***BASE24 Tokens Manual***.

Configuring Fixed-Length Messages

A number of data elements in the ISO message are defined as variable-length, meaning that the size of the element itself can vary depending on the data contained in the element. BASE24 products support an option that allows certain variable-length data elements to be made fixed-length. Under this option, the variable-length elements are simply considered by the BASE24 system to be fixed-length, with the size of the element equal to the maximum-length set for the variable data.

This fixed-length option does not affect how the elements are identified. They are still defined as variable-length elements, and they still require prefixes to indicate their lengths. However, the data in the element is always left-justified and padded out to the maximum allowed length of the data element. For more information on how this option affects external messages, refer to the ***BASE24 External Message Manual***.

The MESSAGE FORMAT fields in the HCF control this option. There is a separate MESSAGE FORMAT field for each product and for network management messages. To invoke this option, the field must be set to a value of 00 for the BASE24 product whose messages are to be affected.

Note: BASE24 network management messages are sent as fixed format messages when the MESSAGE FORMAT field on HCF screen 1 contains the value 00, or when any of the product-specific MESSAGE FORMAT fields (on HCF screens 5, 7, 10, or 22) contain the value 00. That is, if messages for any individual product are sent as fixed format messages, network management messages are also sent as fixed format messages.

Configuring Previous Release Messages

BASE24 products support both current and previous release ISO message formats. Institutions specify which message format the host uses in the RELEASE INDICATOR field on the product-specific HCF screen.

Configuring Transaction Security

The transaction security requirements of any organization can vary greatly according to the environment in which the organization is operating. BASE24 products support transaction security between the BASE24 system and the host in the following areas:

- PIN encryption ensures that PIN information is maintained in a confidential manner.
- PIN verification ensures that the rightful cardholder or account holder is performing the transaction.
- Message authentication ensures that data remains tamper-free while passing between an EFT device and its authorization destination.
- Full message encryption ensures that the entire transaction message is maintained in a confidential manner.

These transaction security measures require that the BASE24 system and the host use keys to encrypt, decrypt, and verify information in the message. These keys should be changed regularly to maintain proper security. The monitoring and changing of these keys is known as key management, and can be performed manually or automatically. Automated key management is often known as dynamic key management (DKM). BASE24 products support DKM between the host and the BASE24 system.

As a part of Dynamic Key Management, BASE24 products also support key synchronization error detection and correction. This feature uses encryption key check digits to detect when the host and the ISO Host Interface process are using different encryption keys and allows the ISO Host Interface process to automatically take steps to resynchronize the keys.

Note: The full message encryption and key synchronization error detection and correction features are part of the BASE24-es Transaction Security Services, which must be implemented in order to use these features. In addition, these features are only available with the BASE24-atm and BASE24-pos products.

For information on establishing transaction security, Dynamic Key Management, and the BASE24-es Transaction Security Services, refer to the ***BASE24 Transaction Security Manual***.

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3: Network Management and Operations

Network management messages can be divided into two categories: those used to manage the operational status of the communications lines between the BASE24 system and the host, and those used to manage PIN, message encryption, and message authentication keys between the BASE24 system and the host.

The ISO Host Interface process supports four types of network management messages:

- Logon messages
- Echo-test messages
- Logoff messages
- Dynamic key management (DKM) messages

Network management messages are sent as 0800 messages and require 0810 messages in response.

This section describes the message flows that occur when network management messages are sent.

Note: ASCII-to-EBCDIC conversion is supported for all network management messages. Message authentication is supported on New Key, Repeat Key, and Verify Key network management messages only. Full message encryption is not performed on any network management messages.

Monitoring DPCs and Stations

The ISO Host Interface process keeps track of the status of all DPCs and stations with which it communicates using its DPC table. This DPC table is built at initialization and is updated continuously during processing.

Stations are marked *up* or *down* in the table depending on whether the ISO Host Interface process can communicate with them. For instance, if a station fails to respond appropriately to a message, the ISO Host Interface process checks the maximum retries allowed for the stations associated with that DPC. If the maximum retries allowed is reached, the ISO Host Interface process marks the station down and takes the necessary network management actions to reestablish the link with the station. Once the link is reestablished, the ISO Host Interface process marks the station up.

The ISO Host Interface process also tracks the status of DPCs. A DPC is marked as up if the ISO Host Interface process can send messages to and receive messages from at least one of the stations associated with that DPC. If all of the stations to a DPC are marked down, the DPC itself is marked down. Additionally, the ISO Host Interface process tracks such things as whether the DPC is logged off, whether store-and-forward processing is in progress for the DPC, and whether a Store-and-Forward File (SAF) extract is in progress for the DPC.

The ISO Host Interface process keeps track of station and DPC information at all times and logs an event message when the status of the DPC changes (for example, from up to down or at the start of store-and-forward processing).

Note: The actual communications link to each DPC station is maintained by the XPNET process. The ISO Host Interface process only tracks the logical ability of BASE24 products to communicate with each station. If a station responds to a message, the ISO Host Interface process considers it to be up. If a station fails to respond to a message for the number of times specified in the MAX TIMEOUTS field on HCF screen 1, the ISO Host Interface process considers the station to be down.

Logon Messages

Logon messages are network management messages used to manage communications access between the BASE24 system and a DPC. They are sent as 0800 messages and require 0810 messages in response.

Logon messages are distinguished by a value of 001 in the Network Management Information Code data element (S-70) of the external message.

Logon Messages from the BASE24 System

The ISO Host Interface process initiates logon messages at initialization and upon receipt of a LOGON command from a network control facility.

At initialization, the ISO Host Interface process sends a logon message to each send/receive station.

Upon receiving a LOGON command from a network control facility, the ISO Host Interface process sends a logon message to each send/receive station associated with the DPC indicated in the command.

Logon Messages from a DPC

After a period of being logged off, DPCs can send logon messages to log on to the BASE24 system.

If the ISO Host Interface process receives a logon message after the DPC has been logged off, the ISO Host Interface process marks the DPC as logged on and begins sending echo-test messages to its stations. The ISO Host Interface process then returns a logon response to the DPC. If the ISO Host Interface process receives a logon message when the DPC is not logged off, it simply returns a logon response.

Note: In its responses, the ISO Host Interface process always sets the Response Code data element (P-39) in the external message to the value 00 to indicate an approval.

Echo-Test Messages

Echo-test messages are network management messages used to establish whether a DPC station is available for message processing. They are sent as 0800 messages and require 0810 messages in response.

Echo-test messages are distinguished by a value of 301 in the Network Management Information Code data element (S-70) of the external message.

Echo-Test Messages from the BASE24 System

The ISO Host Interface process initiates echo-test messages under the following circumstances:

- Upon expiration of an extended network management message timer. This processing is described in section 2.
- Upon expiration of a wait-for-traffic timer. This processing is described in section 2.
- When a DPC initiates a logon message after a period of being logged off. The ISO Host Interface process sends an echo-test message to each send/receive station once the DPC has been marked as logged on.

Echo-Test Messages from a DPC

DPCs can send echo-test messages at any time. The ISO Host Interface process always returns an echo-test response.

Note: In its responses, the ISO Host Interface process always sets the Response Code data element (P-39) in the external message to the value 00 to indicate an approval.

Echo-test messages do not substitute for logon messages. The ISO Host Interface process does not consider the DPC to be logged on and available for processing transactions until it receives a logon message.

Logoff Messages

Logoff messages are network management messages used to stop communications between the BASE24 system and a DPC. They are sent as 0800 messages and require 0810 messages in response.

Logoff messages are distinguished by a value of 002 in the Network Management Information Code data element (S-70) of the external message. They can be sent by either the BASE24 system or a host. In either situation, once a DPC is logged off, the ISO Host Interface process does not send echo-test messages or non-network management messages to any of the stations associated with that DPC. Any non-network management messages that must be forwarded to the DPC are placed in the SAF instead.

The ISO Host Interface process only attempts to reopen the link to a DPC that is logged off if a LOGON command is received from a network control facility. The DPC, on the other hand, can reopen the link at any time by sending a logon message.

Note: When a DPC is logged off, any transactions in progress are allowed to complete.

Logoff Messages from the BASE24 System

The ISO Host Interface process only sends logoff messages to a DPC as a result of a LOGOFF command issued from a network control facility. The ISO Host Interface process then expects a logoff response from the DPC.

If no logoff response is received, the ISO Host Interface process still marks the DPC as logged off, but keeps sending logoff messages to the DPC until a response is received.

Logoff Messages from a DPC

When a host must log off because of a scheduled down or unavailable period, it can transmit a logoff message from any of its stations.

Upon receiving a logoff message, the ISO Host Interface process marks the DPC down and logged off and returns a logoff response to the DPC.

Note: In its responses, the ISO Host Interface process always sets the Response Code data element (P-39) in the external message to the value 00 to indicate an approval.

Dynamic Key Management Messages

Dynamic key management (DKM) messages make up a subset of network management (0800) messages. There are four key management messages:

- Change key
- New key
- Repeat key
- Verify key

The change key and new key messages can be generated automatically when a defined threshold is surpassed. In addition, all of these messages can be generated manually by an operator issuing a command from a network control facility.

This section provides an overview of how dynamic key management messages are used. For more information on dynamic key management, refer to the ***BASE24 Transaction Security Manual***. For more information on using a network control facility to generate key management messages, and for complete syntax of the network control commands involved, refer to the ***BASE24 Text Command Reference Manual***.

Change Key Message

A change key message requests that a key processor generate a new PIN encryption, message encryption, or message authentication key. The receiving processor sends a response to the change key message and then, in a separate action, generates the new key and sends it to the requesting process in a new key message.

The ISO Host Interface process sends this type of message to a host when the host is responsible for generating a particular key and the ISO Host Interface process has determined that that key needs to be changed. The ISO Host Interface process could make that determination based on a DKM threshold being passed or in response to a KEY CHANGE command from a network control facility (the KEY CHANGE command has the same effect as passing a DKM threshold).

The ISO Host Interface process can receive this type of message from a host when the ISO Host Interface process is responsible for generating a particular key and the host wants that key to be changed.

New Key Message

A new key message carries a new PIN encryption, message encryption, or message authentication key between processors. The receiving processor verifies and begins using the new key and returns a response to the key-generating processor indicating that the new key has been implemented.

The ISO Host Interface process sends this type of message to a host when the ISO Host Interface process is responsible for generating a particular key and has either determined that that key needs to be changed or been requested by the host to change or repeat the key. This could occur as a result of a DKM threshold being passed, notification from the host (by way of a change key or repeat key message), or could occur in response to a KEY CHANGE command from a network control facility (the KEY CHANGE command has the same effect as passing a DKM threshold).

The ISO Host Interface process can receive this type of message from a host when the host is responsible for generating a particular key and the host has either determined that that key needs to be changed or been requested by the ISO Host Interface process to change or repeat the key (by way of a change key or repeat key message).

Repeat Key Message

A repeat key message requests a processor that generated a key to resend that key. The processor receiving the repeat key message obtains the requested key from its database and sends it to the requesting process in a new key message. The key itself is not changed, nor are any threshold counters affected.

The ISO Host Interface process only generates a repeat key message as the result of receiving a KEY REPEAT command from a network control facility. It responds to a repeat key message from the host with a new key message carrying the requested key.

Verify Key Message

A verify key message requests a processor that generated a key to verify the key by comparing check digits. The verify key message contains check digits for the key in question, and the processor receiving the message compares the check digits in the message with the check digits it has on file for the particular key. The processor then responds to the verify key message with the results of the check digit comparison. (If the check digits do not match, the processor that sent the verify key message should use a change key message to request a new key.)

The ISO Host Interface process only generates the verify key message as the result of receiving a KEY VERIFY command from a network control facility. It responds to a verify key message from a host by comparing check digits for the specified key and returning the results.

Dynamic Key Management at Initialization

If the BASE24 system is set up for dynamic key management, the ISO Host Interface process must make sure at initialization that its keys are in place prior to marking a DPC up and allowing transactions to be processed. To do this, the ISO Host Interface process enforces the following sequence of events at initialization as it begins to establish its communication with a DPC. If dynamic key management is not used, this sequence is not applicable.

1. Establishes a successful logon with the DPC. At this point, the ISO Host Interface process can begin exchanging keys, but does not mark the DPC up.
2. Exchanges message authentication keys with the DPC if message authentication is performed by the ISO Host Interface process. This allows the message authentication keys to be put in place prior to the exchange of PIN and message encryption keys.
3. Exchanges PIN encryption keys with the DPC if PIN encryption is performed by the ISO Host Interface process. This allows the PIN encryption keys to be put in place prior to transaction processing.
4. Exchanges message encryption keys with the DPC if message encryption is performed by the ISO Host Interface process. This allows the message encryption keys to be put in place prior to transaction processing.
5. Marks the DPC up. At this point, transactions can be accepted from and sent to the DPC.

Note: Since this sequence is only required and only occurs at initialization, it is not included in the logon and key exchange sequences documented in this section.

Interactive Network Management Message Flows

The remainder of this section contains examples of network management and DKM message flows and describes the processing performed by the ISO Host Interface process for each.

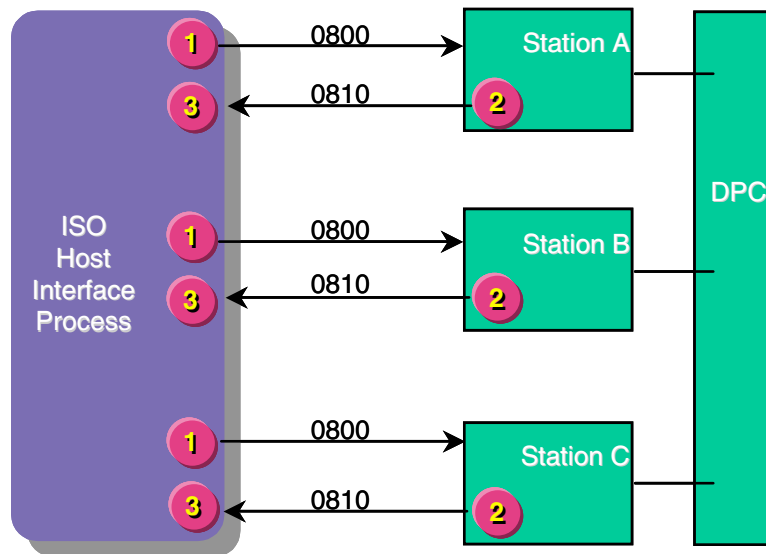
Logon Message Flow

This subsection provides examples of logon sequences initiated by BASE24 and a DPC.

Note: The ISO Host Interface process does not perform message authentication or full message encryption on logon messages.

Logon initiated by BASE24

The following diagram illustrates the processing performed when the ISO Host Interface process logs on to the DPC. The steps corresponding to the diagram are provided below.

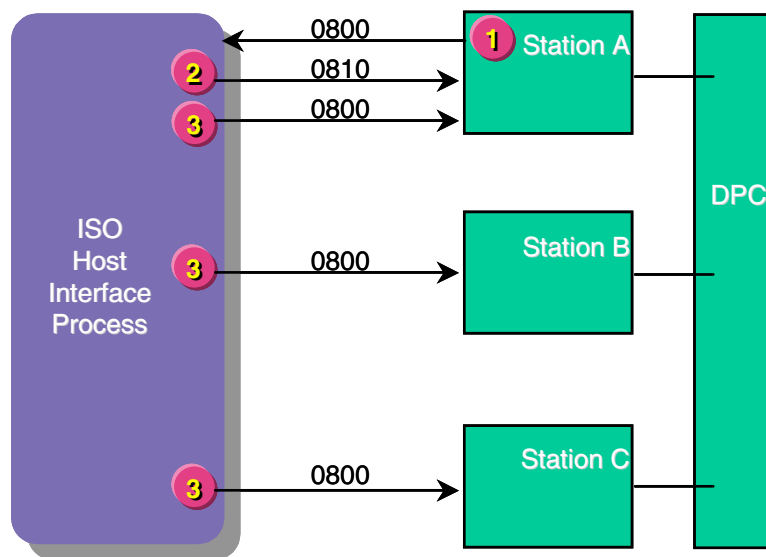


1. If the ISO Host Interface process cannot send network management messages to the host (the NMM ENABLED field on HCF screen 1 is set to the value N), the ISO Host Interface process marks each station up and the DPC as logged

- on. If the ISO Host Interface process can send network management messages to the host (the NMM ENABLED field on HCF screen 1 is set to the value Y), the ISO Host Interface process performs the following steps:
- a. Creates a logon message for each of the send/receive stations associated with that DPC. The logon message is identified by the value 001 in the Network Management Information Code data element (S-70) of the external message.
 - b. Converts the message from ASCII to EBCDIC if the CHARACTER FORMAT field on screen 1 of the HCF is set to E for the DPC.
 - c. Sends the messages to the send/receive stations associated with that DPC.
 - d. Sets a network management message timer for each station.
 - e. Marks each receive-only station up.
2. The host returns a logon response from each station. The logon response is identified by the value 001 in the Network Management Information Code data element (S-70) of the external message.
 3. The ISO Host Interface process receives each logon response and checks the Response Code data element (P-39) in the external message, as follows:
 - If the Response Code data element contains zeros, indicating an approval, the ISO Host Interface process marks the DPC as up and logged on. The ISO Host Interface process also marks the corresponding station up and sets a wait-for-traffic timer for that station.
 - If the Response Code data element does not contain zeros, the ISO Host Interface process logs an event message, marks the DPC as logged off, and ceases all attempts to communicate with the station. In order to reinstate communications with the station, the host must send a message from that station.

Logon initiated by DPC

The following diagram illustrates the processing performed when the DPC logs on to the BASE24 system. The steps corresponding to the diagram are provided below.



1. The host sends a logon message to reestablish communications. The logon message is identified by the value 001 in the Network Management Information Code data element (S-70) of the external message.
2. The ISO Host Interface process receives the message and performs the following steps:
 - a. Marks the DPC as logged on.
 - b. Sets a wait-for-traffic timer for each station defined for the DPC.
 - c. Sends a logon response to the DPC. The logon response is identified by the value 001 in the Network Management Information Code data element (S-70) of the external message.
3. The ISO Host Interface process sends an echo-test message and sets a network management message timer for each send/receive station that is down. The echo-test message is identified by the value 301 in the Network Management Information Code data element (S-70) of the external message.

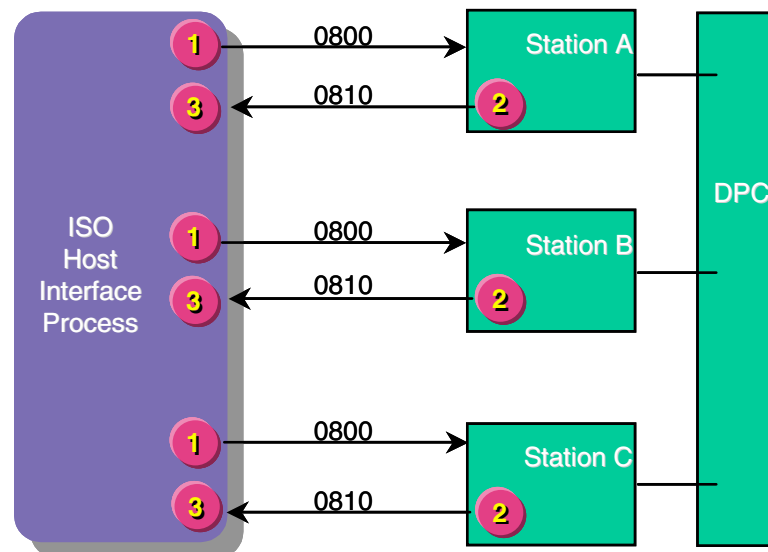
Logoff Message Flow

This subsection provides examples of sequences initiated by BASE24 and a DPC.

Note: The ISO Host Interface process does not perform message authentication or full message encryption on logoff messages.

Logoff Initiated by BASE24

The following diagram illustrates the processing performed when the ISO Host Interface process logs off from the DPC. The steps corresponding to the diagram are provided below.



1. The ISO Host Interface process receives a LOGOFF command from a network control facility. The ISO Host Interface process sends a logoff message to each of the send/receive stations associated with that DPC and starts a network management message timer for each. The logoff message is identified by the value 002 in the Network Management Information Code data element (S-70) of the external message.
2. The host returns a logoff response (from one or each station). The logoff response is identified by the value 002 in the Network Management Information Code data element (S-70) of the external message.

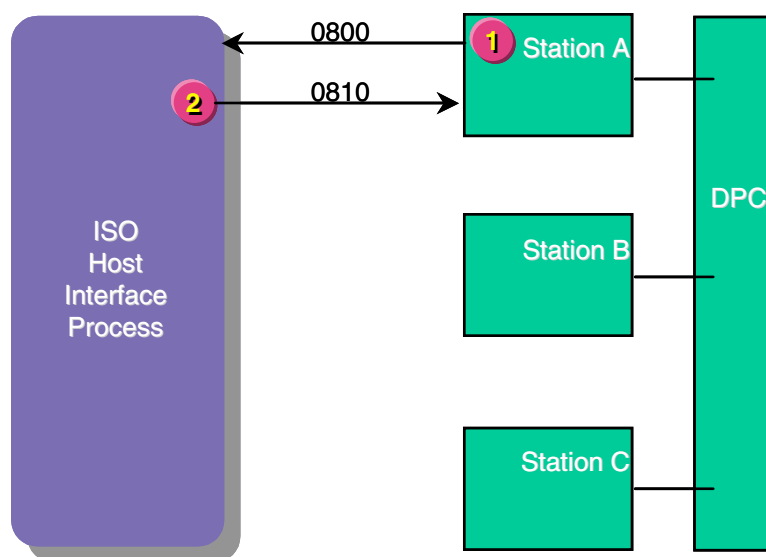
3. The ISO Host Interface process marks the DPC as logged off and clears the following timers running for the DPC and its stations:

- Network management message timers
- Extended network management message timers
- Wait-for-traffic timers

All other timers are left running to ensure that transactions in progress complete.

Logoff Initiated by a DPC

The following diagram illustrates the processing performed when the DPC logs off from the BASE24 system. The steps corresponding to the diagram are provided below.



1. The host sends a logoff message to the BASE24 system. The logoff message is identified by the value 002 in the Network Management Information Code data element (S-70) of the external message.
2. The ISO Host Interface process receives the logoff message and performs the following steps:
 - a. Marks the DPC as logged off.

b. Clears the following timers running for the DPC and its stations:

- Network management message timers
- Extended network management message timers
- Wait-for-traffic timers

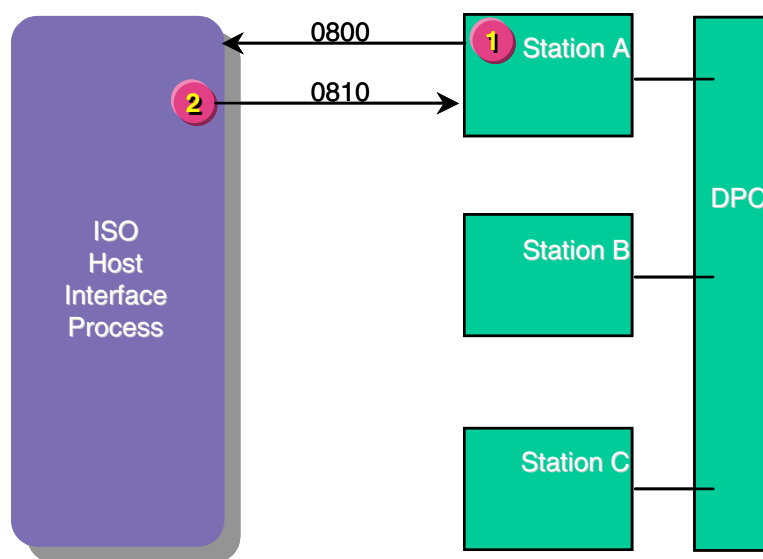
All other timers are left running to ensure that current transactions in progress complete.

c. Sends a logoff response to the DPC. The logoff response is identified by the value 002 in the Network Management Information Code data element (S-70) of the external message.

Echo-Test Message Flow

The following diagram illustrates how the ISO Host Interface process handles echo-test messages from the DPC. In this case, the DPC is logged on, but the ISO Host Interface process considers the station to be down. The steps corresponding to the diagram are provided below.

Note: The ISO Host Interface process does not perform message authentication or full message encryption on echo-test messages.

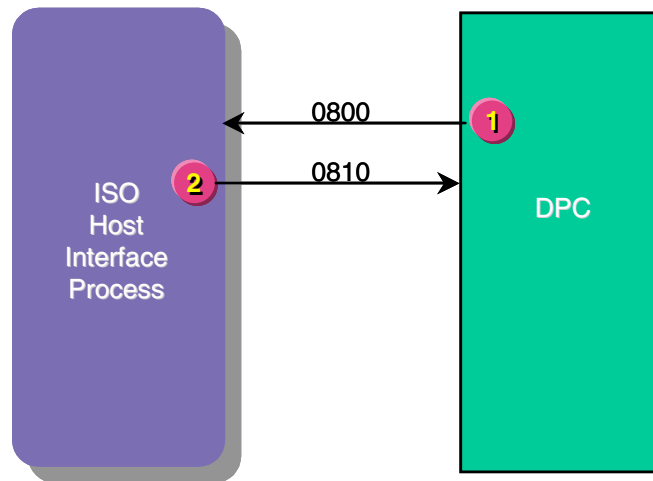


1. The host sends an echo-test message from station A. The echo-test message is identified by the value 301 in the Network Management Information Code data element (S-70) of the external message.
2. The ISO Host Interface process receives the message and performs the following steps:
 - a. Returns an echo-test response. The echo-test response is identified by the value 301 in the Network Management Information Code data element (S-70) of the external message.
 - b. Sends an echo-test message and sets a network management message timer for each of the send/receive stations that are down.

Change Key Message Flow

The diagram below illustrates the message flow between the host and ISO Host Interface process when the host initiates a change key request message. In this case, the ISO Host Interface process is responsible for generating the key requested. The steps corresponding to the diagram are provided below.

Note: The ISO Host Interface process does not perform message authentication or full message encryption on change key messages.



1. The host formats and sends an 0800 message with a network management information code of 161 to the ISO Host Interface process. This message includes the Cryptographic Service Message data element (S-123) containing a Request Service Initiation message (RSI).
2. The ISO Host Interface process receives the 0800 message and validates it before responding. To validate the message, the ISO Host Interface process performs the following steps:
 - a. Verifies that the RSI is in the correct format. If the RSI is not in the correct format, the ISO Host Interface process sets an error flag to F to indicate that a format error was found.
 - b. Verifies that the receiver and originator of the message are correct. To verify the originator and receiver of the message, the ISO Host Interface process compares the receiver ID from the RSI to the value from the ORIGINATING ID field on KEYF screen 3 or KEY 6 screen 4, and the originator ID from the RSI to the value from the RECEIVING ID field on KEYF screen 3 or KEY 6 screen 4. If the receiver or originator does

not match the corresponding KEYF values for this DPC, the ISO Host Interface process sets an error flag to the value of H to indicate that the receiver or originator ID is incorrect.

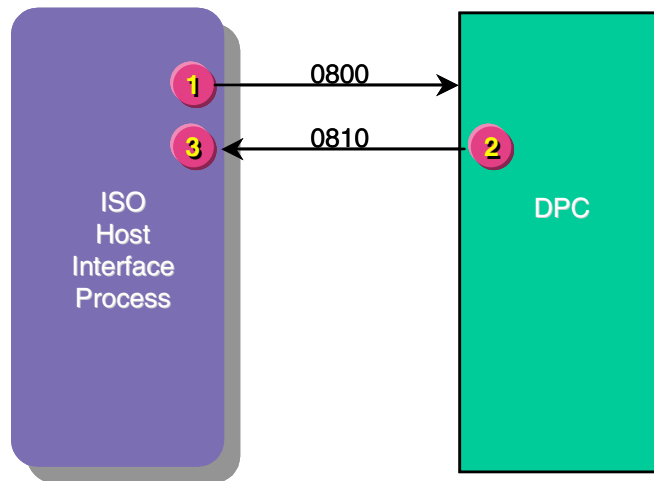
- c. Verifies that there is not already a key change action in progress for the type of key identified in the RSI. The ISO Host Interface process checks an internal flag to verify that a key change is not currently in progress. If a key change is in progress, the ISO Host Interface process sets an error flag to the value of C to indicate that the current change key request cannot be processed.
- d. Formats an 0810 response message and returns it to the host. To format the response message, the ISO Host Interface process performs one of the following steps:
 - If one or more of the above checks failed with an error, the ISO Host Interface process formats an 0810 response message with the response code set to 05 (denied). This message includes the Cryptographic Service Message data element (S-123) containing an Error Service Message (ESM). The ERF field in the ESM includes any error flag values set during the checks. No further processing is performed.
 - If no errors were encountered in the above checks, the ISO Host Interface process formats an 0810 response message with the response code set to 00 (approved) and a network management information code of 161. This message includes the Cryptographic Service Message data element (S-123) containing a copy of the original RSI.
- e. Creates a new key and sends it to the host in a new key message.

For a description of this processing, refer to the “New Key Message Flow” topic provided later in this section.

New Key Message Flow

The diagram below illustrates the message flow between the ISO Host Interface process and the host when the ISO Host Interface process sends a new key message. In this case, the ISO Host Interface process is responsible for generating the key being sent. The steps corresponding to the diagram are provided below.

Note: The ISO Host Interface process does not perform full message encryption on new key messages.



1. The ISO Host Interface process creates a new key message and sends it to the host. To do this, the ISO Host Interface process performs the following steps:
 - a. Generates the new key and check digits.
 - b. Formats a Key Service Message (KSM) containing the new key and the check digits.
 - c. Generates a message authentication code (MAC) for the message, if message authentication is supported and is specified for this message type. If the new key in the message is for message authentication, the new key is used to generate the MAC.
 - d. Converts the message from ASCII to EBCDIC if the CHARACTER FORMAT field on screen 1 of the HCF is set to E for the DPC.
 - e. Sends the 0800 message with a network management information code of 162 to the host. This message includes the Cryptographic Service Message data element (S-123) containing the KSM.
 - f. Sets an internal flag indicating that a key exchange is in progress for the type of key being sent.

- g. Starts a network management message timer for the message. If this timer expires before an 0810 response message is received from the host, the ISO Host Interface process starts an extended network management message timer for the message. If the extended network management message timer expires without receiving a response, the ISO Host Interface process resends the new key message.
2. The host receives the new key message, processes it, and returns an 0810 response message with a network management information code of 162 to the ISO Host Interface process. This message includes the Cryptographic Service Message data element (S-123) containing either a Response Service Message (RSM) or an Error Service Message (ESM).

Note: If the new key message contains a new message authentication key, the host must extract the new key from the message and use that key to verify the MAC for the new key message. The host must also use the new key to generate the MAC for the 0810 response message.

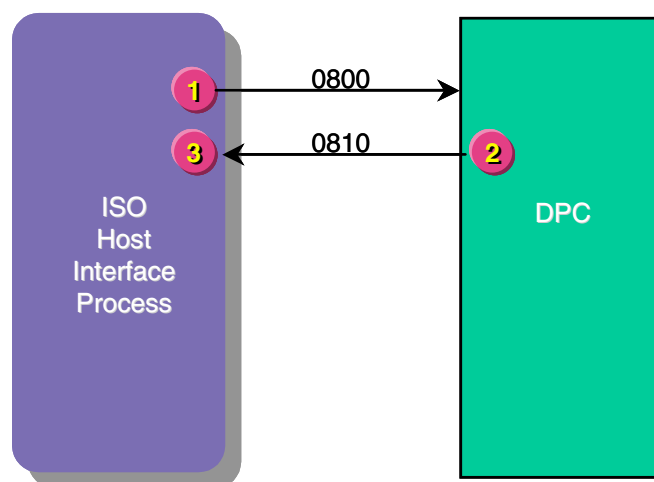
3. The ISO Host Interface process receives the 0810 response message and processes it by performing the following steps:
 - a. Converts the message from EBCDIC to ASCII if the CHARACTER FORMAT field on screen 1 of the HCF is set to E for the DPC.
 - b. Verifies the MAC received with the message, if message authentication is supported and is specified for new key messages. If a message authentication key was sent, the ISO Host Interface process uses this new key to verify the MAC.
 - c. Verifies that the original 0800 message did not time out.
 - If the 0800 message timed out before the 0810 response message was received, the ISO Host Interface process has already set an extended network management message timer for the new key message (see step 1f). No further steps are performed.
 - If the 0800 message has not timed out, processing continues with step 3c.
 - d. Clears the internal flag indicating that a key exchange is in progress.

- e. Continues processing depending on the response code sent in the 0810 response message.
 - If the transaction was declined, the ISO Host Interface process parses the Cryptographic Service Message data element (S-123) to determine why the new key was declined. In this case, the Cryptographic Service Message data element contains an Error Service Message (ESM).
 - If the ESM indicates the message was declined because the BASE24 key counter did not match the key counter maintained by the host, the ISO Host Interface process resets its key counter in the KEYF to the value sent by the host. The ISO Host Interface process then resends the new key to the host. Processing returns to step 1.
 - If the ESM indicates that any other error occurred, the ISO Host Interface process sends an event message indicating the error to the network logging facility and drops the 0810 response message.
 - If the transaction was approved, the ISO Host Interface process updates the KEYF with the new key, check digit, and counter information. The ISO Host Interface process also reinitializes its internal key change threshold counters and timers.

Repeat Key Message Flow

The diagram below illustrates the message flow between the ISO Host Interface process and the host when the ISO Host Interface process initiates a repeat key message. In this case, the host is responsible for generating the key requested. The ISO Host Interface process will only initiate a repeat key message if it receives a KEY REPEAT command from a network control facility. The processing described below starts from the point where the ISO Host Interface process receives the network control command. The steps corresponding to the diagram are provided below.

Note: The ISO Host Interface process does not perform full message encryption on repeat key messages.



1. When the ISO Host Interface process receives the REPEAT KEY command from a network control facility, it performs the following steps:
 - a. Verifies that the syntax is correct and that DKM is supported. If the syntax is incorrect or DKM is not supported, the ISO Host Interface process logs a message and drops the command. If the syntax is correct and DKM is supported, the ISO Host Interface process continues with step 1b.
 - b. Determines whether a key exchange is in progress for the key in the message. If a key exchange is in progress, the ISO Host Interface process logs a message and performs no further steps.
 - c. Creates an 0800 message with a network management information code of 163. This message includes a blank Cryptographic Service Message data element (S-123).

- d. Generates a message authentication code (MAC) for the message, if message authentication is supported and is specified for this message type.
 - e. Converts the message from ASCII to EBCDIC if the CHARACTER FORMAT field on screen 1 of the HCF is set to E for the DPC.
 - f. Sends the message to the host.
 - g. Starts a network management message timer for the message.
2. The host receives the repeat key request and processes the message. The host returns an 0810 response message with a network management information code of 163 to the ISO Host Interface process. This message includes the Key Service Message (KSM) in the Cryptographic Service Message data element (S-123).
3. The ISO Host Interface process receives the 0810 response message and processes it by performing the following steps:
 - a. Converts the message from EBCDIC to ASCII if the CHARACTER FORMAT field on screen 1 of the HCF is set to E for the DPC.
 - b. Verifies the MAC received with the message, if message authentication is supported and is specified for the message type.
 - c. Verifies that the original 0800 message did not time out.
 - If the 0800 message timed out before the 0810 response message was received, the ISO Host Interface process drops the 0810 response message, and no further steps are performed.
 - If the 0800 message has not timed out, processing continues with step 3c.
 - d. Continues processing depending on the response code sent in the 0810 response message.
 - 1) If the transaction was declined, the ISO Host Interface process logs a message and drops the 0810 message. No further steps are performed.
 - 2) If the transaction was approved, the ISO Host Interface process performs the following steps:
 - a) Verifies that the receiver and originator of the message are correct. To verify the originator and receiver of the message, the ISO Host Interface process compares the receiver ID from the KSM to the value from the ORIGINATING ID field on KEYF screen 3 or KEY 6 screen 4, and the originator ID from the KSM to the value from the RECEIVING ID field on

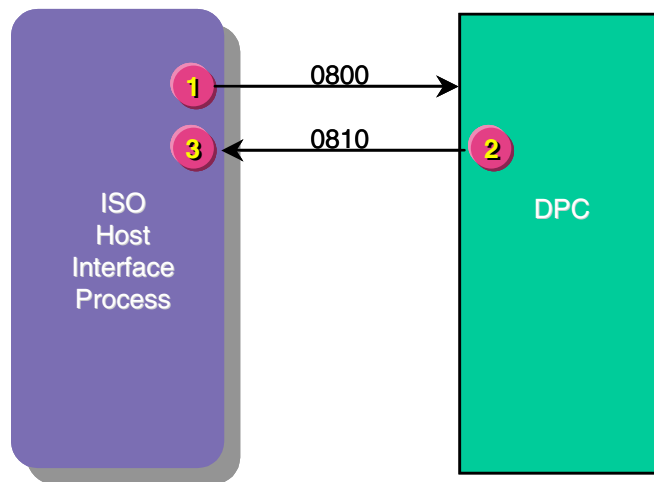
KEYF screen 3 or KEY 6 screen 4. If the receiver or originator does not match the corresponding KEYF values for this DPC, the ISO Host Interface process logs a message and drops the 0810 response message.

- b) Verifies the key. To do this, the ISO Host Interface process translates the key sent by the host and verifies the check digits. The check digits generated as a result of translating the key must match the check digits sent by the host. If the ISO Host Interface process cannot translate the key or the check digits do not match, the ISO Host Interface process logs a message and drops the 0810 response message.
- c) Updates the KEYF with the new key and check digit information.

Verify Key Message Flow

The diagram below illustrates the message flow between the ISO Host Interface process and the host when the ISO Host Interface process initiates a verify key message. In this case, the host is responsible for generating the key being verified. The ISO Host Interface process will only initiate a verify key message if it receives a KEY VERIFY command from a network control facility. The processing described below starts from the point where the ISO Host Interface process receives the network control command. The steps corresponding to the diagram are provided below.

Note: The ISO Host Interface process does not perform full message encryption on verify key messages.



1. When the ISO Host Interface process receives the VERIFY KEY command from a network control facility, it performs the following steps:
 - a. Verifies that the syntax is correct and that DKM is supported. If the syntax is incorrect or DKM is not supported, the ISO Host Interface process logs a message and drops the command. If the syntax is correct and DKM is supported, the ISO Host Interface process continues with step 1b.
 - b. Determines whether a key exchange is in progress for the key in the message. If a key exchange is in progress, the ISO Host Interface process logs a message and performs no further processing.

- c. Creates an 0800 message with a network management information code of 164. This message does not include the Cryptographic Service Message data element (S-123). However, the ISO Host Interface process does include the check digits for the key from the BASE24 database for the host to use in verifying the key.
 - d. Generates a message authentication code (MAC) for the message, if message authentication is supported and is specified for this message type.
 - e. Converts the message from ASCII to EBCDIC if the CHARACTER FORMAT field on screen 1 of the HCF is set to E for the DPC.
 - f. Sends the message to the host.
 - g. Starts a network management message timer for the message.
2. The host receives the verify key request and processes the message. The host returns an 0810 response message with a network management information code of 164 to the ISO Host Interface process.
3. The ISO Host Interface process receives the 0810 response message and processes it by performing the following steps:
 - a. Converts the message from EBCDIC to ASCII if the CHARACTER FORMAT field on screen 1 of the HCF is set to E for the DPC.
 - b. Verifies the MAC received with the message, if message authentication is supported and is specified for the message type.
 - c. Verifies that the original 0800 message did not time out, as follows:
 - If the 0800 message timed out before the 0810 response message was received, the ISO Host Interface process drops the 0810 response message, and no further steps are performed.
 - If the 0800 message has not timed out, processing continues with step 3c.
 - d. Continues processing depending on the response code sent in the 0810 response message, as follows:
 - If the transaction was declined (the Response Code data element (P-39) in the external message contained the value 05), the ISO Host Interface process logs an event message indicating the host declined the request and drops the 0810 message.
 - If the transaction was approved (the Response Code data element (P-39) in the external message contained the value 00), the ISO Host Interface process logs an event message indicating that the command was successful and drops the 0810 message.

- If the Response Code data element (P-39) in the external message contained any other value, the ISO Host Interface process logs an event message indicating that the request failed and a change key request is required. The ISO Host Interface process drops the 0810 message.

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4: Store-and-Forward Processing

Store-and-forward processing allows for storing transactions that the BASE24 system has processed when an issuer host is down or unavailable.

The ISO Host Interface process keeps track of each station and DPC. If a message bound for a DPC cannot be transmitted, the ISO Host Interface process places the message in the Store-and-Forward File (SAF) for later transmission. When communications with the host resume, the messages in the SAF are sent to the host.

This section describes store-and-forward processing. It includes information on what messages can be placed in the SAF, when messages are placed in the SAF, and when messages from the SAF are sent to the host. It also contains message flows that illustrate store-and-forward processing.

The Store-and-Forward File

The Store-and-Forward File (SAF) temporarily stores messages bound for a host until those messages can be sent to the intended DPCs. The SAF is a key-sequenced file, and, in general, the messages are processed as first-in, first-out, as described below.

If the store-and-forward cycle is single-threaded (that is, the MAXIMUM OUTSTANDING SAFS field on Host Configuration File (HCF) screen 1 contains the value 1), the oldest SAF records are sent to the host first. Records arrive at the host in the same order in which they were added to the SAF.

If the store-and-forward cycle is multithreaded (that is, the MAXIMUM OUTSTANDING SAFS field on HCF screen 1 contains a value greater than 1), the oldest SAF records are sent to the host first. However, the records may not arrive at the host in the same order as they were sent. For example, the following events could affect the order in which messages arrive:

- Multiple store-and-forward messages could be sent to different stations, which deliver the messages at different times. For example, if the oldest store-and-forward message is sent to station 1 and the second-oldest message is sent to station 2, the message sent to station 2 may arrive before the message sent to station 1.
- A store-and-forward message could time out, causing it to be resent. If subsequent store-and-forward messages have already been sent, they may be processed by the host before the message that timed out.

Messages in the SAF are stored in the format that they are to be sent to the host.

Issuer hosts can have store-and-forward messages sent to them by way of communications lines, or following long periods of down-time, the SAF can be extracted to tape in order to keep from overloading communications once communications have been restored.

Messages Placed in the SAF

The ISO Host Interface process places the following types of messages in the SAF:

- 0120—Authorization Advice
- 0121—Authorization Advice Repeat
- 0220—Financial Transaction Advice
- 0221—Financial Transaction Advice Repeat
- 0300—File Update Request (BASE24-from host maintenance used with the ACI Smart Chip Manager product)^{*}
- 0310—File Update Request Response (BASE24-from host maintenance used with products other than ACI Smart Chip Manager)[†]
- 0320—File Inquiry/Update Advice
- 0321—File Inquiry/Update Advice Repeat
- 0402—Card Issuer Reversal Request
- 0420—Reversal Advice
- 0421—Reversal Advice Repeat
- 0520—Acquirer Reconciliation Advice
- 0521—Acquirer Reconciliation Advice Repeat
- 0620—Administrative Advice
- 0621—Administrative Advice Repeat

Message types 0120, 0220, 0300^{*}, 0310[†], 0320, 0402, 0420, 0520, and 0620 are placed in the SAF if the intended DPC is marked down at the time the message is to be sent. If the ISO Host Interface process receives one of these message to be sent to a DPC, but discovers the DPC is marked down, the ISO Host Interface process places the message in the SAF without attempting to send it to the DPC.

In addition, these messages can be added to the SAF as the result of a recoverable message authentication error. In this case, the link to the DPC may not be down.

If the ISO Host Interface process sends a 0120, 0220, 0300, 0320, 0402, 0420, 0520, or 0620 message to a DPC and the message times out, the ISO Host Interface process changes the 0120 message to a 0121, 0220 to a 0221, 0300 to a 0301, 0320 to a 0321, 0420 to a 0421, 0520 to a 0521, and 0620 to a 0621 message and places it in the SAF. The 0402 and 0310 messages are not changed before they are placed in the SAF.

^{*} The 0300 message used by the BASE24-teller product is not placed in the SAF.

[†] The 0310 message used by the BASE24-teller product is not placed in the SAF.

Messages Not Placed in the SAF

The ISO Host Interface process never places the following types of messages in the SAF:

- 0100—Authorization Request
- 0110—Authorization Request Response
- 0130—Authorization Advice Response
- 0200—Financial Transaction Request
- 0210—Financial Transaction Request Response
- 0230—Financial Transaction Advice Response
- 0300—File Inquiry/Update Request (BASE24-teller)*
- 0310—File Inquiry/Update Request Response (BASE24-teller)†
- 0330—File Inquiry/Update Advice Response
- 0412—Card Issuer Reversal Request Response
- 0430—Reversal Advice Response
- 0500—Acquirer Reconciliation Request
- 0510—Acquirer Reconciliation Response
- 0530—Acquirer Reconciliation Advice Response
- 0600—Administrative Request
- 0610—Administrative Response
- 0630—Administrative Advice Response

Message types 0100, 0110, 0200, 0210, 0300*, 0310†, 0600, and 0610 are interactive messages and their timing is crucial. If they are not processed within a given period of time, the transactions they represent are declined. Therefore, these messages are not placed in the SAF.

Message types 0130, 0230, 0330, 0412, 0430, 0530, and 0630 are acknowledgments or responses to other messages. Here again, timing is crucial. If a host does not receive a required acknowledgment or response within a given time, the host assumes that the BASE24 system never received the message and might then send it again as a duplicate.

* The 0300 message as used by the BASE24-from host maintenance product to send notification messages to the ACI Smart Chip Manager product can be placed in the SAF.

† The 0310 message as used by the BASE24-from host maintenance product with products other than the ACI Smart Chip Manager can be placed in the SAF.

Deleting Messages from the SAF

Records are deleted from the SAF under the following circumstances:

- When the ISO Host Interface process determines that the SAF record was processed successfully by the host.

If a host must send text-level acknowledgments, the ISO Host Interface process uses the text-level acknowledgments to determine when the host has processed the SAF records. If the host is not required to send text-level acknowledgments, the XPNET process notifies the ISO Host Interface process that the message was sent successfully. In both of these cases, the ISO Host Interface process assumes the record was processed successfully by the host and deletes the record.

- When the ISO Host Interface process determines that the SAF record cannot be processed by the host.

If the message has been rejected by the host (with an error that cannot be retried), or the same store-and-forward message has timed out or failed the number of times specified in the MAX SAF RETRY field on HCF screen 1, the ISO Host Interface process assumes the host cannot process the message. In this case, the ISO Host Interface process writes the message in an event message to the network logging facility and deletes it from the SAF.

- When the record is extracted from the SAF by way of the SAF extract.

Once a record is extracted to tape, the Super Extract process deletes it from the SAF.

Sending Store-and-Forward Messages to a DPC

The ISO Host Interface process checks for the presence of store-and-forward messages at the following times:

- When a text-level or logical acknowledgment is received for a store-and-forward message.
- Each time a message is received from the DPC.
- When a STARTSAF or SAFSTART command is received from a network control facility.

If there are store-and-forward messages to send, the ISO Host Interface process determines whether a SAF extract is in progress for the DPC. If a SAF extract is not in progress, the ISO Host Interface process begins sending the store-and-forward messages.

DPCs can have their store-and-forward messages sent to them all at once or interspersed with live transaction messages. This option is controlled by the STORE AND FORWARD PROCESSING FLAG field on HCF screen1.

Note: SAF pacing, described later in this section, overrides the processing described above. When SAF pacing is enabled, the ISO Host Interface process only checks for and sends store-and-forward messages upon expiration of a SAF pacing timer.

Noninterspersed Store-and-Forward Messages

If the HCF specifies that all store-and-forward transactions must be sent prior to sending any live transaction messages, the ISO Host Interface process stops sending any other types of messages while it sends store-and-forward messages.

Once store-and-forward processing has started, no other message types are sent to the host. Message types 0200, 0300, and 0600 from an Authorization process are returned to the Authorization process as failed; message types 0220, 0320, 0402, 0420, and 0620 are placed in the SAF.

Note: BASE24-teller 0220, 0320, 0420, and 0620 messages with the RQST.ADVC-RESP-REQ field in the Teller Standard Internal Message Header (TSTMH) set to 1 are not placed in the SAF. These messages are declined if they are received during a noninterspersed store-and-forward cycle.

The ISO Host Interface process does, however, accept 0110, 0210, 0310, 0412, and 0610 messages returning from a host. This allows transactions in progress to complete.

Interspersed Store-and-Forward Messages

If the HCF specifies that store-and-forward transactions are interspersed with live transaction messages, the ISO Host Interface process continues sending 0100, 0120, 0200, 0220, 0300, 0320, 0600, and 0620 messages along with its store-and-forward messages. For single-station hosts, the 0100, 0120, 0200, 0220, 0300, 0320, 0600, and 0620 messages always take priority.

The only messages affected in this case are 0402 and 0420 messages going to the host. Instead of being sent to the host, these messages are placed in the SAF to be sent at the end of the store-and-forward processing. This ensures that a reversal will not reach the host before the original transaction.

Single-Threaded or Multithreaded Store-and-Forward Messages

Store-and-forward cycles can be either single-threaded or multithreaded. The institution can specify the number of store-and-forward messages that can be outstanding to the host at any one time.

If the MAXIMUM OUTSTANDING SAFS field on HCF screen 1 contains the value 1, the store-and-forward cycle is single-threaded. In this case, the ISO Host Interface process only allows one outstanding store-and-forward message at a time. Additional store-and-forward messages are not sent until the ISO Host Interface process receives an acknowledgment for the original store-and-forward message or the original store-and-forward message times out.

If the MAXIMUM OUTSTANDING SAFS field on HCF screen 1 contains a value greater than 1, the store-and-forward cycle is multithreaded. In this case, the ISO Host Interface process continues to send store-and-forward messages as long as a station is available and the total number of store-and-forward messages outstanding does not exceed the value in the MAXIMUM OUTSTANDING SAFS field. For example, if the MAXIMUM OUTSTANDING SAFS field contains the value 10, the ISO Host Interface process could have up to 10 store-and-forward messages outstanding to the host at any time. For a multithreaded DPC (that is, for a DPC with the DPC TYPE field on HCF screen 1 set to a value of 0), this may mean that multiple store-and-forward messages are outstanding to a single station.

When the store-and-forward cycle is multithreaded, the host must be able to process the transactions as quickly as they arrive. In addition, the host must be able to handle transactions that arrive out of order. For example, with multithreaded processing it is possible for a 0420 message to arrive at the host before the 0220 message that is being reversed.

Note: SAF pacing, while enabled, overrides multithreading of SAF messages to the DPC. In this case, SAF messages are processed as single-threaded regardless of how the MAXIMUM OUTSTANDING SAFs field is set.

Store-and-Forward Pacing

The ISO Host Interface process allows for store-and-forward messages to be throttled as needed to control the pace at which the messages are sent to a host DPC.

This throttling, referred to as SAF pacing, is achieved by setting a SAF pacing timer each time a SAF message is sent to the DPC and then waiting for the timer to expire before sending another SAF message.

SAF pacing overrides the routine store-and-forward processing used by the ISO Host Interface process. When SAF pacing is enabled, the ISO Host Interface process only checks for and sends store-and-forward messages upon expiration of a SAF pacing timer.

The SAF pacing interval—the length of the SAF pacing timer—is specified in hundredths of a second, with a minimum value of 0 (no throttling) and a maximum value of 9999 (99.99 seconds).

Enabling pacing, as well as setting the SAF pacing interval, is done operationally by entering a SAFPACING command from the network control facility. This command sets the timer interval for a specific DPC.

Disabling pacing is done by entering the SAFPACING command from the network control facility with a interval value of 0.

Operators can check on the status of SAF pacing for a DPC by entering a PACINGSTAT command. This command causes the ISO Host Interface process to generate event messages to report the DPC's pacing interval.

Multithreading Note: SAF pacing, while enabled, overrides multithreading of SAF messages to the DPC.

Nonintersperced Store-and-Forward Processing Note: SAF pacing is not intended for use by hosts set up for receiving nonintersperced store-and-forward messages. Nonintersperced store-and-forward processing overrides the throttling provided by SAF pacing.

Text-Level Acknowledgments for Store-and-Forward Messages

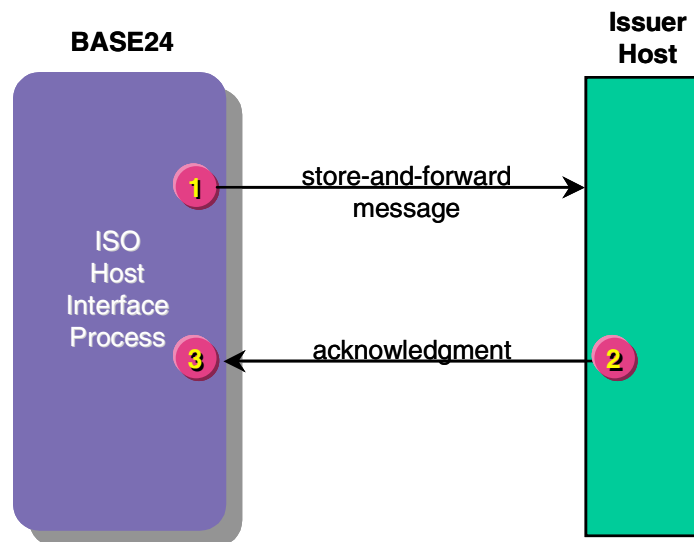
If text-level acknowledgments are required from a host, the host must respond to store-and-forward messages with the following text-level acknowledgments.

Store-and-Forward Message	Text-Level Acknowledgment
0120	0130
0121	0130
0220	0230
0221	0230
0300*	none
0310*	none
0320	0330
0321	0330
0402	0412
0420	0430
0421	0430
0520	0530
0521	0530
0620	0630
0621	0630

* A 0300 or 0310 message can be included in the SAF for the BASE24-from host maintenance product, but not for BASE24-teller. No text-level acknowledgment is ever required or expected from the host for 0300 and 0310 store-and-forward messages.

Store-and-Forward Message Processing

The following diagram illustrates the message exchanges expected by the BASE24 system when store-and-forward messages are sent to an issuer host. The steps corresponding to the diagram are provided below.



If SAF pacing is enabled, the following steps only occur upon expiration of a SAF pacing timer. If SAF pacing is not enabled, these steps occur whenever store-and-forward messages need to be sent to the DPC.

1. The ISO Host Interface process sends a store-and-forward message to the host. To do this, it performs the following steps:
 - a. Reads a record from the SAF.
 The ISO Host Interface process selects the oldest record in the file for the DPC. (The record is already in external message format.)
 - b. Sets a flag for the DPC to indicate that store-and-forward processing is in progress. The ISO Host Interface process performs this step only on the first store-and-forward message processed.
 - c. Locates an available send/receive station and sends the message.
 - d. Sets a store-and-forward message timer for the message.

- e. Sets a SAF pacing timer if SAF pacing is enabled.
- f. Adds one to the count of store-and-forward messages outstanding.

Note: If the store-and-forward cycle is multithreaded and SAF pacing is not enabled, the ISO Host Interface process sends multiple SAF records to the host using the steps described above. In this case, the ISO Host Interface process can send up to the value specified in the MAX OUTSTANDING SAFS field before processing any acknowledgments from the host. If SAF pacing is enabled, only one SAF record can be sent at a time.

- 2. The host processes the store-and-forward message and returns an acknowledgment message if text-level acknowledgments are required. If text-level acknowledgments are not required, the XPNET process notifies the ISO Host Interface process when it has successfully sent the message to the DPC.
- 3. Upon receipt of an acknowledgment message or notification from the XPNET process that the store-and-forward message was sent, the ISO Host Interface process performs the following steps:
 - a. Locates and deletes the corresponding record in the SAF.
 - b. Subtracts one from the number of store-and-forward messages outstanding.
- 4. Determines whether more SAF records need to be sent.
 - If more SAF records need to be sent, the ISO Host Interface process determines whether SAF pacing is enabled. If SAF pacing is not enabled, it begins again with step 1 to send another record. If SAF pacing is enabled, it waits for the SAF pacing timer to expire before starting again with step 1.
 - If no more SAF records need to be sent, the ISO Host Interface process continues processing acknowledgments as they are received until no more store-and-forward messages are outstanding. The ISO Host Interface process then terminates the store-and-forward processing and resets the flag for the DPC, indicating that store-and-forward processing is no longer in progress.

Rejected Store-and-Forward Messages

If a message cannot be processed by a host, the message must be returned as rejected with the first position of the message type set to 9.

The host can reject the message for several reasons, including retrievable message authentication errors (that is, key synchronization errors and security device failures). If the message was rejected because of a retrievable message authentication error, the Status field in the external message header contains one of the following values: 196, 199, 210, 220, or 230. In this case, the ISO Host Interface process resends the message to the host.

When the ISO Host Interface process receives a message rejected by the host for any other reason, it performs the following steps:

1. Writes the message in an event message to the network logging facility.
2. Deletes the record from the SAF.
3. Clears the SAF retries count.
4. Continues store-and-forward processing with the next record in the SAF.

Failed Store-and-Forward Messages

Messages marked as failed by the XPNET process are handled as if the messages timed out. For a description of this processing, refer to the discussion on the store-and-forward message timer in section 2.

5: BASE24-atm Financial Message Handling

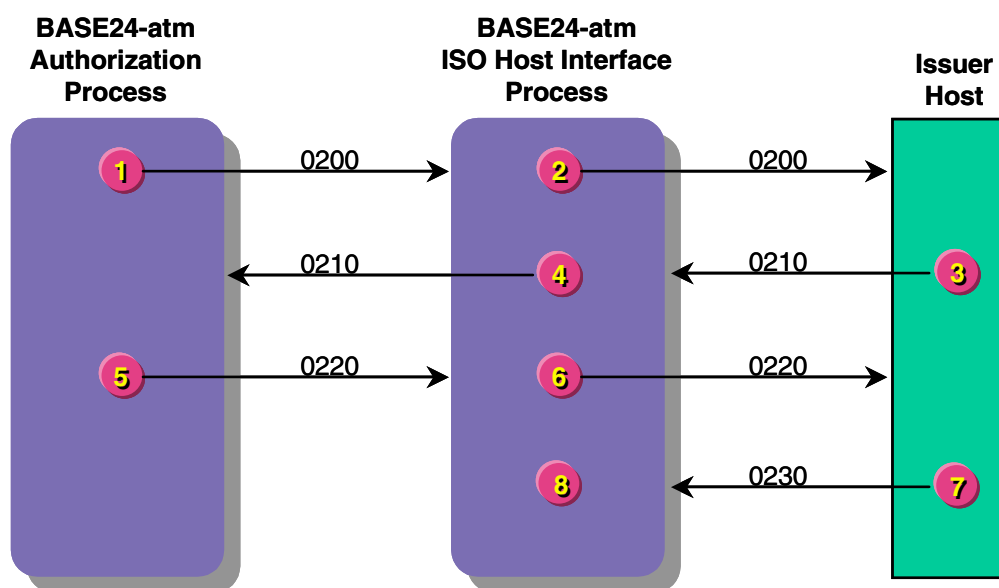
This section contains an overview of how BASE24-atm financial transaction messages are handled by the ISO Host Interface process. It includes message flows for both issuer and acquirer host processing.

Issuer Host Message Handling

This section describes the processing involved for messages to and from an issuer host.

Financial Transaction Message Flow

The following diagram illustrates the message exchange expected for financial transactions sent to an issuer host by the BASE24-atm ISO Host Interface process. The steps corresponding to the diagram are provided below.



1. The BASE24 system receives a 0200 financial transaction request message, which is sent to the BASE24-atm Authorization process.

The BASE24-atm Authorization process determines that the request must be sent to an issuer host for authorization and sends it to the ISO Host Interface process for forwarding to that host.

The BASE24-atm Authorization process retrieves the name of the ISO Host Interface process to which to send the message from the ATM ROUTING TABLE fields on IDF screen 9.

2. The ISO Host Interface process receives the 0200 message and performs the following steps:
 - a. Locates the appropriate DPC processing information in its internal table using the DPC number from the DPC-NUM field in the Standard Internal Message (STM).

- b. Creates an external message from the STM.
- c. Updates the EXIT-TIM field in the STM to the current time.
- d. Determines its ability to send the message. The ISO Host Interface process cannot send the message if any of the following conditions are true:
 - The DPC is marked down.
 - There is a noninterspersed store-and-forward cycle in progress.
 - The host is single-threaded and no send/receive stations are available.
 - The current number of outstanding messages to the DPC equals the value in the MAXIMUM OUTSTANDING REQUESTS OUTBOUND field on HCF screen 1.

If any of the above conditions exist, the ISO Host Interface process performs the following steps:

- 1) Sets the RTE.STAT field in the STM to a value of 12 to indicate that the DPC was marked down.
- 2) Adjusts the RQST.COMPL-REQ field in the STM, if necessary.
 - If the RQST.COMPL-REQ field in the STM contains a value of 0 (completions not required) or 1 (approved completions required), the ISO Host Interface process sets the RQST.COMPL-REQ field in the STM to a value of 2 to indicate that an advice is required if the request is approved offline.
 - If the RQST.COMPL-REQ field in the STM contains a value of 4 (completions required), 5 (completions required for approved BASE24-authorized transactions), or 6 (completions required for all BASE24-authorized transactions - approved and denied), the ISO Host Interface process does not change the value.
- 3) Returns the 0200 message (STM) to the process indicated in the ORIG-PRO-NAME field of the STM.

If none of the above conditions exist, the ISO Host Interface process continues with step 2e.

- e. Saves a copy of the STM in extended memory. As part of saving the message, the ISO Host Interface process stores the current time in a user buffer in memory.

- f. Sets an outbound request timer for the number of seconds specified by the OUTBOUND field on HCF screen 5.

If this timer expires before a 0210 message is received, the ISO Host Interface process checks the value of the EXTRN-RQST-TIMEOUT param in the LCONF to determine which actions are performed.

- If the param is set to a value of 0, the ISO Host Interface process returns the original 0200 message to the BASE24-atm Authorization process as failed for alternate or stand-in routing. If the ISO Host Interface process receives a late 0210 message, it generates an 0420 message and sends it to the host.
 - If the param is set to a value of 1, the ISO Host Interface process returns the original 0200 message to the BASE24-atm Authorization process as failed for alternate or stand-in routing and generates an 0420 message that it sends to the host. If the ISO Host Interface process receives a late 0210 message, it drops the message.
- g. Updates its internal counters. The ISO Host Interface process performs the following steps:
- 1) Adds one to the number of messages currently outstanding to the DPC.
 - 2) Adds one to the total number of messages currently outstanding from the ISO Host Interface process.
- h. Generates a message authentication code (MAC) for the message, if message authentication is supported and is specified for this message type.
- i. Converts the message from ASCII to EBCDIC if the CHARACTER FORMAT field on screen 1 of the HCF is set to E for the DPC.
- j. Encrypts the message if full message encryption is supported and is specified for the DPC.
- k. Sends the 0200 message to the DPC.

Along with the message, the ISO Host Interface process also sends the maximum number of seconds the XPNET process can allow the message to queue to the host. This number of seconds is derived using the values in the QUEUE SUBTRACT and OUTBOUND fields on HCF screen 5. The value in the QUEUE SUBTRACT field is subtracted from the value in the OUTBOUND field and the result is used.

3. The host validates the request, processes the transaction, and returns a 0210 message either approving or declining the transaction.

4. The ISO Host Interface process receives the 0210 message and performs the following steps:
 - a. Locates the DPC in its internal table, identifying the necessary information for processing.
 - b. Decrypts the message if full message encryption is supported and is specified for the DPC.
 - c. Converts the message from EBCDIC to ASCII if the CHARACTER FORMAT field on screen 1 of the HCF is set to E for the DPC.
 - d. Verifies the MAC received with the message, if message authentication is supported and is specified for the message type.
 - e. Retrieves the stored STM from extended memory.
 - f. Deletes the outbound request timer.
 - g. Updates its counters. The ISO Host Interface process performs the following steps:
 - 1) Subtracts one from the number of messages currently outstanding to the DPC.
 - 2) Adds one to the number of requests processed for the DPC.
 - 3) Adds the amount of time taken to process this request to the total time taken for all requests to the DPC. (For performance statistics, this value is divided by the number of requests processed for the DPC.)
 - h. Updates the EXIT-TIM field in the STM using the time stored in the user buffer when the request message was saved in extended memory (see step 2e).
 - i. Sets the RE-ENTRY-TIM field in the STM to the current time.
 - j. Sets the TYP field in the STM to 0210.
 - k. Updates the STM from the external message using only that information which may have been changed by the host.
 - l. Sends the 0210 message (STM) to the Authorization process.
5. The BASE24-atm Authorization process finishes processing the transaction, logs it to the Transaction Log File (TLF), and returns it to its originator.
 If the host requires advices, the BASE24-atm Authorization process generates the appropriate 0220 message and sends it to the ISO Host Interface process.

6. The ISO Host Interface process receives the 0220 message and performs the following steps:
 - a. Locates the appropriate DPC processing information in its internal table using the DPC number from the DPC-NUM field in the STM.
 - b. Creates an external message from the STM.
 - c. Checks the RQST.COMPL-REQ field in the STM. If the value in that field is 2 (host is offline—completion required), the ISO Host Interface process writes the external version of the 0220 message to the Store and Forward File (SAF) and takes no further action with the message until a store-and-forward cycle is started.
 - d. Determines whether the DPC is marked down. If so, the ISO Host Interface process writes the external version of the 0220 message to the SAF and takes no further action with the message until a store-and-forward cycle is started.
 - e. Determines whether a noninterspersed store-and-forward cycle is in progress. If so, the ISO Host Interface process writes the external version of the 0220 message to the SAF, and the message is handled in sequential order.
 - f. Locates an available send/receive station. If no send/receive stations are available, the ISO Host Interface process writes the external version of the 0220 message to the SAF and takes no further action with the message until a store-and-forward cycle is started.
 - g. Generates a MAC for the message, if message authentication is supported and is specified for this message type.
 - h. Converts the message from ASCII to EBCDIC if the CHARACTER FORMAT field on screen 1 of the HCF is set to E for the DPC.
 - i. Encrypts the message if full message encryption is supported and is specified for the DPC.
 - j. Sends the 0220 message to the DPC.

Along with the message, the ISO Host Interface process also sends the maximum number of seconds the XPNET process can allow the message to queue to the host. This number of seconds is derived using the values in the QUEUE SUBTRACT and COMPLETION ACK fields on HCF screen 5. The value in the QUEUE SUBTRACT field is subtracted from the value in the COMPLETION ACK field and the result is used.

- k. Stores a copy of the external message.

- l. Sets a completion acknowledgment timer.
- m. Adds one to the number of messages currently outstanding to the DPC.
- 7. The host processes the 0220 message and returns a 0230 message if text-level acknowledgments are required.

DPCs are required to return text-level acknowledgments if the ACK FROM DPC field on HCF screen 1 contains the value Y.

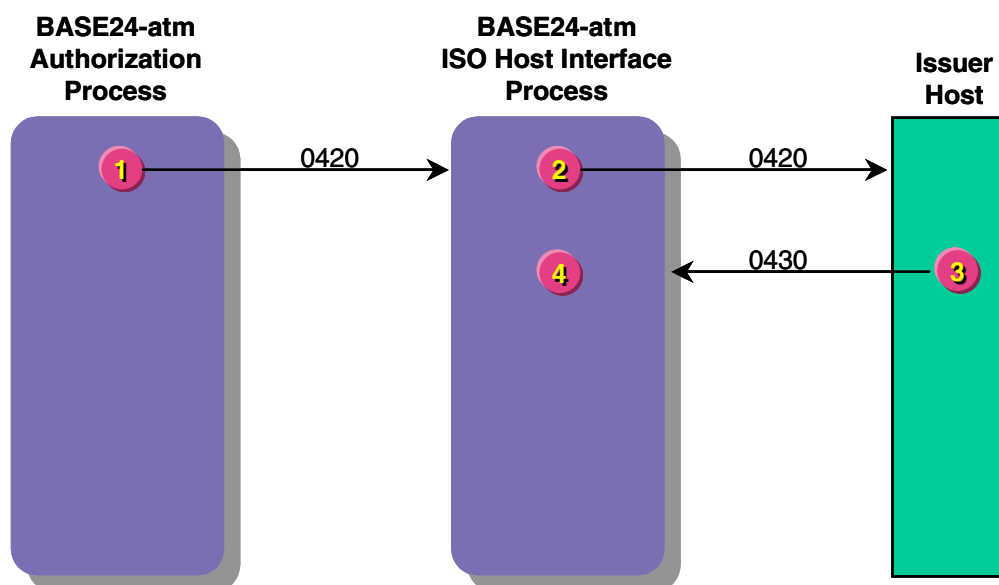
- 8. Upon receipt of a 0230 message, the ISO Host Interface process performs the following steps:
 - a. Locates the DPC in its internal table, identifying the necessary information for processing.
 - b. Decrypts the message if full message encryption is supported and is specified for the DPC.
 - c. Converts the message from EBCDIC to ASCII if the CHARACTER FORMAT field on screen 1 of the HCF is set to E for the DPC.
 - d. Verifies the MAC received with the message, if message authentication is supported and is specified for the message type.
 - e. Deletes the stored external message from extended memory.
 - f. Deletes the completion acknowledgment timer.
 - g. Subtracts one from the number of messages currently outstanding to the DPC.

Processing for the 0230 message is now complete.

Reversal Message Flow

The following diagram illustrates the message exchange expected for reversal transactions sent to an issuer host by the BASE24-atm ISO Host Interface process. The steps corresponding to the diagram are provided below.

A reversal indicates that a previously-approved transaction did not complete as intended.



1. The BASE24 system is advised that a transaction did not complete as previously authorized. The notice may be from an ATM controlled by the BASE24 system, an acquirer host, or an interchange.

The BASE24-atm Authorization process backs out the original transaction, posts the revised transaction, logs the transaction to the TLF, and sends a 0420 message in STM format to the ISO Host Interface process.

2. The ISO Host Interface process receives the 0420 message and processes it by performing the following steps:
 - a. Locates the DPC record in its internal table using the DPC-NUM field in the STM.
 - b. Checks the value in the RQST.RESP-CDE field in the STM. If it is greater than 09, meaning the transaction was declined, the ISO Host Interface process drops the message.
 - c. Creates an external message from the STM.

- d. Determines whether the DPC is marked down. If so, the ISO Host Interface process writes the external version of the 0420 message to the SAF and takes no further action with the message until a store-and-forward cycle is started.
 - e. Determines whether a store-and-forward cycle (either interspersed or noninterspersed) is in progress for the DPC. If so, the ISO Host Interface process writes the external version of the 0420 message to the SAF, and the message is handled in sequential order.
 - f. Checks the RQST.COMPL-REQ field in the STM. If it equals 2 (host offline, completion required), the ISO Host Interface process writes the external version of the 0420 message to the SAF and takes no further action with the message until a store-and-forward cycle is started.
 - g. Locates an available send/receive station. If no send/receive stations are available, the ISO Host Interface process writes the external version of the 0420 message to the SAF and takes no further action with the message until a store-and-forward cycle is started.
 - h. Generates a MAC for the message, if message authentication is supported and is specified for this message type.
 - i. Converts the message from ASCII to EBCDIC if the CHARACTER FORMAT field on screen 1 of the HCF is set to E for the DPC.
 - j. Encrypts the message if full message encryption is supported and is specified for the DPC.
 - k. Sends the 0420 message to the DPC.

Along with the message, the ISO Host Interface process also sends the maximum number of seconds the XPNET process can allow the message to queue to the host. This number of seconds is derived using the values in the QUEUE SUBTRACT and COMPLETION ACK fields on HCF screen 5. The value in the QUEUE SUBTRACT field is subtracted from the value in the COMPLETION ACK field and the result is used.
 - l. Stores a copy of the external message.
 - m. Sets a completion acknowledgment timer.
 - n. Adds one to the number of messages currently outstanding to the DPC.
3. The host processes the 0420 message and returns a 0430 message if text-level acknowledgments are required.

DPCs are required to return text-level acknowledgments if the ACK FROM DPC field on HCF screen 1 contains the value Y.

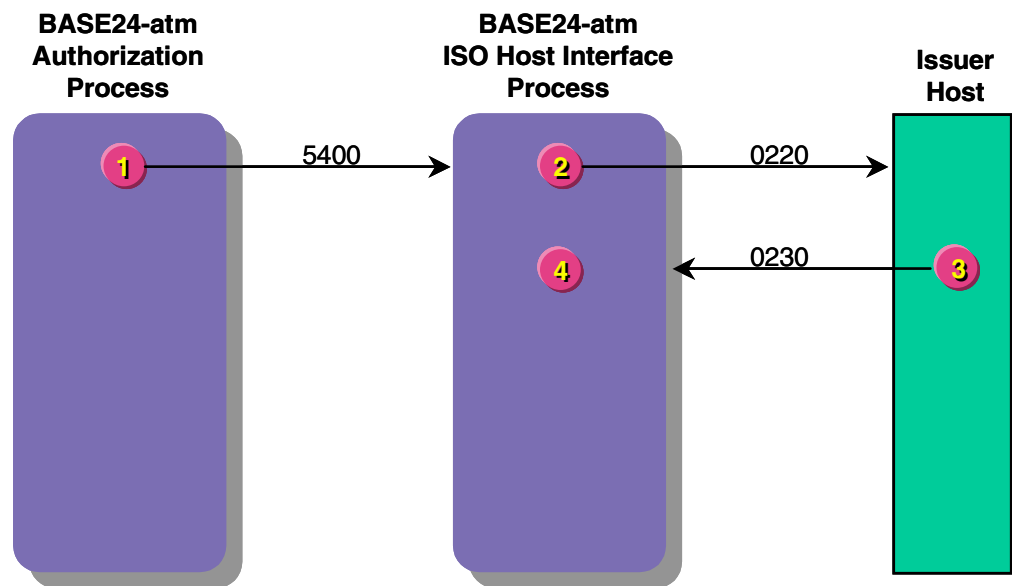
4. Upon receipt of a 0430 message, the ISO Host Interface process performs the following steps:
 - a. Locates the DPC in its internal table, identifying the necessary information for processing.
 - b. Decrypts the message if full message encryption is supported and is specified for the DPC.
 - c. Converts the message from EBCDIC to ASCII if the CHARACTER FORMAT field on screen 1 of the HCF is set to E for the DPC.
 - d. Verifies the MAC received with the message, if message authentication is supported and is specified for the message type.
 - e. Deletes the stored external message from extended memory.
 - f. Deletes the completion acknowledgment timer.
 - g. Subtracts one from the number of messages currently outstanding to the DPC.

Processing for the 0430 message is now complete.

Interchange Adjustment Transaction Message Flow

The following diagram illustrates the message exchange expected when the BASE24 system receives an interchange adjustment from a regional or national interchange that must be transmitted to an issuer host. The steps corresponding to the diagram are provided below.

Note: Adjustments are used by some interchanges to advise the BASE24 system of previously-authorized financial transactions that do not complete as authorized. They are only allowed from interchanges.



1. Adjustment transactions are sent to the BASE24-atm Authorization process as 5400 messages by the Interchange Interface process.

The Authorization process determines whether the 5400 message must be sent to an issuer host using the value from the HOST ADJ. PROCESSING field on IDF screen 3. If the first position of the HOST ADJ. PROCESSING field is set to the value 2, the Authorization process forwards the message to the ISO Host Interface process.

2. The ISO Host Interface process receives the 5400 message and performs the following steps:
 - a. Locates the appropriate DPC processing information in its internal table using the DPC number from the DPC-NUM field in the STM.
 - b. Creates a 0220 external message from the STM.

- c. Determines whether the DPC is marked down. If so, the ISO Host Interface process writes the external version of the 0220 message to the SAF and takes no further action with the message until a store-and-forward cycle is started.
 - d. Determines whether a noninterspersed store-and-forward cycle is in progress. If so, the ISO Host Interface process writes the external version of the 0220 message to the SAF, and the message is handled in sequential order.
 - e. Locates an available send/receive station. If no send/receive stations are available, the ISO Host Interface process writes the external version of the 0220 message to the SAF and takes no further action with the message until a store-and-forward cycle is started.
 - f. Generates a MAC for the message, if message authentication is supported and is specified for this message type.
 - g. Converts the message from ASCII to EBCDIC if the CHARACTER FORMAT field on screen 1 of the HCF is set to E for the DPC.
 - h. Encrypts the message if full message encryption is supported and is specified for the DPC.
 - i. Sends the 0220 message to the DPC.

Along with the message, the ISO Host Interface process also sends the maximum number of seconds the XPNET process can allow the message to queue to the host. This number of seconds is derived using the values in the QUEUE SUBTRACT and COMPLETION ACK fields on HCF screen 5. The value in the QUEUE SUBTRACT field is subtracted from the value in the COMPLETION ACK field and the result is used.
 - j. Stores a copy of the external message.
 - k. Sets a completion acknowledgment timer.
 - l. Adds one to the number of messages currently outstanding to the DPC.
3. The host processes the 0220 message and returns a 0230 message if text-level acknowledgments are required.

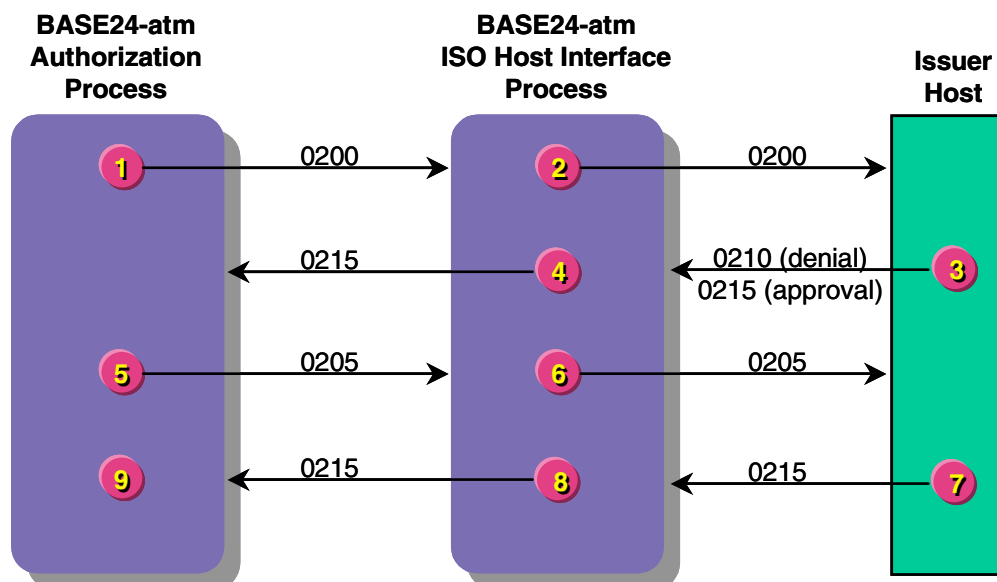
DPCs are required to return text-level acknowledgments if the ACK FROM DPC field on HCF screen 1 contains the value Y.

4. Upon receipt of a 0230 message, the ISO Host Interface process performs the following steps:
 - a. Locates the DPC in its internal table, identifying the necessary information for processing.
 - b. Decrypts the message if full message encryption is supported and is specified for the DPC.
 - c. Converts the message from EBCDIC to ASCII if the CHARACTER FORMAT field on screen 1 of the HCF is set to E for the DPC.
 - d. Verifies the MAC received with the message, if message authentication is supported and is specified for the message type.
 - e. Deletes the stored external message from extended memory.
 - f. Deletes the completion acknowledgment timer.
 - g. Subtracts one from the number of messages currently outstanding to the DPC.

Processing for the 0230 message is now complete.

Statement Print Transaction Message Flow

The following diagram illustrates the message exchange expected when the BASE24 system sends a statement print transaction to the host. The steps corresponding to the diagram are provided below.



1. The BASE24 system receives a request for statement information, which is sent to the BASE24-atm Authorization process as a 0200 message.

The BASE24-atm Authorization process determines that the transaction is a request for statement information, which must be provided by an issuer host, and sends it to the ISO Host Interface process for forwarding to that host.

The BASE24-atm Authorization process retrieves the name of the ISO Host Interface process to which to send the message from the ATM ROUTING TABLE fields on IDF screen 9.

2. The ISO Host Interface process receives the 0200 message and treats it the same as a financial transaction. The ISO Host Interface process converts the message to an external message format and sends it to the host.

ASCII-to-EBCDIC conversions, message authentication, and full message encryption are supported throughout this transaction flow in the same manner as for financial transactions.

3. The host processes the 0200 message and returns a 0215 message approving the transaction or a 0210 message declining the transaction. The 0215 message carries the statement information.

4. The ISO Host Interface process receives the 0215 message, converts the message to STM format using the STMT-INFO definition fields of the STM, and sends the STM to the Authorization process.
5. The BASE24-atm Authorization process finishes processing the 0215 message, logs it to the TLF, and returns it to its originator.

The originator can request additional pages of statement information, in which case it will send the BASE24-atm Authorization process a 0205 message. When the BASE24-atm Authorization process receives the 0205 message, it passes the 0205 message on directly to the ISO Host Interface process.

6. The ISO Host Interface process receives the 0205 message, converts it to an external message, and sends it to the host.

For purposes of ASCII-to-EBCDIC conversions, message authentication, full message encryption, and resetting timers and internal counters, the ISO Host Interface process treats the 0205 message the same as a 0200 message.

7. The host processes the 0205 message and returns a 0215 message.
8. The ISO Host Interface process receives the 0215 message, converts it to STM format, sets the TYP field in the STM to the value 0215, and sends it to the Authorization process.
9. The BASE24-atm Authorization process then passes the 0215 message to the originator (only the first 0215 message is logged to the TLF).

Late Response from a DPC

If a DPC responds late to a request, the ISO Host Interface process performs the following steps for the 0210 message:

1. Checks the Response Code data element (P-39) in the external message.

If the transaction was declined, the ISO Host Interface process drops the message.

2. Checks the Processing Code data element (P-3) in the external message.

If the transaction was a balance inquiry, statement print transaction, or log-only transaction, the ISO Host Interface process drops the message. For all other messages, processing continues with step 3.

Note: If the transaction is a balance inquiry and the REVERSE-BAL-INQ param in the Logical Network Configuration File (LCONF) is set to a true value, the ISO Host Interface process does not drop the transaction, but rather continues with step 4.

3. Checks the value of the EXTRN-RQST-TIMEOUT param in the LCONF. If this param is set to a value of 0, processing continues with step 4. If this param is set to 1, the ISO Host Interface process drops the message.
4. Creates a 0420 message. In doing so, the ISO Host Interface process performs the following steps:
 - a. Sets the Response Code data element (P-39) in the external message to 68 (timeout).
 - b. Sets the Replacement Amounts data element (S-95) in the external message to zeros.
 - c. Sets the Original Data Elements data element (S-90) in the external message to 0200.
5. Selects the station to which to send the message.

To do this, the ISO Host Interface process first checks the station from which the 0210 was received. If it is a send/receive station and is available, the ISO Host Interface process uses that one. Otherwise, it checks for any available send/receive stations.

If a station is available, the ISO Host Interface process uses it. If no stations are available, the ISO Host Interface process writes the 0420 message to the SAF and takes no further action with the message until a store-and-forward cycle is started.

6. Generates a MAC for the message, if message authentication is supported and is specified for this message type.
7. Converts the message from ASCII to EBCDIC if the CHARACTER FORMAT field on screen 1 of the HCF is set to E for the DPC.
8. Encrypts the message if full message encryption is supported and is specified for the DPC.
9. Sends the 0420 message to the selected station.

Along with the message, the ISO Host Interface process also sends the maximum number of seconds the XPNET process can allow the message to queue to the host. This number of seconds is derived using the values in the QUEUE SUBTRACT and COMPLETION ACK fields on HCF screen 5. The value in the QUEUE SUBTRACT field is subtracted from the value in the COMPLETION ACK field and the result is used.

10. Stores a copy of the external message.
11. Sets a completion acknowledgment timer.
12. Adds one to the number of messages currently outstanding to the DPC.

Failed Messages to Stations

If a message is forwarded to the XPNET process, but cannot be sent to a station, the XPNET process returns the message to the ISO Host Interface process as failed.

Upon receipt of a message failure, the ISO Host Interface process marks the station down and sets an extended network management message timer. It then performs the following steps based on the message type.

Failed 0200 Messages

The ISO Host Interface process takes the following steps when it receives a failed 0200 message back from the XPNET process. This condition may be due to an unavailable station or the transaction becoming stale while queued by the XPNET process.

1. Locates the DPC in its internal table, identifying the necessary information for processing.
2. Retrieves the stored STM from extended memory.
3. Deletes the outbound request timer.
4. Updates its counters. The ISO Host Interface process performs the following steps:
 - a. Subtracts one from the number of messages currently outstanding to the DPC.
 - b. Adds one to the number of requests processed for the DPC.
 - c. Adds the amount of time taken to process this request to the total time taken for all requests to the DPC. (For performance statistics, this value is divided by the number of requests processed for the DPC.)
5. Sets the RTE.STAT field in the STM to a value of 12 to indicate the line was down.
6. Sets the RQST.COMPL-REQ field in the STM to a value of 2 to indicate that a 0220 message is required if the transaction is subsequently approved.
7. Returns the STM (as a failed 0200 message) to the BASE24-atm Authorization process.

Failed 0220/0420 Messages

The ISO Host Interface process takes the following steps when it receives a failed 0220 or 0420 message back from the XPNET process. This condition may be due to an unavailable station or the transaction becoming stale while queued by the XPNET process.

1. Locates the DPC in its internal table, identifying the necessary information for processing.
2. Retrieves the stored external message from extended memory.
3. Checks for a completion acknowledgment timer for the message and deletes it if found.
4. Subtracts one from the number of messages currently outstanding to the DPC.
5. Adds the message to the SAF.

Unintelligible Messages from a DPC

If the ISO Host Interface process receives an external message that it cannot process or reformat for internal use (because of bad data or a security-related failure), it rejects the message by performing the following steps:

1. Changes the first position of the message type to a value of 9 (for example, a 0200 message would be changed to a 9200 message, and a 0420 message would be changed to a 9420 message).
2. Indicates the reason the message was rejected in the status field of the external message header. If the message was rejected because of a security failure, the status field is set to an indicative reject reason code. If the message was rejected because of bad data, this field is set to the bit number of the data element causing rejection (for example, if the Track 2 data in the P-35 data element cannot be parsed, the status field in the header would be set to 035).
3. Returns the message to the host.

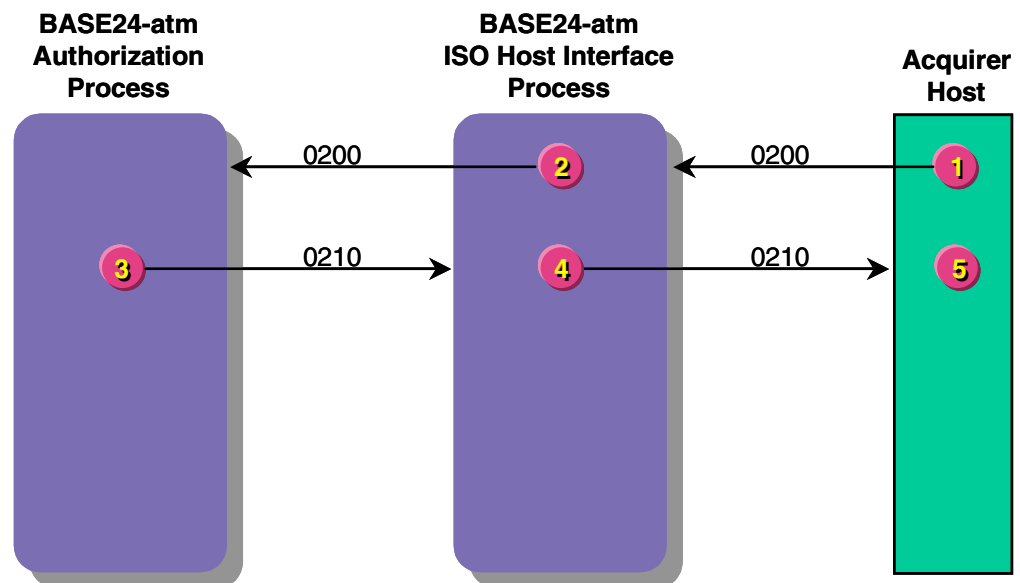
The above actions are taken on any message type that cannot be processed, not just those that require a response.

Acquirer Host Message Handling

This section describes the processing involved for messages to and from an acquirer host.

Financial Transaction Message Flow

The following diagram illustrates the message exchanges expected by BASE24-atm for financial transactions sent to the BASE24 system by an acquirer host. The steps corresponding to the diagram are provided below.



1. A cardholder initiates a financial transaction at a terminal connected to and controlled by an acquirer host. The acquirer host then formats a 0200 message and sends it to the BASE24 system.
2. The ISO Host Interface process receives the 0200 message and performs the following steps:
 - a. Locates the DPC in its internal table, identifying the necessary information for processing.
 - b. Decrypts the message if full message encryption is supported and is specified for the DPC.
 - c. Converts the message from EBCDIC to ASCII if the CHARACTER FORMAT field on screen 1 of the HCF is set to E for the DPC.

- d. Verifies the MAC received with the message, if message authentication is supported and is specified for the message type.
 - e. Creates an STM from the external message.
 - f. Updates the following fields:
 - 1) Sets the value in the RTE.ORIGINATOR field to indicate the transaction was initiated through an acquirer host.
 - 2) Sets the value in the DPC-NUM field to the number of the DPC originating the transaction.
 - 3) Sets the value in the ENTRY-TIM field to the current time.
 - g. Checks the current number of outstanding 0200 messages to the Authorization process.

If the number of 0200 messages outstanding to the Authorization process equals the value in the MAXIMUM OUTSTANDING REQUESTS INBOUND field on HCF screen 1, the ISO Host Interface process creates a 0210 message with a response code of 91 (issuer inoperative) and sends it to the originating DPC.
 - h. Saves a copy of the external message in extended memory.
 - i. Sets an inbound request timer for the number of seconds specified by the value in the INBOUND field on HCF screen 5.

If this timer expires before a 0210 message is received, the ISO Host Interface process declines the transaction.
 - j. Adds one to the number of requests currently outstanding from the DPC.
 - k. Sends the 0200 message (STM) to the Authorization process indicated in the AUTH PROCESS field on HCF screen 5.
3. The BASE24-atm Authorization process processes the transaction, logs it to the TLF, and returns a 0210 message to the ISO Host Interface process.
- Note:** The transaction may also have been authorized by an interchange or issuer host, depending on the cardholder and the type of authorization processing performed.
4. The ISO Host Interface process receives the 0210 message and performs the following steps:
- a. Locates the stored external message.
 - b. Deletes the inbound request timer.
 - c. Subtracts one from the number of requests currently outstanding from the DPC.

- d. Creates a 0210 external message from the STM.
- e. Generates a MAC for the message, if message authentication is supported and is specified for this message type.
- f. Converts the message from ASCII to EBCDIC if the CHARACTER FORMAT field on screen 1 of the HCF is set to E for the DPC.
- g. Encrypts the message if full message encryption is supported and is specified for the DPC.
- h. Sends the 0210 external message to the station from which the 0200 message was originally received.

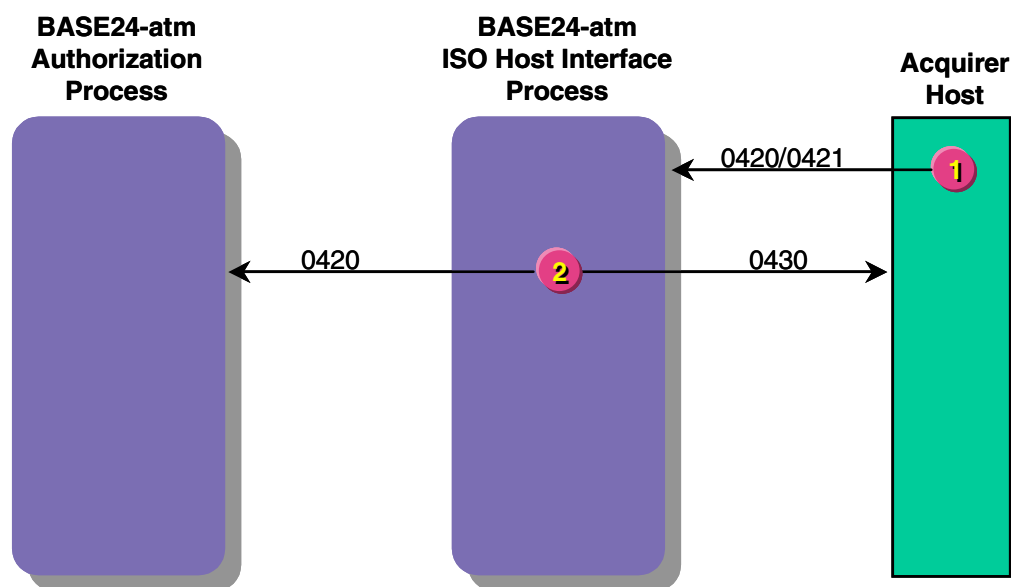
If that station cannot be sent messages, the ISO Host Interface process selects another send/receive station. If no send/receive stations are available, the ISO Host Interface process creates a reversal (0420) message using the STM and returns it to the Authorization process. A reversal is only created if the transaction is approved and not a balance inquiry, statement print transaction, or log-only transaction.

Note: If the transaction is an approved balance inquiry and the REVERSE-BAL-INQ param in the LCONF is set to a true value, the ISO Host Interface process returns a reversal to the Authorization process for the transaction.

- 5. The acquirer host then completes the transaction with the cardholder at the ATM.

Reversal Transaction Message Flow

The following diagram illustrates the message exchanges expected by BASE24-atm for reversals sent to BASE24-atm by an acquirer host. A reversal occurs when a BASE24-authorized transaction does not complete as authorized. The steps corresponding to the diagram are provided below.



1. The transaction-originating terminal notifies the acquirer host that the transaction did not complete as authorized. The acquirer host then formats a 0420 message and sends it to the BASE24 system (0421 messages may be used to indicate a possible duplicate).
2. The ISO Host Interface process receives the 0420 or 0421 message and performs the following steps:
 - a. Locates the DPC in its internal table, identifying the necessary information for processing.
 - b. Decrypts the message if full message encryption is supported and is specified for the DPC.
 - c. Converts the message from EBCDIC to ASCII if the CHARACTER FORMAT field on screen 1 of the HCF is set to E for the DPC.
 - d. Verifies the MAC received with the message, if message authentication is supported and is specified for the message type.
 - e. Creates an STM from the external message.

- f. Sends the 0420 message (STM) to the Authorization process indicated in the AUTH PROCESS field on HCF screen 5.
- g. Generates a 0430 message if the ACK TO DPC field on HCF screen 1 contains the value Y.

The ISO Host Interface process performs ASCII-to-EBCDIC conversions, MAC creation, and full message encryption on the 0430 message as required.

Late Response from the Authorization Process

If the ISO Host Interface process receives a late 0210 message from the Authorization process, it performs the following steps:

1. Checks the value in the RQST.RESP field in the STM to determine whether the transaction was declined. If it was, the ISO Host Interface process drops the message.
2. Checks the value in the TRAN-CDE field in the STM to determine whether the transaction is a balance inquiry, statement print transaction, or log-only transaction. If it is, the ISO Host Interface process drops the message.

Note: If the transaction is a balance inquiry and the REVERSE-BAL-INQ param in the LCONF is set to a true value, the ISO Host Interface process does not drop the transaction, but rather continues with step 3.

3. Changes the TYP field in the STM to 0420.
4. Sets the RQST.RVSL-CDE field in the STM to a value of 01 to indicate that the Authorization process timed out.
5. Sets the RQST.AMT-2 and RQST.AMT-3 fields in the STM to values of zeros.
6. Sends the 0420 message back to the Authorization process that sent it.

Failed Messages to Stations

If a message is forwarded to the XPNET process, but cannot be sent to a station, the XPNET process returns the message to the ISO Host Interface process as failed.

Upon receipt of a message failure, the ISO Host Interface process marks the station down and sets an extended network management message timer. It then performs the following steps based on message type.

Failed 0210 Messages

The ISO Host Interface process takes the following steps when it receives a failed 0210 message back from the XPNET process.

1. Locates the DPC in its internal table, identifying the necessary information for processing.
2. Checks the Response Code data element (P-39) in the external message to determine whether the transaction was declined. If it was, the ISO Host Interface process drops the message.
3. Checks the Processing Code data element (P-3) in the external message to determine whether the transaction is a balance inquiry, statement print transaction, or log-only transaction. If it is, the ISO Host Interface process drops the message.

Note: If the transaction is a balance inquiry and the REVERSE-BAL-INQ param in the LCONF is set to a true value, the ISO Host Interface process does not drop the transaction, but rather continues with step 4.
4. Creates an STM from the external message and sets the TYP field to a value of 0420.
5. Sets the RQST.RVSL-CDE field in the STM to a value of 03 to indicate that the destination was not available.
6. Sets the RQST.AMT-2 and RQST.AMT-3 fields in the STM to values of zeros.
7. Sends the 0420 message to the Authorization process indicated in the AUTH PROCESS field on HCF screen 5.

Failed 0230/0430 Messages

If the ISO Host Interface process receives a failed 0230 or 0430 message back from the XPNET process, it drops the message.

Unintelligible Messages from a DPC

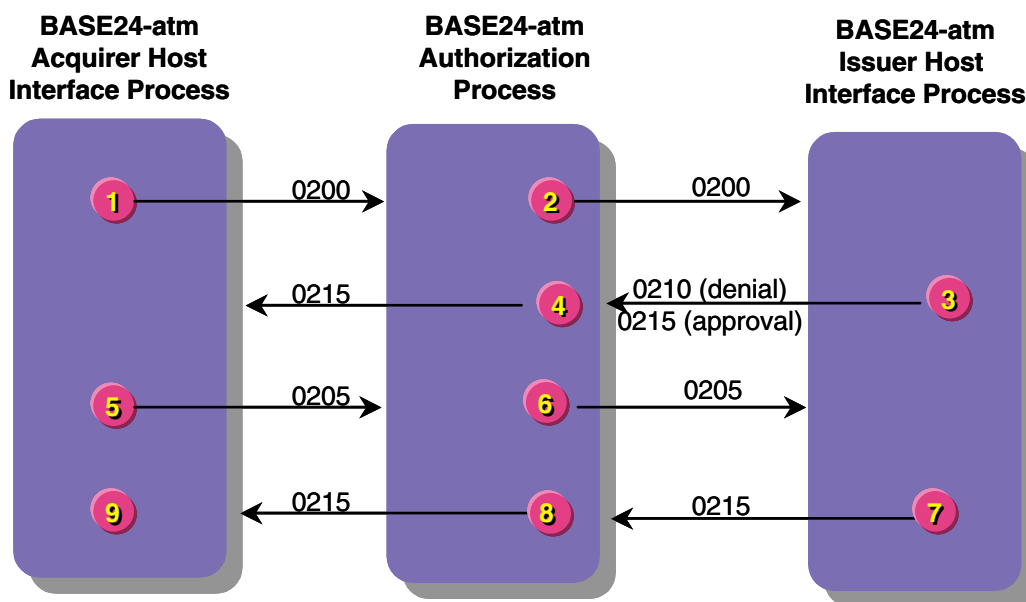
If the ISO Host Interface process receives a message that it cannot process, it rejects it, returning the message to the DPC that sent it.

To reject a message, the ISO Host Interface process performs the following:

- Changes the first position of the message type to a value of 9. For example, a 0200 message would be changed to a 9200 message, and a 0420 message would be changed to a 9420 message.
- Places the number of the data element that could not be processed in the status field of the external message header. For example, if the Track 2 data in the P-35 data element could not be parsed, the status field in the header would be set to the value 035.

Statement Print Transaction Message Flow

The following diagram illustrates the message exchange expected when an acquirer host sends a statement print transaction to the issuer host through the BASE24-atm system. The steps corresponding to the diagram are provided below.



1. The BASE24 system receives a request for statement information, which is sent to the BASE24-atm Authorization as a 0200 message from the ISO Acquirer Host Interface process.

The BASE24-atm Authorization process determines that the transaction is a request for statement information, which must be provided by an issuer host, and sends it to the ISO Issuer Host Interface process for forwarding to that host.

The BASE24-atm Authorization process retrieves the name of the ISO Issuer Host Interface process to which to send the message from the ATM ROUTING TABLE fields on IDF screen 9.

2. The ISO Issuer Host Interface process receives the 0200 message and treats it the same as a financial transaction. The Issuer Host Interface process converts the message to an external message format and sends it to the issuer host.

ASCII-to-EBCDIC conversions, message authentication, and full message encryption are supported throughout this transaction flow in the same manner as for financial transactions.

3. The issuer host processes the 0200 message and returns a 0215 message approving the transaction or a 0210 message declining the transaction. The 0215 message carries the statement information.

The ISO Issuer Host Interface process receives the 0215 message, converts the message to STM format using the STMT-INFO definition fields of the STM, and sends the STM to the BASE24-atm Authorization process.

4. The BASE24-atm Authorization process finishes processing the 0215 message, logs it to the TLF, and sends it to the ISO Acquirer Host Interface process. The Acquirer Host Interface sends the 0215 reply to the acquirer host.
5. The originator can request additional pages of statement information, in which case it sends the BASE24-atm Authorization process a 0205 message. When the BASE24-atm Authorization process receives the 0205 message, it sends the 0205 message to the Issuer Host Interface process.
6. The Issuer Host Interface process receives the 0205 message, converts it to an external message, and sends it to the issuer host.

For purposes of ASCII-to-EBCDIC conversions, message authentication, full message encryption, and resetting timers and internal counters, the ISO Host Interface process treats the 0205 message the same as a 0200 message.
7. The issuer host processes the 0205 message and returns a 0215 message.

The Issuer Host Interface process receives the 0215 message, converts it to STM format, sets the TYP field in the STM to the value 0215, and sends it to the Authorization process.
8. The BASE24-atm Authorization process then passes the 0215 message to the Acquirer Host Interface process (only the first 0215 message is logged to the TLF).
9. The Acquirer Host Interface process receives the 0215 message, converts it to an external message, and sends it to the acquirer host.

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6: BASE24-pos Financial Message Handling

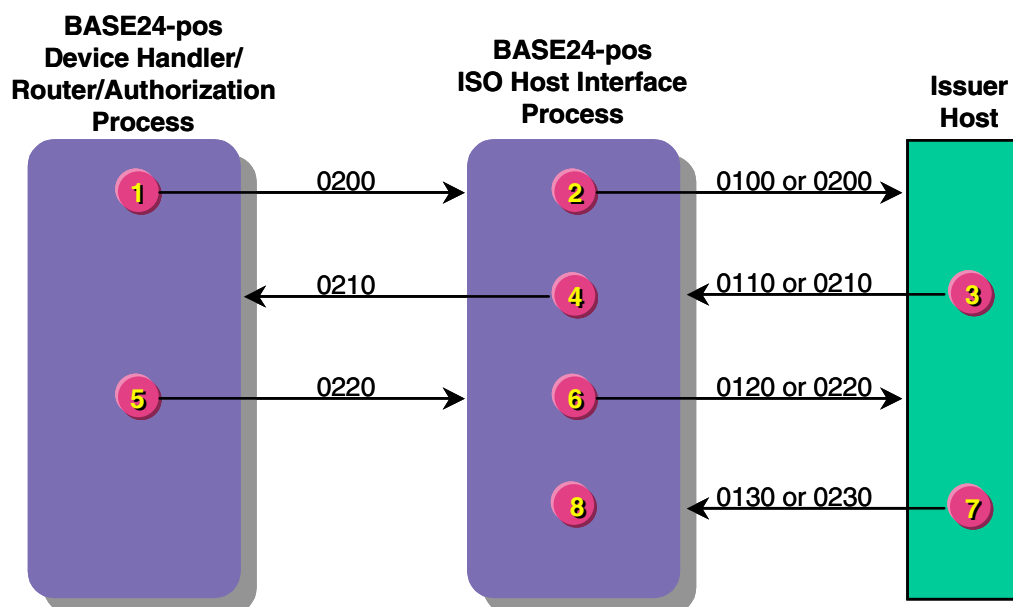
This section contains an overview of how BASE24-pos financial transaction messages are handled by the ISO Host Interface process. It includes message flows for both issuer and acquirer host processing.

Issuer Host Message Handling

This section describes the processing involved for messages sent to and from an issuer host.

Financial Transaction Message Flow

The following diagram illustrates the message exchange expected when the ISO Host Interface process sends financial transactions to an issuer host. The steps corresponding to the diagram are provided below.



1. The BASE24 system receives a 0200 financial transaction request message, which is sent to the BASE24-pos Device Handler/Router/Authorization process.

The Authorization module determines that the request must be sent to an issuer host for authorization and sends the transaction to the ISO Host Interface process to be forwarded to that host.

2. The ISO Host Interface process receives the 0200 message and performs the following steps:
 - a. Locates the appropriate DPC processing information in its internal table using the DPC number from the DPC-NUM field in the POS Standard Internal Message (PSTM).
 - b. Creates an external message from the PSTM.

- c. Determines its ability to send the message. The ISO Host Interface process cannot send the message if any one of the following conditions exist:
- The DPC is marked down.
 - A noninterspersed store-and-forward cycle is in progress.
 - The host is single-threaded and no send/receive stations are available.
 - The current number of outstanding messages to the DPC equals the value in the MAXIMUM OUTSTANDING REQUESTS OUTBOUND field on Host Configuration File (HCF) screen 1.

If any of the above conditions exist, the ISO Host Interface process performs the following steps:

- 1) Determines whether an alternate routing destination has been configured. If so, sets the ORIG-PRO-NAME field of the PSTM to that destination process name, and continues processing with step 4. If there is no alternate destination, ensures that the current authorizer is set to AUTHH.
- 2) Sets the RTE-STAT field in the PSTM to a value of 11, indicating that the DPC was unavailable, or to a value of 12 if the DPC was marked down.
- 3) Adjusts the RQST.COMPL-REQ field in the PSTM, if necessary.
 - If the RQST.COMPL-REQ field in the PSTM contains a value of 0 (completions not required) or 1 (approved completions required), the ISO Host Interface process sets the RQST.COMPL-REQ field in the PSTM to a value of 2 to indicate that an advice is required if the request is approved offline.
 - If the RQST.COMPL-REQ field in the PSTM contains a value of 4 (completions required), 5 (completions required for approved BASE24-authorized transactions), or 6 (completions required for all BASE24-authorized transactions - approved and denied), the ISO Host Interface process does not change the value.
- 4) Returns the 0200 message (PSTM) to the process indicated in the ORIG-PRO-NAME field of the PSTM.

If none of the above conditions exist, the ISO Host Interface process continues with step 2d.

- d. Saves a copy of the PSTM in extended memory. As part of saving the message, the ISO Host Interface process stores the current time in a user buffer in memory.
- e. Checks the TRAN-CDE.TC field of the PSTM for an 11. If the value is 11, the ISO Host Interface process changes the message type to 0100, indicating that this transaction does not affect the cardholder's application account. Otherwise, the message type remains 0200.
- f. Sets an outbound request timer for the number of seconds specified by the OUTBOUND field on HCF screen 7.

If this timer expires before a 0110 or 0210 message is received, the ISO Host Interface process checks the value of the EXTRN-RQST-TIMEOUT param to determine which actions are performed.

- If the param is set to a value of 0, the ISO Host Interface process returns the original 0200 message to the BASE24-pos Device Handler/Router/Authorization process as failed for alternate or stand-in routing. If the ISO Host Interface process receives a late 0210 message, it generates an 0420 message and sends it to the host.
 - If the param is set to a value of 1, the ISO Host Interface process returns the original 0200 message to the BASE24-pos Device Handler/Router/Authorization process as failed for alternate or stand-in routing and generates an 0420 message that it sends to the host. If the ISO Host Interface process receives a late 0210 message, it drops the message.
- g. Updates two of its internal counters as follows:
 - 1) Adds one to the number of messages currently outstanding to the DPC.
 - 2) Adds one to the total number of messages currently outstanding from the ISO Host Interface process.
 - h. Generates a MAC for the message, if message authentication is supported and is specified for this message type.
 - i. Converts the message from ASCII to EBCDIC if the CHARACTER FORMAT field on screen 1 of the HCF is set to E for the DPC.
 - j. Encrypts the message if full message encryption is supported and is specified for the DPC.

- k. Sends the 0100 or 0200 message to the DPC.

Along with the message, the ISO Host Interface process also sends the maximum number of seconds the XPNET process can allow the message to queue to the host. This number of seconds is derived using the values in the QUEUE SUBTRACT and OUTBOUND fields on HCF screen 7. The value in the QUEUE SUBTRACT field is subtracted from the value in the OUTBOUND field and the result is used.

- 3. The host validates the request, processes the transaction, and returns a 0110 or 0210 message either approving or declining the transaction.
- 4. The ISO Host Interface process receives the 0110 or 0210 message and performs the following steps:
 - a. Locates the DPC in its internal table, identifying the necessary information for processing.
 - b. Decrypts the message if full message encryption is supported and is specified for the DPC.
 - c. Converts the message from EBCDIC to ASCII if the CHARACTER FORMAT field on screen 1 of the HCF is set to E for the DPC.
 - d. Verifies the MAC received with the message, if message authentication is supported and is specified for the message type.
 - e. Retrieves the stored PSTM from extended memory.
 - f. Deletes the outbound request timer.
 - g. Updates its counters as follows:
 - 1) Subtracts one from the number of messages currently outstanding to the DPC.
 - 2) Subtracts one from the number of messages currently outstanding from the ISO Host Interface process.
 - 3) Adds one to the number of requests processed for the DPC for performance statistics.
 - 4) Adds the amount of time taken to process this request to the total time taken for all requests to the DPC. For performance statistics, this value is divided by the number of requests processed for the DPC.
 - h. Updates the value in the EXIT-TIM field in the PSTM using the time stored in the user buffer when the request message was saved in extended memory in step 2d.

- i. Sets the value in the RE-ENTRY-TIM field in the PSTM to the current time.
 - j. Sets the value in the TYP field in the PSTM to 0210.
 - k. Updates the PSTM from the external message using only that information which may have been changed by the host.
 - l. Sends the 0210 message (PSTM) to the Device Handler/Router/Authorization process.
5. The BASE24-pos Device Handler/Router/Authorization process finishes processing the transaction, logs it to the POS Transaction Log File (PTLF), and returns it to its originator.

If the host requires advices, the BASE24-pos Device Handler/ Router/ Authorization process generates the appropriate 0220 message and sends it to the ISO Host Interface process.

6. The ISO Host Interface process receives the 0220 message and performs the following steps:
- a. Locates the appropriate DPC processing information in its internal table using the DPC number from the DPC-NUM field in the PSTM.
 - b. Creates an external message from the PSTM.
 - c. Checks the TRAN-CDE.TC field in the PSTM. If the value is 11, the ISO Host Interface process changes the message type to 0120. Otherwise, the message type remains 0220.
 - d. Checks the COMPL-REQ field in the PSTM. If the value is 2 (host is offline—completion required), the ISO Host Interface process writes the external version of the 0120 or 0220 message to the Store-and-Forward File (SAF) and takes no further action.
 - e. Determines whether the DPC is marked down. If so, the ISO Host Interface process writes the external version of the 0120 or 0220 message to the SAF and takes no further action.
 - f. Determines whether a noninterspersed store-and-forward cycle is in progress. If so, the ISO Host Interface process writes the external version of the 0120 or 0220 message to the SAF and takes no further action.
 - g. Locates an available send/receive station. If no send/receive stations are available, the ISO Host Interface process writes the external version of the 0120 or 0220 message to the SAF and takes no further action.
 - h. Generates a MAC for the message, if message authentication is supported and is specified for this message type.

- i. Converts the message from ASCII to EBCDIC if the CHARACTER FORMAT field on screen 1 of the HCF is set to E for the DPC.
- j. Encrypts the message if full message encryption is supported and is specified for the DPC.
- k. Sends the 0120 or 0220 message to the DPC.

Along with the message, the ISO Host Interface process also sends the maximum number of seconds the XPNET process can allow the message to queue to the host. This number of seconds is derived using the values in the QUEUE SUBTRACT and COMPLETION ACK fields on HCF screen 7. The value in the QUEUE SUBTRACT field is subtracted from the value in the COMPLETION ACK field and the result is used.

- l. Stores a copy of the external message.
 - m. Sets a completion acknowledgment timer.
 - n. Adds one to the number of messages currently outstanding to the DPC.
7. The host processes the 0120 or 0220 message and returns a 0130 or 0230 message if text-level acknowledgments are required.

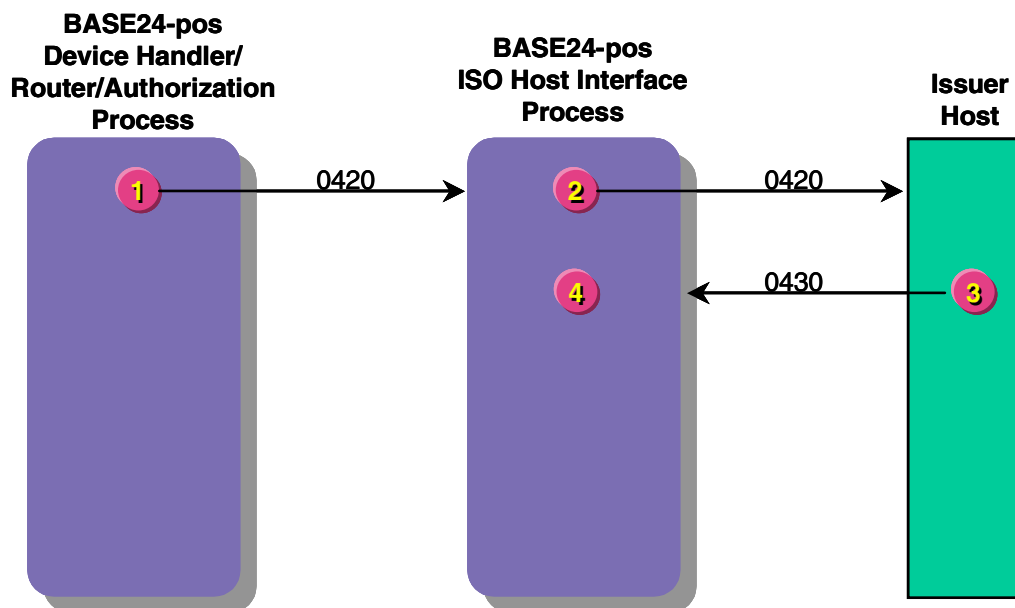
DPCs must return text-level acknowledgments if the ACK FROM DPC field in the HCF contains the value Y.

8. Upon receiving a 0130 or 0230 message, the ISO Host Interface process performs the following steps:
- a. Locates the DPC in its internal table, identifying the necessary information for processing.
 - b. Decrypts the message if full message encryption is supported and is specified for the DPC.
 - c. Converts the message from EBCDIC to ASCII if the CHARACTER FORMAT field on screen 1 of the HCF is set to E for the DPC.
 - d. Verifies the MAC received with the message, if message authentication is supported and is specified for the message type.
 - e. Deletes the stored external message from extended memory.
 - f. Deletes the completion acknowledgment timer.
 - g. Subtracts one from the number of messages currently outstanding to the DPC.

Processing for the 0130 or 0230 message is now complete.

Reversal Message Flow

The following diagram illustrates the message exchange expected when the ISO Host Interface process sends reversal transactions to an issuer host. A reversal indicates that a previously-approved transaction did not complete as authorized. The steps corresponding to the diagram are provided below.



1. The BASE24 system is notified that a transaction did not complete as previously authorized. The notice can be from a POS device controlled by the BASE24 system, an acquirer host, or an interchange.

The BASE24-pos Device Handler/Router/Authorization process backs out the original transaction, posts the revised transaction, logs the transaction to the PTLF, and sends a 0420 message in PSTM format to the ISO Host Interface process.

2. The ISO Host Interface process receives the 0420 message and processes it by performing the following steps:
 - a. Locates the DPC record in its internal table using the DPC-NUM field in the PSTM.
 - b. Checks the TRAN.RESP-CDE field in the PSTM. If the value is greater than or equal to 50, meaning the transaction was declined, the ISO Host Interface process drops the message.
 - c. Creates an external message from the PSTM.

- d. Determines whether the DPC is marked down. If so, the ISO Host Interface process writes the external version of the 0420 message to the SAF and performs no further steps.
- e. Determines whether a store-and-forward cycle (either interspersed or noninterspersed) is in progress for the DPC. If so, the ISO Host Interface process writes the external version of the 0420 message to the SAF and performs no further steps.
- f. Checks the COMPL-REQ field in the PSTM. If it contains the value 2 (host offline—completion required), the ISO Host Interface process writes the external version of the 0420 message to the SAF and performs no further steps.
- g. Locates an available send/receive station. If no send/receive stations are available, the ISO Host Interface process writes the external version of the 0420 message to the SAF and performs no further steps.
- h. Generates a MAC for the message, if message authentication is supported and is specified for this message type.
- i. Converts the message from ASCII to EBCDIC if the CHARACTER FORMAT field on screen 1 of the HCF is set to E for the DPC.
- j. Encrypts the message if full message encryption is supported and is specified for the DPC.
- k. Sends the 0420 message to the DPC.

Along with the message, the ISO Host Interface process also sends the maximum number of seconds the XPNET process can allow the message to queue to the host. This number of seconds is derived using the values in the QUEUE SUBTRACT and COMPLETION ACK fields on HCF screen 7. The value in the QUEUE SUBTRACT field is subtracted from the value in the COMPLETION ACK field and the result is used.

- l. Stores a copy of the external message.
 - m. Sets a completion acknowledgment timer.
 - n. Adds one to the number of messages currently outstanding to the DPC.
3. The host processes the 0420 message and returns a 0430 message if text-level acknowledgments are required.

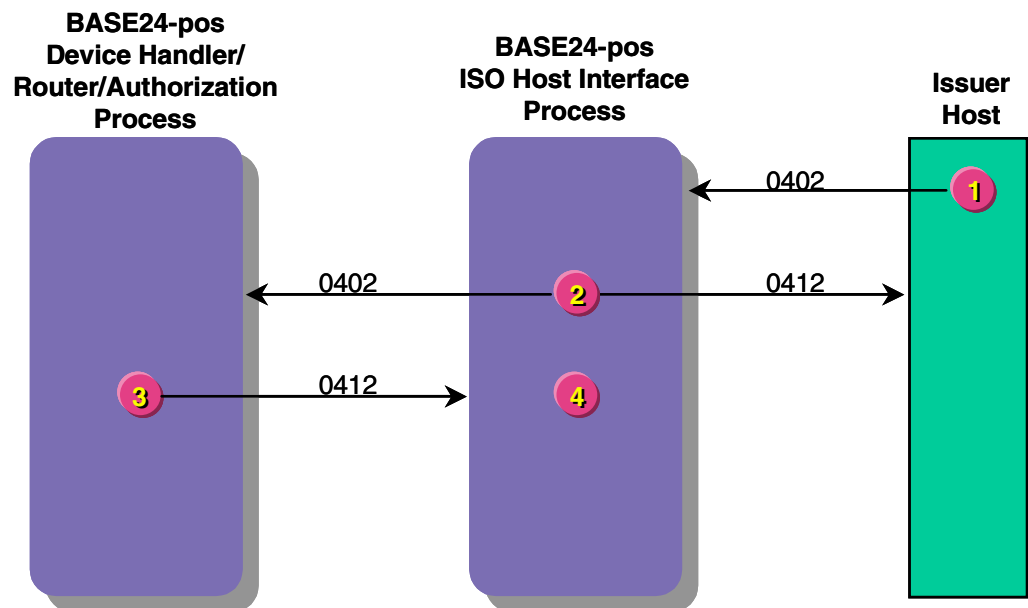
DPCs must return text-level acknowledgments if the ACK FROM DPC field from the HCF contains the value Y.

4. Upon receipt of a 0430 message, the ISO Host Interface process performs the following steps:
 - a. Locates the DPC in its internal table, identifying the necessary information for processing.
 - b. Decrypts the message if full message encryption is supported and is specified for the DPC.
 - c. Converts the message from EBCDIC to ASCII if the CHARACTER FORMAT field on screen 1 of the HCF is set to E for the DPC.
 - d. Verifies the MAC received with the message, if message authentication is supported and is specified for the message type.
 - e. Deletes the stored external message from extended memory.
 - f. Deletes the completion acknowledgment timer.
 - g. Subtracts one from the number of messages currently outstanding to the DPC.

Processing for the 0430 message is now complete.

Chargeback Transaction Message Flow

The following diagram illustrates the message exchange expected when BASE24-pos receives a chargeback transaction from the issuer of a card for which a transaction is being disputed. The steps corresponding to the diagram are provided below



1. The issuer host sends a chargeback transaction to the ISO Host Interface process as a 0402 message.
2. The ISO Host Interface process receives the 0402 message and performs the following steps:
 - a. Locates the appropriate DPC processing information in its internal table.
 - b. Decrypts the message if full message encryption is supported and is specified for the DPC.
 - c. Converts the message from EBCDIC to ASCII if the CHARACTER FORMAT field on screen 1 of the HCF is set to E for the DPC.
 - d. Verifies the MAC received with the message, if message authentication is supported and is specified for the message type.

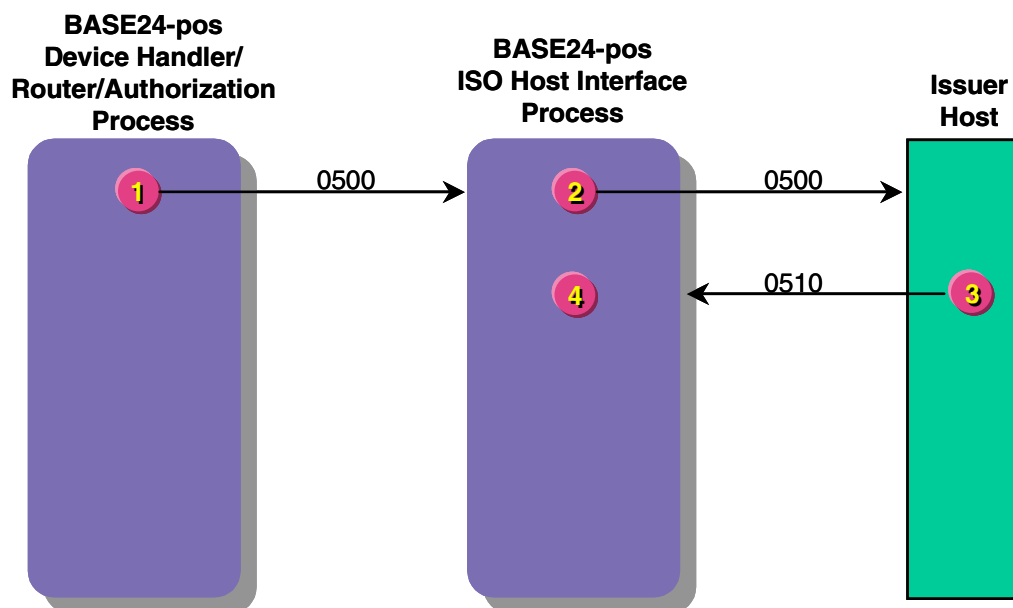
- e. Creates a PSTM from the external message.
If the acquirer of the original transaction was an interchange, the ISO Host Interface process puts the original transaction settlement date from the external message in the ACQUIRER-INTERCHANGE-SETTLEMENT-DATE field in the PSTM.
 - f. Formats a 0412 response.
The ISO Host Interface process performs ASCII-to-EBCDIC conversions, MAC creation, and full message encryption as required.
 - g. Locates an available send/receive station and sends the 0412 message to the issuer host. This message indicates to the host that the ISO Host Interface process received the 0402 message. If no send/receive stations are available, the ISO Host Interface process drops the external version of the 0412 message.
 - h. Sends the 0402 message (PSTM) to the BASE24-pos Device Handler/Router/Authorization process indicated in the AUTH PROCESS field on HCF screen 7.
- 3. The Device Handler/Router/Authorization process processes the chargeback and logs the message to the PTLF. The BASE24-pos Device Handler/Router/Authorization process also formats a 0412 message and sends it to the ISO Host Interface process, indicating it received the 0402 message.
 - 4. When the ISO Host Interface process receives the 0412 message it drops the message.

Representment Transaction Message Flow

BASE24-pos processes a representment transaction the same as a financial transaction; however, the value in the REPRESENTMENT UPDATE field on Card Prefix File (CPF) screen 7 controls whether a representment transaction updates the BASE24 database. Refer to the “Financial Transaction Message Flow” topic presented earlier in this section. BASE24-pos identifies a representment transaction with a value of 2 in the TRAN-CDE.C (transaction code category) field of the PSTM.

Reconciliation Message Flow

The following diagram illustrates the message exchange expected when the ISO Host Interface process sends reconciliation transactions to an issuer host. The steps corresponding to the diagram are provided below.



1. The BASE24 system receives a 0500 acquirer reconciliation request message, which is sent to the BASE24-pos Device Handler/Router/Authorization process.

The Authorization module determines that the request must be sent to an issuer host for processing and sends the transaction to the ISO Host Interface process to be forwarded to that host.

2. The ISO Host Interface process receives the 0500 message and performs the following steps:
 - a. Locates the appropriate DPC processing information in its internal table using the DPC number from the SETL.SERVICES.SET-REC1.DPC-NUM field in the internal message.
 - b. Creates an external message from the internal message. The external message can contain information for up to ten services. If the internal message contains information for more than ten services, multiple 0500 external messages are created and sent to the host.

- c. Determines its ability to send the message. The ISO Host Interface process cannot send the message if any one of the following conditions exist:
- The DPC is marked down.
 - A noninterspersed store-and-forward cycle is in progress.
 - The host is single-threaded and no send/receive stations are available.
 - The current number of outstanding messages to the DPC equals the value in the MAXIMUM OUTSTANDING REQUESTS OUTBOUND field on HCF screen 1.
 - A security device error occurred while attempting to generate a MAC for the message.
 - The transaction could not be suspended in memory.

If any of the above conditions exist, the ISO Host Interface process performs the following steps:

- 1) Changes the message type identifier in the external message header to 0520.
- 2) Adds the message to the SAF and takes no further action.

If none of the above conditions exist, the ISO Host Interface process continues with step 2d.

- d. Saves a copy of the internal message in extended memory. As part of saving the message, the ISO Host Interface process stores the current time in a user buffer in memory.
- e. Sets a completion acknowledgment timer.

Normally, completion acknowledgment timers are set for the amount of time specified in the COMPLETION ACK field on HCF screen 7. However, for 0500 messages, the timer is set for the amount of time specified in the OUTBOUND field on HCF screen 7. If this timer expires before a 0510 message is received, the ISO Host Interface process stores the message as a 0520 message in the SAF.

- f. Updates two of its internal counters as follows:
- 1) Adds one to the number of messages currently outstanding to the DPC.
 - 2) Adds one to the total number of messages currently outstanding from the ISO Host Interface process.

- g. Generates a MAC for the message, if message authentication is supported and is specified for this message type.
 - h. Converts the message from ASCII to EBCDIC if the CHARACTER FORMAT field on screen 1 of the HCF is set to E for the DPC.
 - i. Encrypts the message if full message encryption is supported and is specified for the DPC.
 - j. Sends the 0500 message to the DPC.
- 3. The host validates the request, processes the transaction, and returns a 0510 message either approving or declining the transaction.
- 4. Upon receipt of a 0510 message, the ISO Host Interface process performs the following steps:
 - a. Locates the DPC in its internal table, identifying the necessary information for processing.
 - b. Decrypts the message if full message encryption is supported and is specified for the DPC.
 - c. Converts the message from EBCDIC to ASCII if the CHARACTER FORMAT field on screen 1 of the HCF is set to E for the DPC.
 - d. Verifies the MAC received with the message, if message authentication is supported and is specified for the message type.
 - e. Deletes the stored external message from extended memory.
 - f. Deletes the completion acknowledgment timer.
 - g. Subtracts one from the number of messages currently outstanding to the DPC.

Processing for the 0510 message is now complete.

Late Response from a DPC

If a DPC responds late to a request, the ISO Host Interface process performs the following steps for the 0110 or 0210 message:

- 1. Checks the Response Code data element (P-39) in the external message. If the transaction was declined, the ISO Host Interface process drops the message.
- 2. Checks the Processing Code data element (P-3) in the external message. If the transaction was a balance inquiry transaction, the ISO Host Interface process drops the message.

3. Checks the value of the EXTRN-RQST-TIMEOUT param in the LCONF. If this param is set to a value of 0, processing continues with step 4. If this param is set to 1, the ISO Host Interface process drops the message.
4. Creates a 0420 message. In doing so, the ISO Host Interface process performs the following steps:
 - a. Sets the Response Code data element (P-39) in the external message to a value of 68 (timeout).
 - b. Sets the Original Data Elements data element (S-90) in the external message from the 0110 or 0210 message.
5. Selects the station to which to send the message.

The ISO Host Interface process checks the station that sent the 0110 or 0210 message. If it is a send/receive station and is available, the ISO Host Interface process sends the 0420 message to that station. Otherwise, it uses any available send/receive station. If a station is not available, the ISO Host Interface process writes the 0420 message to the SAF and performs no further steps.
6. Generates a MAC for the message, if message authentication is supported and is specified for this message type.
7. Converts the message from ASCII to EBCDIC if the CHARACTER FORMAT field on screen 1 of the HCF is set to E for the DPC.
8. Encrypts the message if full message encryption is supported and is specified for the DPC.
9. Sends the 0420 message to the selected station.

Along with the message, the ISO Host Interface process also sends the maximum number of seconds the XPNET process can allow the message to queue to the host. This number of seconds is derived using the values in the QUEUE SUBTRACT and COMPLETION ACK fields on HCF screen 7. The value in the QUEUE SUBTRACT field is subtracted from the value in the COMPLETION ACK field and the result is used.
10. Stores a copy of the external message.
11. Sets a completion acknowledgment timer.
12. Adds one to the number of messages currently outstanding to the DPC.

Failed Messages to Stations

If the ISO Host Interface process forwards a message to the XPNET process, but the message cannot be sent to a station, the XPNET process returns the message to the ISO Host Interface process as failed.

Upon receiving a message failure, the ISO Host Interface process marks the station down and sets an extended network management message timer. It then performs the following steps based on message type.

Failed 0100 or 0200 Messages

The ISO Host Interface process takes the following steps when it receives a failed 0100 or 0200 message from the XPNET process. Failed messages occur when a message cannot reach a station or the transaction becomes stale while queued by the XPNET process.

1. Locates the DPC in its internal table, identifying the necessary information for processing.
2. Retrieves the stored PSTM from extended memory.
3. Deletes the outbound request timer.
4. Updates its counters. The ISO Host Interface process performs the following steps:
 - a. Subtracts one from the number of messages currently outstanding to the DPC.
 - b. Adds one to the number of requests processed for the DPC.
 - c. Adds the amount of time taken to process this request to the total time taken for all requests to the DPC. For performance statistics, this value is divided by the number of requests processed for the DPC.
5. Determines whether an alternate routing destination has been configured. If so, sets the ORIG-PRO-NAME field of the PSTM to that destination process name, and continues processing with step 9. If there is no alternate destination, ensures that the current authorizer is set to AUTHH.
6. Sets the RTE-STAT field in the PSTM to a nonzero value, indicating a failed message.
7. Sets the COMPL-REQ field in the PSTM to a value of 2, indicating that a 0120 or 0220 message is required if the transaction is subsequently approved.

8. Returns the 0200 message (PSTM) to the process indicated in the ORIG-PRO-NAME field of the PSTM.

Failed 0220, 0402 or 0420 Messages

The ISO Host Interface process takes the following steps when it receives a failed 0220, 0402, or 0420 message from the XPNET process. Failed messages occur when a station is unavailable or the transaction becomes stale while queued by the XPNET process.

1. Locates the DPC in its internal table, identifying the necessary information for processing.
2. Retrieves the stored external message from extended memory.
3. Checks for a completion acknowledgment timer for the message. If the ISO Host Interface process finds this timer, it is deleted.
4. Subtracts one from the number of messages currently outstanding to the DPC.
5. Adds the message to the SAF.

Unintelligible Messages from a DPC

If the ISO Host Interface process receives a message that it cannot process, the ISO Host Interface process rejects the message, returning the message to the DPC that sent it.

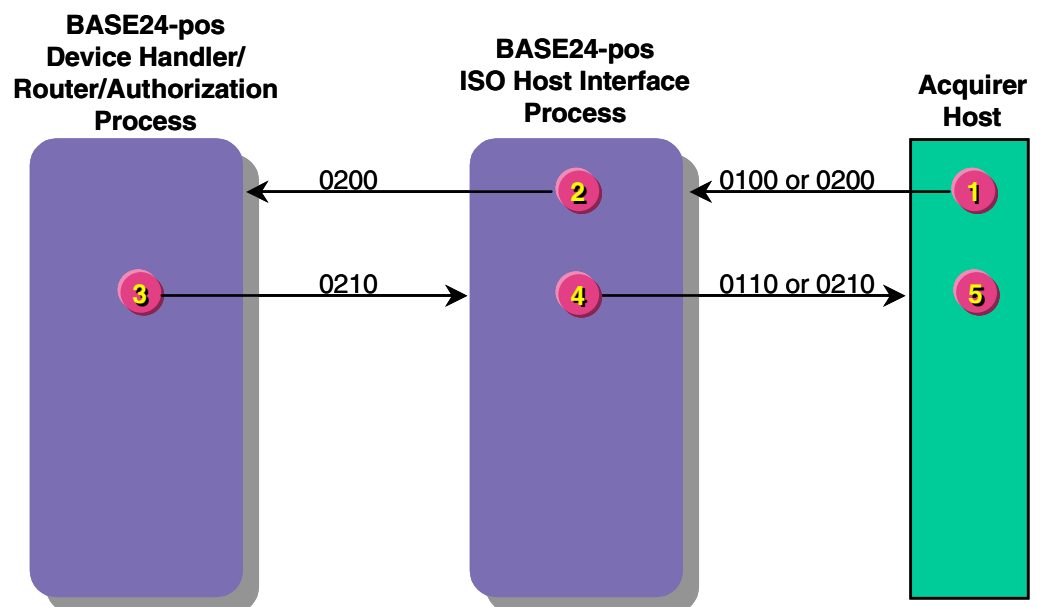
To reject a message, the ISO Host Interface process changes the first position of the message type to a value of 9 (for example, a 0200 message would be changed to a 9200 message, and a 0420 message would be changed to a 9420 message) and places the number of the data element that could not be processed in the status field of the external message header (for example, if the Track 2 data in the P-35 data element could not be parsed, the status field in the header would be set to the value 035).

Acquirer Host Message Handling

This section describes the processing involved for messages sent to and from an acquirer host.

Financial Transaction Message Flow

The following diagram illustrates the message exchanges expected when an acquirer host sends financial transactions to BASE24-pos. The steps corresponding to the diagram are provided below.



1. A cardholder initiates a financial transaction at a terminal connected to and controlled by an acquirer host. The acquirer host then formats a 0100 or a 0200 message and sends it to the BASE24 system.
2. The ISO Host Interface process receives the 0100 or the 0200 message and performs the following steps:
 - a. Locates the DPC in its internal table, identifying the necessary information for processing.
 - b. Decrypts the message if full message encryption is supported and is specified for the DPC.
 - c. Converts the message from EBCDIC to ASCII if the CHARACTER FORMAT field on screen 1 of the HCF is set to E for the DPC.

- d. Verifies the MAC received with the message, if message authentication is supported and is specified for the message type.
- e. Creates a PSTM from the external message. If the original message was a 0100 message, the ISO Host Interface process changes the message type to a 0200 since only 0200 messages can be transmitted in the PSTM. The ISO Host Interface process sets the TRAN-CDE.TC field to the value 11 in order to distinguish the message as originally being a 0100 message.
- f. Updates the following fields:
 - Sets the RESPONDER field to a value of 5, indicating the transaction was initiated through an acquirer host.
 - Sets the DPC-NUM field to the number of the DPC originating the transaction.
 - Sets the ENTRY-TIM field to the current time.
- g. Checks the current number of outstanding 0200 messages to the Device Handler/Router/Authorization process.

If the current number of outstanding 0200 messages equals the value in the MAXIMUM OUTSTANDING REQUESTS INBOUND on HCF screen 1, the ISO Host Interface process creates a 0210 or 0110 message with a response code of 05 (issuer inoperative) and sends it to the originating DPC. No further steps are performed.

If the current number of outstanding 0200 messages is less than the value in the MAXIMUM OUTSTANDING REQUESTS INBOUND on HCF screen 1, the ISO Host Interface process proceeds with step 2e.
- h. Saves a copy of the external message in extended memory.
- i. Sets an inbound request timer for the number of seconds specified in the INBOUND field on HCF screen 7.

If this timer expires before a 0210 message is received, the ISO Host Interface process declines the transaction.
- j. Adds one to the number of requests currently outstanding from the DPC.
- k. Sends the 0200 message (PSTM) to the Device Handler/Router/Authorization process indicated in the AUTH PROCESS field on HCF screen 7.

3. The BASE24-pos Device Handler/Router/Authorization process processes the transaction, logs it to the PTLF, and returns a 0210 message to the ISO Host Interface process.

Note: The transaction could also have been authorized by an interchange or issuer host, depending on the cardholder and the type of authorization processing performed.

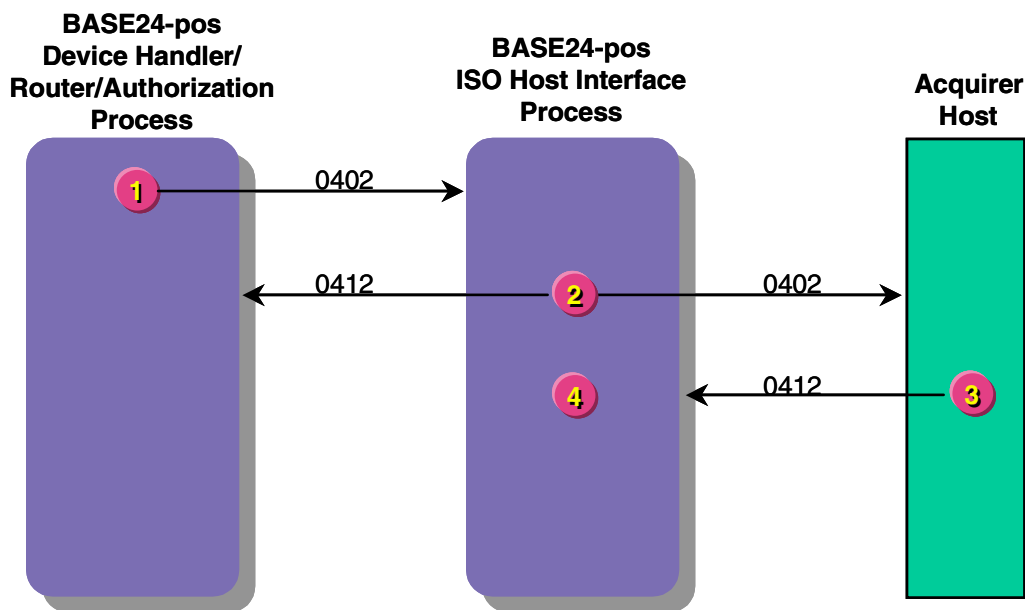
4. The ISO Host Interface process receives the 0210 message and performs the following steps:
 - a. Locates the stored external message.
 - b. Deletes the inbound request timer.
 - c. Subtracts one from the number of requests currently outstanding from the DPC.
 - d. Checks the TRAN-CDE.TC field for the value 11. If it contains the value 11, the ISO Host Interface process changes the 0210 message to a 0110 message.
 - e. Creates a 0110 or a 0210 external message from the PSTM.
 - f. Generates a MAC for the message, if message authentication is supported and is specified for this message type.
 - g. Converts the message from ASCII to EBCDIC if the CHARACTER FORMAT field on screen 1 of the HCF is set to E for the DPC.
 - h. Encrypts the message if full message encryption is supported and is specified for the DPC.
 - i. Sends the 0110 or the 0210 external message to the station that originally sent the 0100 or 0200 message.

If the message cannot be sent to that station, the ISO Host Interface process selects another send/receive station. If no send/receive stations are available, the ISO Host Interface process creates a reversal (0420) message using the PSTM and returns it to the Device Handler/Router/Authorization process. A reversal is created only if the transaction is approved and is not a balance inquiry transaction.

5. The acquirer host then completes the transaction at the POS device.

Chargeback Transaction Message Flow

The following diagram illustrates the message exchange expected when the BASE24 system sends a chargeback to an acquirer host. A chargeback is a transaction resulting from a disputed transaction. The steps corresponding to the diagram are provided below.



1. The BASE24-pos Device Handler/Router/Authorization process backs out the original transaction, posts the revised transaction, logs the transaction to the PTLF, and sends a 0402 message in PSTM format to the ISO Host Interface process.
2. The ISO Host Interface process receives the 0402 message and performs the following steps:
 - a. Locates the appropriate DPC processing information in its internal table using the DPC number from the DPC-NUM field in the PSTM.
 - b. Creates a 0402 external message from the PSTM.
 - c. Formats a 0412 response message and sends it to the Device Handler/Router/Authorization process.
 - d. Determines whether the DPC is marked down. If so, the ISO Host Interface process writes the external version of the 0402 message to the SAF and no further steps are taken.
 - e. Determines whether a store-and-forward cycle is in progress. If so, the ISO Host Interface process writes the external version of the 0402 message to the SAF and no further steps are taken.

- f. Locates an available send/receive station. If no send/receive stations are available, the ISO Host Interface process writes the external version of the 0402 message to the SAF and no further steps are taken.
- g. Generates a MAC for the message, if message authentication is supported and is specified for this message type.
- h. Converts the message from ASCII to EBCDIC if the CHARACTER FORMAT field on screen 1 of the HCF is set to E for the DPC.
- i. Encrypts the message if full message encryption is supported and is specified for the DPC.
- j. Sends the 0402 message to the DPC.

Along with the message, the ISO Host Interface process also sends the maximum number of seconds the XPNET process can allow the message to queue to the host. This number of seconds is derived using the values in the QUEUE SUBTRACT and COMPLETION ACK fields on HCF screen 7. The value in the QUEUE SUBTRACT field is subtracted from the value in the COMPLETION ACK field and the result is used.

- k. Stores a copy of the external message.
 - l. Sets a completion acknowledgment timer.
 - m. Adds one to the number of messages currently outstanding to the DPC.
3. The host processes the 0402 message and returns a 0412 message.
 4. Upon receipt of a 0412 message, the ISO Host Interface process performs the following steps:
 - a. Locates the DPC in its internal table, identifying the necessary information for processing.
 - b. Decrypts the message if full message encryption is supported and is specified for the DPC.
 - c. Converts the message from EBCDIC to ASCII if the CHARACTER FORMAT field on screen 1 of the HCF is set to E for the DPC.
 - d. Verifies the MAC received with the message, if message authentication is supported and is specified for the message type.
 - e. Deletes the stored external message from extended memory.
 - f. Deletes the completion acknowledgment timer.
 - g. Subtracts one from the number of messages currently outstanding to the DPC.

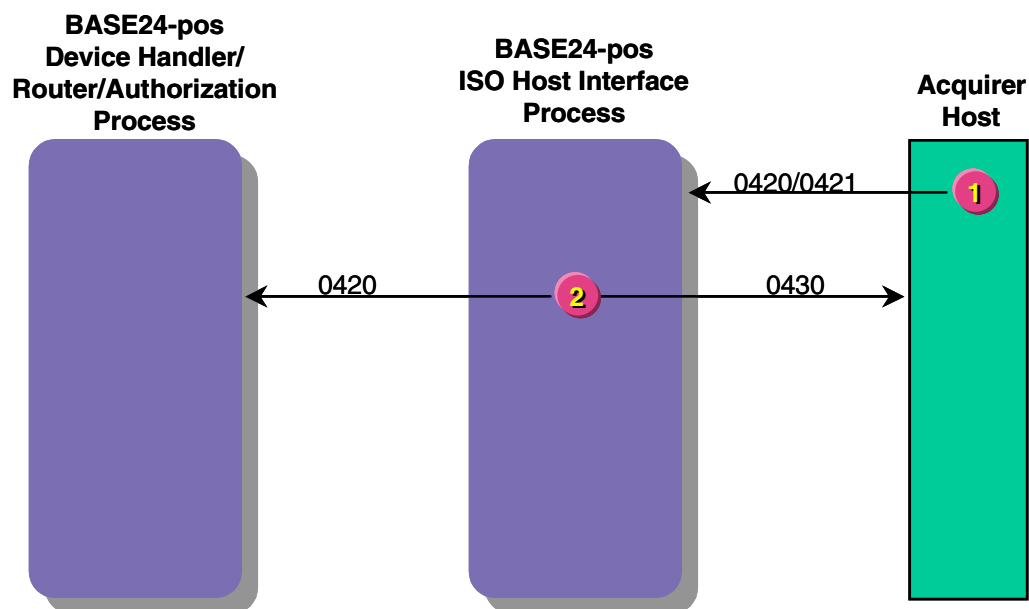
Processing for the 0412 message is now complete.

Representment Transaction Message Flow

BASE24-pos processes a representment transaction the same as a financial transaction; however, the value in the REPRESENTMENT UPDATE field on Card Prefix File (CPF) screen 7 controls whether a representment transaction updates the BASE24 database. Refer to the “Financial Transaction Message Flow” diagram presented earlier in this section. BASE24-pos identifies a representment transaction with a value of 2 in the TRAN-CDE.C (transaction code category) field of the PSTM.

Reversal Transaction Message Flow

The following diagram illustrates the message exchange expected when an acquirer host sends reversals to BASE24-pos. A reversal occurs when a transaction authorized by BASE24 does not complete as authorized. The steps corresponding to the diagram are provided below.



1. The terminal that originated the transaction notifies the acquirer host that the transaction did not complete as authorized. The acquirer host then formats a 0420 or 0421 message and sends it to the BASE24 system. A 0421 message indicates a possible duplicate.

2. The ISO Host Interface process receives the 0420 or 0421 message and performs the following steps:
 - a. Locates the DPC in its internal table, identifying the necessary information for processing.
 - b. Decrypts the message if full message encryption is supported and is specified for the DPC.
 - c. Converts the message from EBCDIC to ASCII if the CHARACTER FORMAT field on screen 1 of the HCF is set to E for the DPC.
 - d. Verifies the MAC received with the message, if message authentication is supported and is specified for the message type.
 - e. Creates a PSTM from the external message.
 - f. Sends the 0420 message in PSTM format to the Device Handler/Router/Authorization process indicated in the AUTH PROCESS field on HCF screen 7.
 - g. Generates a 0430 message if the ACK TO DPC field on HCF screen 1 contains the value Y.

Late Response from the Authorization Module

If the ISO Host Interface process receives a late 0210 message from the Device Handler/Router/Authorization process, it performs the following steps:

1. Checks the RESP-CDE field in the PSTM to determine whether the transaction was declined. If it was declined, the ISO Host Interface process drops the message.
2. Checks the TRAN.TRAN-CDE field in the PSTM to determine whether the transaction is a balance inquiry transaction. If it is a balance inquiry transaction, the ISO Host Interface process drops the message.
3. Changes the TYP field in the PSTM to a value of 0420.
4. Sets the RVSL-CDE field in the PSTM to a value of 01, indicating that the Device Handler/Router/Authorization process did not complete processing before the message timed out.
5. Sets the AMT-2 field in the PSTM to a value of zeros, unless the original transaction was a purchase with cash back or an adjustment.
6. Returns the 0420 message to the Device Handler/Router/Authorization process that sent the late response.

Failed Messages to Stations

If a message is forwarded to the XPNET process, but cannot be sent to a station, the XPNET process returns the message to the ISO Host Interface process as failed.

Upon receipt of a message failure, the ISO Host Interface process marks the station down and sets an extended network management message timer. It then performs the following steps based on message type.

Failed 0210 Messages

The ISO Host Interface process takes the following steps when it receives a failed 0210 message from the XPNET process.

1. Locates the DPC in its internal table, identifying the necessary information for processing.
2. Checks the Response Code data element (P-39) in the external message to determine whether the transaction was declined. If it was declined, the ISO Host Interface process drops the message.
3. Checks the Processing Code data element (P-3) in the external message to determine whether the transaction is a balance inquiry transaction. If it is a balance inquiry transaction, the ISO Host Interface process drops the message.
4. Creates a PSTM from the external message and sets the TYP field to a value of 0420.
5. Sets the RVSL-CDE field in the PSTM to the value 03 to indicate that the destination was not available.
6. Sets the AMT-2 field in the PSTM to zeros unless the original transaction was a purchase with cash back or an adjustment.
7. Sends the 0420 message to the Device Handler/Router/Authorization process indicated in the AUTH PROCESS field on HCF screen 7.

Failed 0220, 0402, 0420, or 0520 Messages

The ISO Host Interface process takes the following steps when it receives a failed 0220, 0402, 0420, or 0520 message from the XPNET process. Failed messages occur when a station is unavailable or the transaction becomes stale while queued by the XPNET process.

1. Locates the DPC in its internal table, identifying the necessary information for processing.
2. Retrieves the stored external message from extended memory.
3. Checks for a completion acknowledgment timer for the message. If the ISO Host Interface process finds this timer, it is deleted.
4. Subtracts one from the number of messages currently outstanding to the DPC.
5. Adds the message to the SAF.

Failed 0230/0430 Messages

If the ISO Host Interface process receives a failed 0230 or 0430 message from the XPNET process, it drops the message.

Failed 0500 Messages

The ISO Host Interface process takes the following steps when it receives a failed 0500 message from the XPNET process. Failed messages occur when a station is unavailable or the transaction becomes stale while queued by the XPNET process.

1. Locates the DPC in its internal table, identifying the necessary information for processing.
2. Retrieves the stored external message from extended memory.
3. Checks for a completion acknowledgment timer for the message. If the ISO Host Interface process finds this timer, it is deleted.
4. Subtracts one from the number of messages currently outstanding to the DPC.
5. Changes the message type to 0520.
6. Adds the message to the SAF.

Unintelligible Messages from a DPC

If the ISO Host Interface process receives a message that it cannot process, it rejects the message and returns it to the DPC that sent it.

To reject a message, the ISO Host Interface process performs the following:

- Changes the first position of the message type to a value of 9. For example, a 0200 message would be changed to a 9200 message, and a 0420 message would be changed to a 9420 message.
- Places the number of the data element that could not be processed in the status field of the external message header. For example, if the Track 2 data in the P-35 data element could not be parsed, the status field in the header would be set to the value 035.

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7: BASE24-telebanking Financial Message Handling

This section contains an overview of how BASE24-telebanking and BASE24-billpay financial transaction messages are handled by the ISO Host Interface process. It includes message flows for both issuer and acquirer host processing.

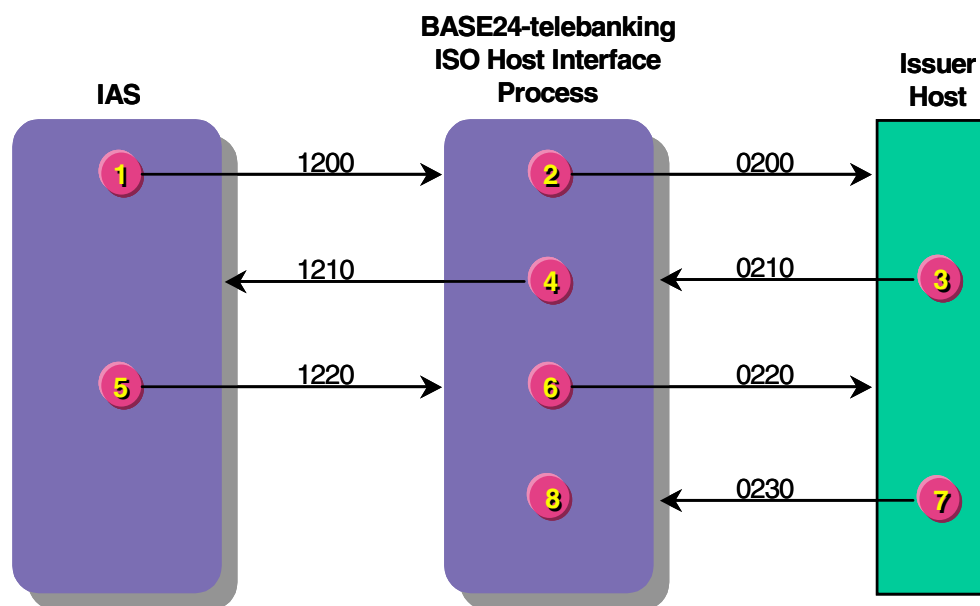
Note: The ISO Host Interface does not support full message encryption for BASE24-telebanking and BASE24-billpay messages to or from a host.

Issuer Host Message Handling

This section describes the processing involved for messages to and from an issuer host.

Financial Transaction Message Flow

The following diagram illustrates the message exchange expected for financial transactions sent to an issuer host by the BASE24-telebanking ISO Host Interface process. A key to the abbreviations and the steps corresponding to the diagram are provided below.



Key

IAS = Integrated Authorization Server process

1. The BASE24-telebanking Integrated Authorization Server process determines that the request must be sent to an issuer host for authorization and sends it to the ISO Host Interface process for forwarding to that host.

The BASE24-telebanking Integrated Authorization Server process retrieves the name of the ISO Host Interface process to which to send the message from the PROCESS NAME field on Internal Message Translation File (IMTF) screen 1.

The BASE24-telebanking Integrated Authorization Server process also retrieves the number of the institution data processing center (DPC) to which the message is sent from the DPC NUMBER field on IMTF screen 1.

2. The ISO Host Interface process receives the 1200 message and performs the following steps:
 - a. Locates the appropriate DPC processing information in its internal table using the DPC number from the DPC-NUM field in the Telebanking Standard Internal Message (BSTM).
 - b. Obtains the compressed Internal Transaction Data (ITD) in the BSTM, expands it, and places it in an internal ITD structure.
 - c. Creates an external message from the BSTM and the internal ITD structure. The internal 1200 message is mapped into an external 0200 message.
 - d. Determines its ability to send the message. The ISO Host Interface process cannot send the message if any of the following conditions exist:
 - The DPC is marked down.
 - Store-and-forward transactions are sent to the host and the store-and-forward method is noninterspersed.
 - The host is single-threaded and no send/receive stations are available.
 - The current number of outstanding messages to the DPC equals the value in the MAXIMUM OUTSTANDING REQUESTS OUTBOUND field on Host Configuration File (HCF) screen 1.

If any of the above conditions exist, the ISO Host Interface process returns the 1200 message (BSTM) to the process indicated in the SRC-SYM-NAM field of the BSTM.

If none of the above conditions exist, the ISO Host Interface process continues with step 2e.

- e. Saves a copy of the BSTM in extended memory. As part of saving the message, the ISO Host Interface process stores the current time in a user buffer in memory.
- f. Sets an outbound request timer for the number of seconds specified in the OUTBOUND field on HCF screen 22.

If this timer expires before a 0210 message is received, the ISO Host Interface process checks the value of the EXTRN-RQST-TIMEOUT param to determine which actions are performed.

- If the param is set to a value of 0, the ISO Host Interface process returns the original 1200 message to the BASE24-telebanking Integrated Authorization Server process as failed for alternate or stand-in routing. If the ISO Host Interface process receives a late 0210 message, it generates an 0420 message and sends it to the host.
 - If the param is set to a value of 1, the ISO Host Interface process returns the original 1200 message to the BASE24-telebanking Integrated Authorization Server process as failed for alternate or stand-in routing and generates an 0420 message that it sends to the host. If the ISO Host Interface process receives a late 0210 message, it drops the message.
- g. Updates its internal counters. The ISO Host Interface process performs the following steps:
 - 1) Adds one to the number of messages currently outstanding to the DPC.
 - 2) Adds one to the total number of messages currently outstanding from the ISO Host Interface process.
 - h. Generates a message authentication code (MAC) for the message, if message authentication is supported and is specified for this message type.
 - i. Converts the message from ASCII to EBCDIC if the CHARACTER FORMAT field on screen 1 of the HCF is set to E for the DPC.
 - j. Sends the 0200 message to the DPC.
- Along with the message, the ISO Host Interface process also sends the maximum number of seconds the XPNET process can allow the message to queue to the host. This number of seconds is derived using the values in the QUEUE SUBTRACT and OUTBOUND fields on HCF screen 22. The value in the QUEUE SUBTRACT field is subtracted from the value in the OUTBOUND field and the result is used.
3. The host validates the request, processes the transaction, and returns a 0210 message either approving or declining the transaction.
 4. The ISO Host Interface process receives the 0210 message and performs the following steps:
 - a. Locates the DPC in its internal table, identifying the necessary information for processing.
 - b. Converts the message from EBCDIC to ASCII if the CHARACTER FORMAT field on screen 1 of the HCF is set to E for the DPC.

- c. Verifies the MAC received with the message, if message authentication is supported and is specified for the message type.
 - d. Retrieves the stored BSTM from extended memory.
 - e. Deletes the outbound request timer.
 - f. Updates its counters. The ISO Host Interface process performs the following steps:
 - 1) Subtracts one from the number of messages currently outstanding to the DPC.
 - 2) Subtracts one from the total number of messages currently outstanding from the ISO Host Interface process.
 - 3) Adds one to the number of requests processed for the DPC.
 - 4) Adds the amount of time taken to process this request to the total time taken for all requests to the DPC. (For performance statistics, this value is divided by the number of requests processed for the DPC.)
 - g. Sets the TYP field in the BSTM to the value 1210.
 - h. Updates the BSTM from the external message using only the information that was changed by the host.
 - i. Sends the 1210 message (BSTM) to the Integrated Authorization Server process.
5. The BASE24-telebanking Integrated Authorization Server process finishes processing the transaction, logs it to the ITS Transaction Log File (ITLF), and returns it to its originator.

If the host requires advices, the BASE24-telebanking Integrated Authorization Server process generates the appropriate 1220 message and sends it to the ISO Host Interface process.

The BASE24-telebanking Integrated Authorization Server process retrieves the name of the ISO Host Interface process to which to send the advice message from the PROCESS NAME field on IMTF screen 1.

The BASE24-telebanking Integrated Authorization Server process also retrieves the number of the institution data processing center (DPC) to which the advice message is sent from the DPC NUMBER field on IMTF screen 1.

6. The ISO Host Interface process receives the 1220 message and performs the following steps:
 - a. Locates the appropriate DPC processing information in its internal table using the DPC number from the DPC-NUM field in the BSTM.
 - b. Obtains the compressed ITD in the BSTM, expands it, and places it in an internal ITD structure.
 - c. Creates an external message from the BSTM and the internal ITD structure. The internal 1220 message is mapped into an external 0220 message.
 - d. Determines whether the DPC is marked down. If so, the ISO Host Interface process writes the 0220 message to the Store-and-Forward File (SAF) and takes no further action with the message until a store-and-forward cycle is started.
 - e. Determines whether a noninterspersed store-and-forward cycle is in progress. If so, the ISO Host Interface process writes the 0220 message to the SAF, and the message is handled in sequential order.
 - f. Locates an available send/receive station. If no send/receive stations are available, the ISO Host Interface process writes the 0220 message to the SAF and takes no further action with the message until a store-and-forward cycle is started.
 - g. Generates a MAC for the message, if message authentication is supported and is specified for this message type.
 - h. Converts the message from ASCII to EBCDIC if the CHARACTER FORMAT field on screen 1 of the HCF is set to E for the DPC.
 - i. Sends the 0220 message to the DPC.

Along with the message, the ISO Host Interface process also sends the maximum number of seconds the XPNET process can allow the message to queue to the host. This number of seconds is derived using the values in the QUEUE SUBTRACT and COMPLETION ACK fields on HCF screen 22. The value in the QUEUE SUBTRACT field is subtracted from the value in the COMPLETION ACK field and the result is used.
 - j. Stores a copy of the external message.
 - k. Sets a completion acknowledgment timer.
 - l. Adds one to the number of messages currently outstanding to the DPC.

7. The host processes the 0220 message and returns a 0230 message if text-level acknowledgments are required.

DPCs are required to return text-level acknowledgments if the ACK FROM DPC field on HCF screen 1 contains the value Y.

8. Upon receipt of a 0230 message, the ISO Host Interface process performs the following steps:
 - a. Locates the DPC in its internal table, identifying the necessary information for processing.
 - b. Converts the message from EBCDIC to ASCII if the CHARACTER FORMAT field on screen 1 of the HCF is set to E for the DPC.
 - c. Verifies the MAC received with the message, if message authentication is supported and is specified for the message type.
 - d. Deletes the stored external message from extended memory.
 - e. Deletes the completion acknowledgment timer.
 - f. Subtracts one from the number of messages currently outstanding to the DPC.

Processing for the 0230 message is now complete.

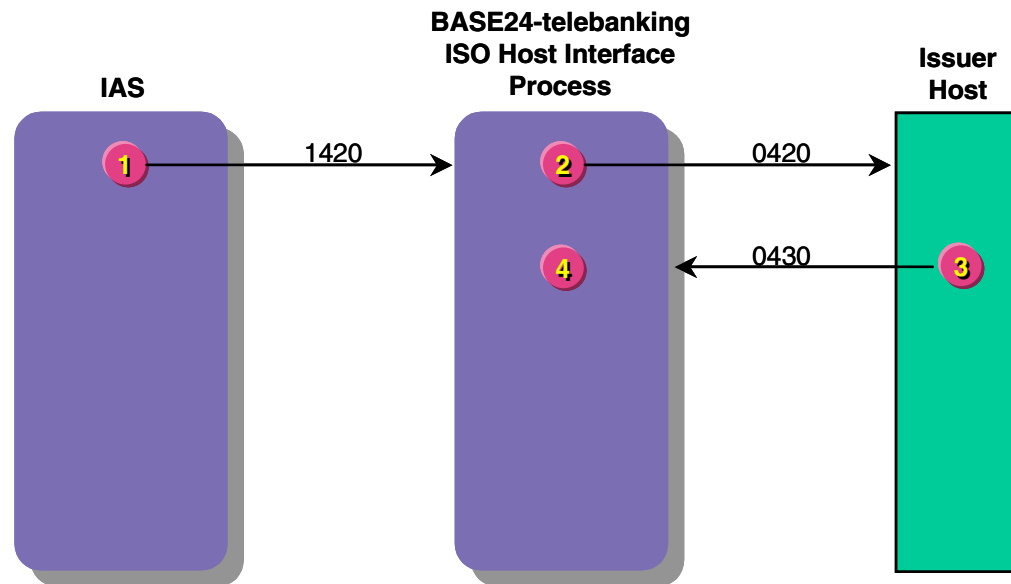
Nonfinancial Transactions Message Flow

BASE24-telebanking nonfinancial transactions include balance inquiries, history inquiries, PIN verifications, and PIN changes. Nonfinancial transactions are handled the same as financial transactions, except nonfinancial transactions use different message types, as shown in the following table. User-defined transactions can be either financial or nonfinancial transactions.

Financial Transaction Message Types	Nonfinancial Transaction Message Types
External Message Types	
0200	0100
0210	0110
0220	0120
0230	0130
Internal Message Types	
1200	1100
1210	1110
1220	1120

Reversal Message Flow

The following diagram illustrates the message exchange expected for reversal transactions sent to a host by the BASE24-telebanking ISO Host Interface process. A reversal indicates that a previously-approved transaction did not complete as intended. A key to the abbreviations and the steps corresponding to the diagram are provided below



Key

IAS = Integrated Authorization Server process

1. The Integrated Authorization Server process determines that a transaction did not complete as previously authorized, then processes it as follows:
 - a. Checks whether a reversal transaction should be sent to the host. The Integrated Authorization Server process does not process the message any further if either of the following conditions exist:
 - The ITD.ACT.CDE field in the BSTM indicates that the transaction was declined.
 - The ITD.PROC-CDE.TXN-CDE field in the BSTM identifies a nonfinancial transaction other than a PIN change and the NON-FINANCIAL REVERSAL INDICATOR field on HCF screen 22 is set to a value of N (do not reverse approved nonfinancial transactions).

- b. Backs out the original transaction, posts the revised transaction, logs the transaction to the ITLF, and sends a 1420 message in BSTM format to the ISO Host Interface process.

The BASE24-telebanking Integrated Authorization Server process retrieves the name of the ISO Host Interface process to which to send the reversal message from the PROCESS NAME field on IMTF screen 1.

The BASE24-telebanking Integrated Authorization Server process also retrieves the number of the institution data processing center (DPC) to which the reversal message is sent from the DPC NUMBER field on IMTF screen 1.

- 2. The ISO Host Interface process receives the 1420 message and processes it by performing the following steps:

- a. Locates the DPC record in its internal table using the value in the DPC-NUM field in the BSTM.
- b. Creates an external message from the BSTM. The internal 1420 message is mapped into an external 0420 message.
- c. Determines whether the DPC is marked down. If so, the ISO Host Interface process writes a 0420 message to the SAF and takes no further action with the message until a store-and-forward cycle is started.
- d. Determines whether a store-and-forward cycle (either interspersed or noninterspersed) is in progress for the DPC. If so, the ISO Host Interface process writes a 0420 message to the SAF, and the message is handled in sequential order.
- e. Locates an available send/receive station. If no send/receive stations are available, the ISO Host Interface process writes a 0420 message to the SAF and takes no further action with the message until a store-and-forward cycle is started.
- f. Generates a MAC for the message, if message authentication is supported and is specified for this message type.
- g. Converts the message from ASCII to EBCDIC if the CHARACTER FORMAT field on screen 1 of the HCF is set to E for the DPC.
- h. Sends the 0420 message to the DPC.

Along with the message, the ISO Host Interface process also sends the maximum number of seconds the XPNET process can allow the message to queue to the host. This number of seconds is derived using the values in the QUEUE SUBTRACT and COMPLETION ACK fields on HCF screen 22. The value in the QUEUE SUBTRACT field is subtracted from the value in the COMPLETION ACK field and the result is used.

- i. Stores a copy of the external message.
 - j. Sets a completion acknowledgment timer.
 - k. Adds one to the number of messages currently outstanding to the DPC.
- 3. The host processes the 0420 message and returns a 0430 message if text-level acknowledgments are required.

DPCs are required to return text-level acknowledgments if the ACK FROM DPC field on HCF screen 1 contains the value Y.
- 4. Upon receipt of a 0430 message, the ISO Host Interface process performs the following steps:
 - a. Locates the DPC in its internal table, identifying the necessary information for processing.
 - b. Converts the message from EBCDIC to ASCII if the CHARACTER FORMAT field on screen 1 of the HCF is set to E for the DPC.
 - c. Verifies the MAC received with the message, if message authentication is supported and is specified for the message type.
 - d. Deletes the stored external message from extended memory.
 - e. Deletes the completion acknowledgment timer.
 - f. Subtracts one from the number of messages currently outstanding to the DPC.

Processing for the 0430 message is now complete.

Late Response from a DPC

If a DPC responds late to a request, the ISO Host Interface process performs the following steps for the 0110 or 0210 message:

1. Checks the Response Code data element (P-39) in the external message.
If the transaction was declined, the ISO Host Interface process drops the message.
2. Checks the Processing Code data element (P-3) in the external message.
If the transaction is a nonfinancial transaction other than a PIN change or BASE24-billpay transaction and the DISCARD NON-FINANCIAL REVERSALS field on HCF screen 22 is set to a value of Y (discard reversals for approved nonfinancial transactions), the ISO Host Interface process drops the message. Otherwise, processing continues with step 3.
3. Checks the value of the EXTRN-RQST-TIMEOUT param in the LCONF. If this param is set to a value of 0, processing continues with step 4. If this param is set to a value of 1, the ISO Host Interface process drops the message.
4. Creates a 0420 message. In doing so, the ISO Host Interface process performs the following:
 - a. Sets the Response Code data element (P-39) in the external message to a value of 68 (timeout).
 - b. Sets the Original Data Elements data element (S-90) in the external message.

The ORIG-TYP field is set to a value of 0100 or 0200, according to the original transaction request.

The ORIG-SEQ-NUM, ORIG-TRAN-DAT, ORIG-TRAN-TIM, and ORIG-B24-POST-DAT fields are set to the appropriate values from the original transaction request.
5. Selects the station to which to send the message.

To do this, the ISO Host Interface process first checks the station from which the 0210 message was received. If it is a send/receive station and is available, the ISO Host Interface process uses that one. Otherwise, it checks for any available send/receive stations.

If a station is available, the ISO Host Interface process uses it. If no stations are available, the ISO Host Interface process writes the 0420 message to the SAF and takes no further action with the message until a store-and-forward cycle is started.

6. Generates a MAC for the message, if message authentication is supported and is specified for this message type.
7. Converts the message from ASCII to EBCDIC if the CHARACTER FORMAT field on screen 1 of the HCF is set to E for the DPC.
8. Sends the 0420 message to the selected station.

Along with the message, the ISO Host Interface process also sends the maximum number of seconds the XPNET process can allow the message to queue to the host. This number of seconds is derived using the values in the QUEUE SUBTRACT and COMPLETION ACK fields on HCF screen 22. The value in the QUEUE SUBTRACT field is subtracted from the value in the COMPLETION ACK field and the result is used.

9. Stores a copy of the external message.
10. Sets a completion acknowledgment timer.
11. Adds one to the number of messages currently outstanding to the DPC.
12. Upon receipt of a 0430 message, the ISO Host Interface process performs the following steps:
 - a. Locates the DPC in its internal table, identifying the necessary information for processing.
 - b. Converts the message from EBCDIC to ASCII if the CHARACTER FORMAT field on screen 1 of the HCF is set to E for the DPC.
 - c. Verifies the MAC received with the message, if message authentication is supported and is specified for the message type.
 - d. Deletes the stored external message from extended memory.
 - e. Deletes the completion acknowledgment timer.
 - f. Subtracts one from the number of messages currently outstanding to the DPC.

Processing for the 0430 message is now complete.

Failed Messages to Stations

If a message is forwarded to the XPNET process, but cannot be sent to a station, the XPNET process returns the message to the ISO Host Interface process as failed.

Upon receipt of a message failure, the ISO Host Interface process marks the station down and sets an extended network management message timer. It then performs the following steps based on message type.

Failed 0100 and 0200 Messages

The ISO Host Interface process takes the following steps when it receives a failed 0100 or 0200 message back from the XPNET process. A failed message may be due to an unavailable station or the transaction becoming stale while queued by the XPNET process.

1. Locates the DPC in its internal table, identifying the necessary information for processing.
2. Retrieves the stored BSTM from extended memory.
3. Deletes the outbound request timer.
4. Updates its counters. The ISO Host Interface process performs the following steps:
 - a. Subtracts one from the number of messages currently outstanding to the DPC.
 - b. Adds one to the number of requests processed for the DPC.
 - c. Adds the amount of time taken to process this request to the total time taken for all requests to the DPC. (For performance statistics, this value is divided by the number of requests processed for the DPC.)
5. Sets the FAILURE-FLG field of the Telebanking Standard Internal Message (BSTM) to a value of 1 to indicate a failure.

This identifies the 0100 or 0200 message as a failed message.

6. Returns the BSTM (as a failed 1100 or 1200 message) to the BASE24-telebanking Integrated Authorization Server process.

Failed 0120, 0220, and 0420 Messages

The ISO Host Interface process takes the following steps when it receives a failed 0120, 0220 or 0420 message back from the XPNET process. This condition may be due to an unavailable station or the transaction becoming stale while queued by the XPNET process.

1. Locates the DPC in its internal table, identifying the necessary information for processing.
2. Retrieves the stored external message from extended memory.
3. Checks for a completion acknowledgment timer for the message and deletes it if found.
4. Subtracts one from the number of messages currently outstanding to the DPC.
5. Adds the message to the SAF.

Unintelligible Messages from a DPC

If the ISO Host Interface process receives an external message that it cannot process or reformat for internal use (because of bad data or a security-related failure), it rejects the message by performing the following steps:

1. Changes the first position of the message type to 9 (for example, a 0200 message would be changed to a 9200 message, and a 0420 message would be changed to a 9420 message).
2. Indicates the reason the message was rejected in the status field of the external message header. If the message was rejected because of a security failure, the status field is set to an indicative reject reason code. If the message was rejected because of bad data, this field is set to the bit number of the data element causing rejection.
3. Returns the message to the host.

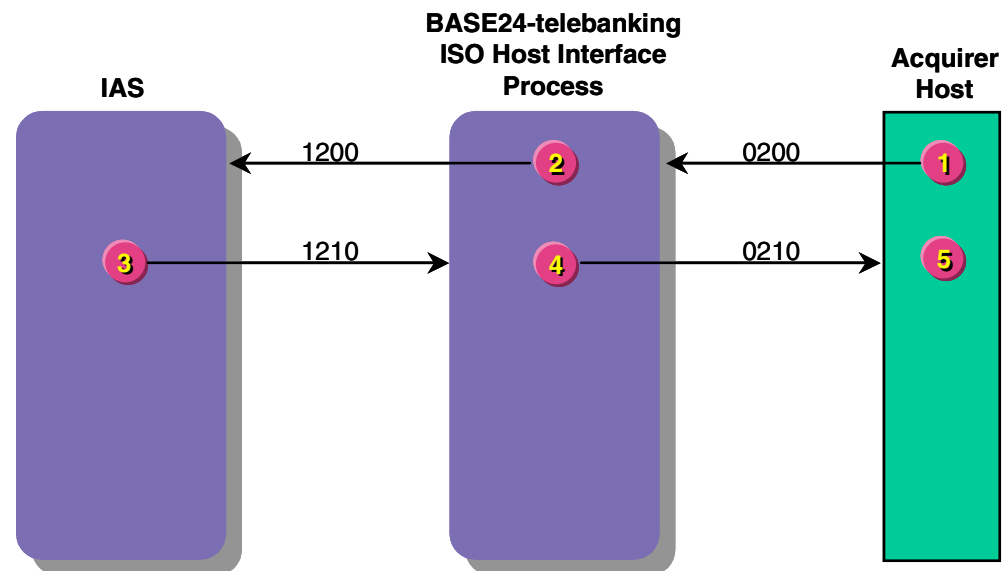
The above actions are taken on any message type that cannot be processed, not just those that require a response.

Acquirer Host Message Handling

This section describes the processing involved for messages to and from an acquirer host.

Financial Transaction Message Flow

The following diagram illustrates the message exchanges expected by BASE24-telebanking for financial transactions sent to the BASE24 system from an acquirer host. A key to the abbreviations and the steps corresponding to the diagram are provided below



Key

IAS = Integrated Authorization Server process

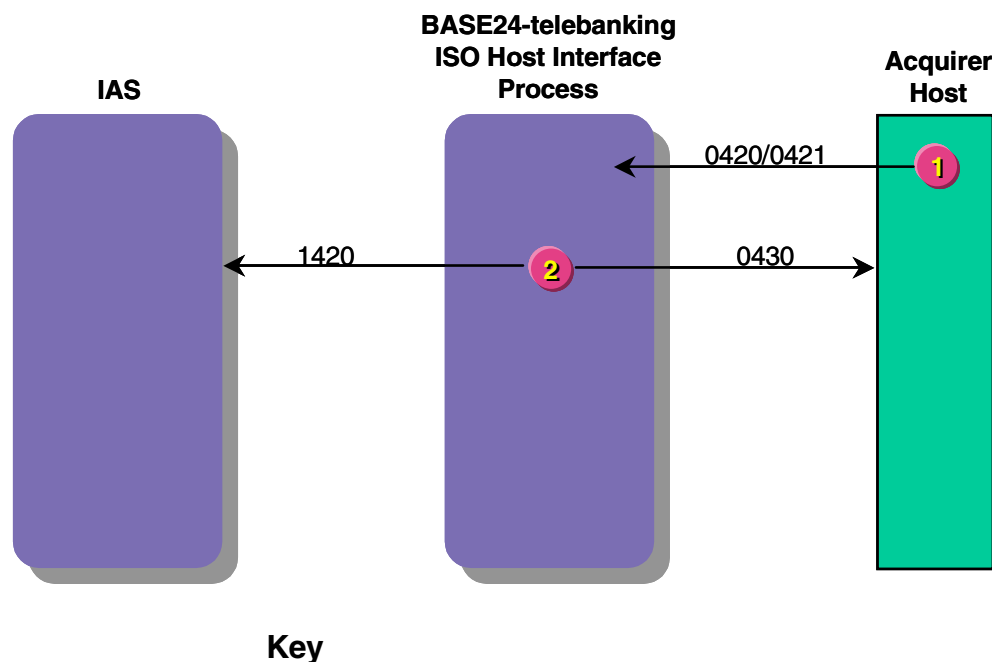
1. An account holder initiates a financial transaction at a device connected to and controlled by an acquirer host. The acquirer host then formats a 0200 message and sends it to the BASE24 system.

2. The ISO Host Interface process receives the 0200 message and performs the following steps:
 - a. Locates the DPC in its internal table, identifying the necessary information for processing.
 - b. Converts the message from EBCDIC to ASCII if the CHARACTER FORMAT field on screen 1 of the HCF is set to E for the DPC.
 - c. Verifies the MAC received with the message, if message authentication is supported and is specified for the message type.
 - d. Creates a BSTM from the external 0200 message. The external 0200 message is mapped into the internal 1200 message.
 - e. Updates the following fields:
 - 1) Sets the value in the RTE.ORIGINATOR field to indicate the transaction was initiated through an acquirer host.
 - 2) Sets the value in the DPC-NUM field to the number of the DPC originating the transaction.
 - 3) Sets the value in the ENTRY-TIM field to the current time.
 - f. Checks the current number of outstanding 1200 messages to the Integrated Authorization Server process.
 If the number of 1200 messages outstanding to the Integrated Authorization Server process equals the value in the MAXIMUM OUTSTANDING REQUESTS INBOUND field on HCF screen 1, the ISO Host Interface process creates a 0210 message with a response code of 91 (issuer inoperative) and sends it to the originating DPC.
 - g. Saves a copy of the external message in extended memory.
 - h. Sets an inbound request timer for the number of seconds specified in the INBOUND field on HCF screen 22.
 If this timer expires before a 1210 message is received, the ISO Host Interface process declines the transaction.
 - i. Adds one to the number of requests currently outstanding from the DPC.
 - j. Sends the 0200 message (BSTM) to the Integrated Authorization Server process indicated in the AUTH PROCESS field on HCF screen 22.
3. The BASE24-telebanking Integrated Authorization Server process processes the transaction, logs it to the ITLF, and returns a 1210 message to the ISO Host Interface process.

4. The ISO Host Interface process receives the 1210 message and performs the following steps:
 - a. Locates the stored external message.
 - b. Deletes the inbound request timer.
 - c. Subtracts one from the number of requests currently outstanding from the DPC.
 - d. Creates a 0210 external message from the BSTM.
 - e. Generates a MAC for the message, if message authentication is supported and is specified for this message type.
 - f. Converts the message from ASCII to EBCDIC if the CHARACTER FORMAT field on screen 1 of the HCF is set to E for the DPC.
 - g. Sends the 0210 external message to the station from which the 0200 message was originally received.
5. The acquirer host then completes the transaction with the account holder at the remote banking endpoint device or customer service representative (CSR) terminal.

Reversal Transaction Message Flow

The following diagram illustrates the message exchanges expected by BASE24-telebanking for reversals sent to BASE24-telebanking by an acquirer host. A reversal occurs when a BASE24-authorized transaction does not complete as authorized. A key to the abbreviations and the steps corresponding to the diagram are provided below.



IAS = Integrated Authorization Server process

1. The transaction-originating terminal notifies the acquirer host that the transaction did not complete as authorized. The acquirer host then formats a 0420 message and sends it to the BASE24 system (a 0421 message may be used to indicate a possible duplicate).
2. The ISO Host Interface process receives the 0420 or 0421 message and performs the following steps:
 - a. Locates the DPC in its internal table, identifying the necessary information for processing.
 - b. Converts the message from EBCDIC to ASCII if the CHARACTER FORMAT field on screen 1 of the HCF is set to E for the DPC.
 - c. Verifies the MAC received with the message, if message authentication is supported and is specified for the message type.

- d. Creates a BSTM from the external 0420 message. The external 0420 message is converted to an internal 1420 message.
- e. Sends the 1420 message (BSTM) to the Integrated Authorization Server process indicated in the AUTH PROCESS field on HCF screen 22.
- f. Generates a 0430 message if the ACK TO DPC field on HCF screen 1 contains the value Y.

The ISO Host Interface process performs ASCII-to-EBCDIC conversions and message authentication on the 0430 message as required.

Late Response from the Integrated Authorization Server Process

If the ISO Host Interface process receives a late 1210 message from the Integrated Authorization Server process, it performs the following steps:

1. Checks the ITD.ACT.CDE field in the BSTM to determine whether the transaction was declined. If it was, the ISO Host Interface process drops the message.
2. Checks the ITD.PROC-CDE.TXN-CDE field in the BSTM.

If the transaction is a nonfinancial transaction other than a PIN change or BASE24-billpay transaction and the DISCARD NON-FINANCIAL REVERSALS field on HCF screen 22 is set to a value of Y, the ISO Host Interface process drops the message.
3. Changes the TYP field in the BSTM to a value of 1420.
4. Sets the ITD.RSN-CDE-MSG field in the BSTM to a value of 4006 to indicate that the Integrated Authorization Server process sent a late response.
5. Sends the 1420 message back to the Integrated Authorization Server process that sent it.

Failed Messages to Stations

If a message is forwarded to the XPNET process, but cannot be sent to a station, the XPNET process returns the message to the ISO Host Interface process as failed.

Upon receipt of a message failure, the ISO Host Interface process marks the station down and sets an extended network management message timer. It then performs the following steps based on message type.

Failed 0210 Messages

The ISO Host Interface process takes the following steps when it receives a failed 0210 message back from the XPNET process.

1. Locates the DPC in its internal table, identifying the necessary information for processing.
2. Checks the Response Code data element (P-39) in the external message to determine whether the transaction was declined. If it was, the ISO Host Interface process drops the message.
3. Checks the Processing Code data element (P-3) in the external message.
If the transaction is a nonfinancial transaction other than a PIN change or BASE24-billpay transaction and the DISCARD NON-FINANCIAL REVERSALS field on HCF screen 22 is set to a value of Y, the ISO Host Interface process drops the message.
4. Creates a BSTM from the external message and sets the TYP field to a value of 1420.
5. Sets the ITD.RSN-CDE-MSG field in the BSTM to a value of 4013 to indicate that the destination was not available.
6. Sends the 1420 message to the Integrated Authorization Server process indicated in the AUTH PROCESS field on HCF screen 22.

Failed 0230/0430 Messages

If the ISO Host Interface process receives a failed 0230 or 0430 message back from the XPNET process, it drops the message.

Unintelligible Messages from a DPC

If the ISO Host Interface process receives a message that it cannot process, it rejects it, returning the message to the DPC that sent it.

To reject a message, the ISO Host Interface process performs the following:

- Changes the first position of the message type to a value of 9. For example, a 0200 message would be changed to a 9200 message, and a 0420 message would be changed to a 9420 message.
- Places the number of the data element that could not be processed in the status field of the external message header. For example, if the Track 2 data in the P-35 data element could not be parsed, the status field in the header would be set to the value 035.

8: BASE24-teller Financial Message Handling

This section contains an overview of how BASE24-teller financial transaction messages are handled by the ISO Host Interface process. Although both issuer and host acquirer host processing are supported, this section includes message flows for issuer host processing only.

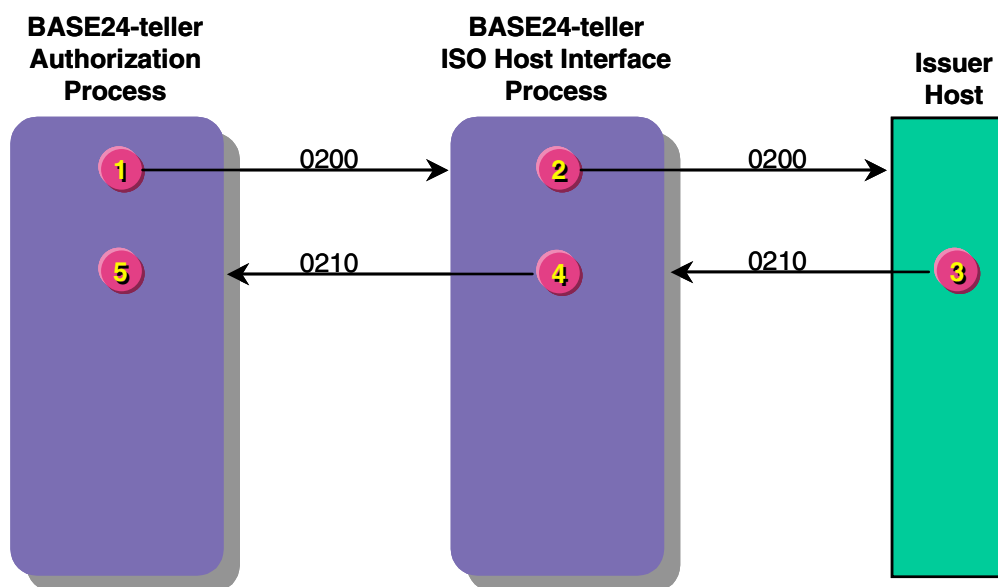
Note: The ISO Host Interface does not support full message encryption for BASE24-teller messages to or from a host.

Issuer Host Message Handling

This section describes the processing involved for messages to and from an issuer host.

Financial Transaction Message Flow

The following diagram illustrates the message exchange expected for financial transactions sent to an issuer host by the BASE24-teller ISO Host Interface process. The steps corresponding to the diagram are provided below



1. The BASE24 system receives a 0200 financial transaction request message, which is sent to the BASE24-teller Authorization process.

The BASE24-teller Authorization process determines that the request must be sent to an issuer host for authorization and sends it to the ISO Host Interface process for forwarding to that host.

The BASE24-teller Authorization process retrieves the name of the ISO Host Interface process to which to send the message from the TELLER ROUTING INFORMATION table fields on IDF screen 28.

2. The ISO Host Interface process receives the 0200 message and performs the following steps:
 - a. Verifies that the message type is a valid message type for a Teller Standard Internal Message (TSTM). If the message type is not valid, the ISO Host Interface process logs an event message and drops the 0200 message.
 - b. Locates the appropriate DPC processing information in its internal table using the DPC number from the DPC-NUM field in the TSTM.
 - c. Determines its ability to send the message to the host. The ISO Host Interface process cannot send the message if any of the following conditions are true:
 - The DPC is marked down.
 - There is a noninterspersed store-and-forward cycle in progress.
 - The host is single-threaded and no send/receive stations are available.
 - The current number of outstanding messages to the DPC equals the value in the MAXIMUM OUTSTANDING REQUESTS OUTBOUND field on HCF screen 1.

If any of the above conditions exist, the ISO Host Interface process performs the following steps:

- 1) Sets the RTE.STAT field in the TSTM to a value of 12 to indicate that the DPC was marked down.
- 2) Sends the TSTM to the process identified as the originator of the message in the system message header.

If none of the above conditions exist, the ISO Host Interface process continues with step 2d.

- d. Creates an external message from the internal message.
- e. Saves a copy of the TSTM in extended memory. As part of saving the message, the ISO Host Interface process stores the current time in a user buffer in memory.
- f. Sets an outbound request timer for the number of seconds specified in the OUTBOUND field on HCF screen 5.

If this timer expires before a 0210 message is received, the ISO Host Interface process checks the value of the EXTRN-RQST-TIMEOUT param to determine which actions are performed.

- If the param is set to a value of 0, the ISO Host Interface process returns the original 0200 message to the BASE24-teller Authorization process as failed for alternate or stand-in routing. If the ISO Host Interface process receives a late 0210 message, it generates an 0420 message and sends it to the host.
 - If the param is set to a value of 1, the ISO Host Interface process returns the original 0200 message to the BASE24-teller Authorization process as failed for alternate or stand-in routing and generates an 0420 message that it sends to the host. If the ISO Host Interface process receives a late 0210 message, it drops the message.
- g. Updates its internal counters. The ISO Host Interface process performs the following steps:
- 1) Adds one to the number of messages currently outstanding to the DPC.
 - 2) Adds one to the total number of messages currently outstanding from the ISO Host Interface process.
- h. Generates a MAC for the message, if message authentication is supported and is specified for this message type.
- i. Converts the message from ASCII to EBCDIC if the CHARACTER FORMAT field on screen 1 of the HCF is set to E for the DPC.
- j. Sends the 0200 message to the DPC.
- Along with the message, the ISO Host Interface process also sends the maximum number of seconds the XPNET process can allow the message to queue to the host. This number of seconds is derived using two values from fields on HCF screen 10: QUEUE SUBTRACT and OUTBOUND. The value in the QUEUE SUBTRACT field is subtracted from the value in the OUTBOUND field and the result is used.
3. The host validates the request, processes the transaction, and returns a 0210 message either approving or declining the transaction.
4. The ISO Host Interface process receives the 0210 message and performs the following steps:
- a. Locates the DPC in its internal table, identifying the necessary information for processing.
 - b. Converts the message from EBCDIC to ASCII if the CHARACTER FORMAT field on screen 1 of the HCF is set to E for the DPC.
 - c. Verifies the MAC received with the message, if message authentication is supported and is specified for the message type.

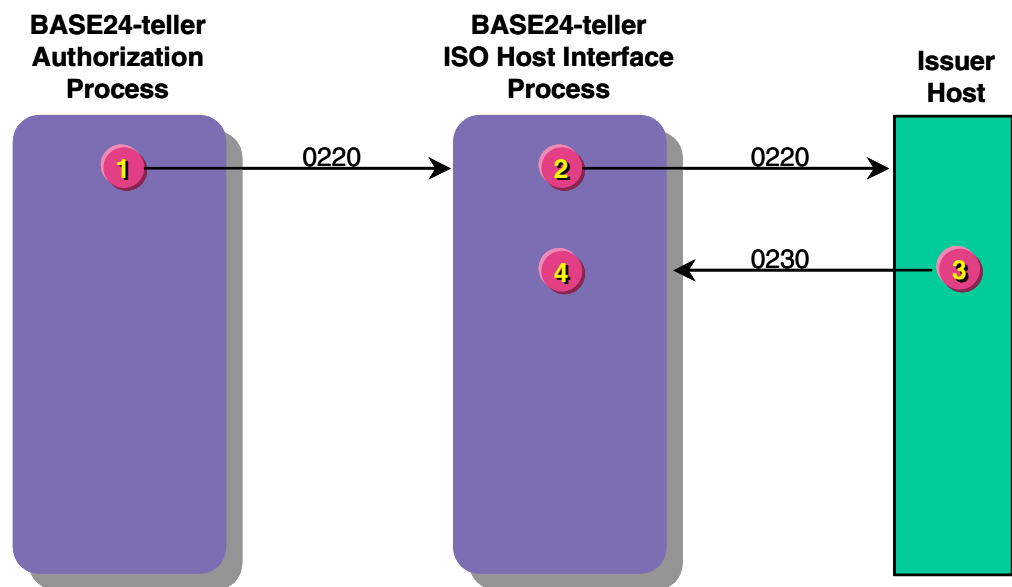
- d. Retrieves the stored TSTM from extended memory.
 - e. Deletes the outbound request timer.
 - f. Updates its counters. The ISO Host Interface process performs the following steps:
 - 1) Subtracts one from the number of messages currently outstanding to the DPC.
 - 2) Adds one to the number of requests processed for the DPC.
 - 3) Adds the amount of time taken to process this request to the total time taken for all requests to the DPC. (For performance statistics, this value is divided by the number of requests processed for the DPC.)
 - g. Updates the value in the EXIT-TIM field in the TSTM using the time stored in the user buffer when the request message was saved in extended memory (see step 2e).
 - h. Sets the value in the RE-ENTRY-TIM field in the TSTM to the current time.
 - i. Sets the value in the TYP field in the TSTM to 0210.
 - j. Updates the TSTM from the external message using only the information that was changed by the host.
 - k. Sends the 0210 message (TSTM) to the Authorization process.
5. The BASE24-teller Authorization process finishes processing the transaction, logs it to the TTLF, and returns it to its originator.

File Inquiry/Update and Administrative Transactions

The message exchange expected for file inquiry/update transactions (0300 messages) and for administrative messages (0600 messages) sent to an issuer host by the BASE24-teller ISO Host Interface process is the same as the message exchange for financial transactions (0200 messages).

Financial Transaction Advice Message Flow

When a transaction is approved by the terminal or terminal controller, the device sends an advice of the transaction to the BASE24 system. The Device Handler process formats this message into a 0220 message and passes it on to the Authorization process. Advices can also be generated by the Authorization process when the BASE24 system is set up to approve transactions and completions are required by the host. The following diagram illustrates the message exchange expected for financial transaction advices sent to an issuer host by the BASE24-teller ISO Host Interface process. The steps corresponding to the diagram are provided below.



1. The Authorization process receives or generates a 0220 message and sends it to the ISO Host Interface process.
2. The ISO Host Interface process receives the 0220 message and performs the following steps:
 - a. Locates the appropriate DPC processing information in its internal table using the DPC number from the DPC-NUM field in the TSTM.
 - b. Creates an external message from the TSTM.
 - c. Checks the RQST.ADVC-RESP-REQ field in the TSTM and performs the following steps depending on the value in that field:
 - 1) If the RQST.ADVC-RESP-REQ field contains a value of 1 (text-level acknowledgments required), the ISO Host Interface process performs the following checks:

- a) Determines whether the DPC is marked down.
- b) Determines whether a noninterspersed store-and-forward cycle is in progress.
- c) Determines whether a send/receive station is available.

If the DPC is marked down, a noninterspersed store-and-forward cycle is in progress, or there are no available stations, the ISO Host Interface process creates a 0230 message, sets the response code to the value 12 (host is down), and sends the 0230 message to the Authorization process. Otherwise, processing continues with step 2d.

- 2) If the RQST.ADVC-RESP-REQ field contains a value of 0 (text-level acknowledgments are not required), the ISO Host Interface process performs the following checks:

- a) Determines whether the DPC is marked down.
- b) Determines whether a noninterspersed store-and-forward cycle is in progress.
- c) Determines whether a send/receive station is available.

If the DPC is marked down, a noninterspersed store-and-forward cycle is in progress, or there are no available stations, the ISO Host Interface process writes the external version of the 0220 message to the Store-and-Forward File (SAF) and takes no further action with the message until a store-and-forward cycle is started. Otherwise, processing continues with step 2d.

- d. Generates a MAC for the message, if message authentication is supported and is specified for this message type.
- e. Converts the message from ASCII to EBCDIC if the CHARACTER FORMAT field on screen 1 of the HCF is set to E for the DPC.
- f. Sends the 0220 message to the DPC.

Along with the message, the ISO Host Interface process also sends the maximum number of seconds the XPNET process can allow the message to queue to the host. This number of seconds is derived using two values from fields on HCF screen 10: QUEUE SUBTRACT and COMPLETION ACK. The value in the QUEUE SUBTRACT field is subtracted from the value in the COMPLETION ACK field and the result value is used.

- g. Stores a copy of the external message.

- h. Sets a completion acknowledgment timer.
 - i. Adds one to the number of messages currently outstanding to the DPC.
3. The host processes the 0220 message and returns a 0230 message if text-level acknowledgments are required.

DPCs are required to return text-level acknowledgments if the ACK FROM DPC field on HCF screen 1 contains the value Y or if the value in the RQST.ADVC-RESP-REQ field is equal to 1.

4. Upon receipt of a 0230 message, the ISO Host Interface process performs the following steps:
- a. Locates the DPC in its internal table, identifying the necessary information for processing.
 - b. Converts the message from EBCDIC to ASCII if the CHARACTER FORMAT field on screen 1 of the HCF is set to E for the DPC.
 - c. Verifies the MAC received with the message, if message authentication is supported and is specified for the message type.
 - d. Deletes the stored external message from extended memory.
 - e. Deletes the completion acknowledgment timer.
 - f. Subtracts one from the number of messages currently outstanding to the DPC.
 - g. If the RQST.ADVC-RESP-REQ field contains the value 1 (text-level acknowledgments required), the ISO Host Interface process sends the 0230 message to the Authorization process. Otherwise (RQST.ADVC-RESP-REQ field contains the value 0), processing for the 0230 message is complete.

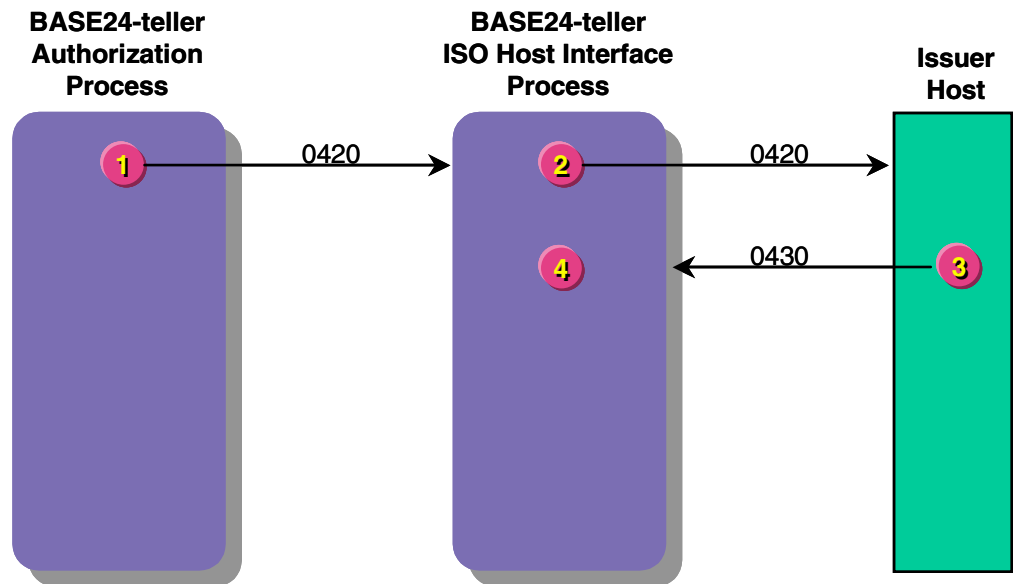
File Inquiry/Update and Administrative Advice Transactions

The message exchange expected for file inquiry/update advice transactions (0320 messages) and for administrative advice transactions (0620 messages) sent to an issuer host by the BASE24-teller ISO Host Interface process is the same as the message exchange for financial advice transactions (0220 messages).

Reversal Message Flow

The following diagram illustrates the message exchange expected for reversal transactions sent to an issuer host by the BASE24-teller ISO Host Interface process. The steps corresponding to the diagram are provided below.

A reversal indicates that a previously approved transaction did not complete as was intended.



1. The BASE24 system is advised that a transaction did not complete as previously authorized.

The BASE24-teller Authorization process backs out the original transaction, posts the revised transaction, logs the transaction to the TTLF, and sends a 0420 message in TSTM format to the ISO Host Interface process.

2. The ISO Host Interface process receives the 0420 message and processes it by performing the following steps:
 - a. Locates the DPC record in its internal table using the value in the DPC-NUM field in the TSTM.
 - b. Creates an external message from the TSTM.
 - c. Checks the RQST.ADVC-RESP-REQ field in the TSTM and performs the following steps depending on the value in that field.
 - 1) If the RQST.ADVC-RESP-REQ field contains a value of 1 (online completion required), the ISO Host Interface process performs the following checks:

- a) Determines whether the DPC is marked down.
- b) Determines whether a noninterspersed store-and-forward cycle is in progress.
- c) Determines whether a send/receive station is available.

If the DPC is marked down, a noninterspersed store-and-forward cycle is in progress, or there are no available stations, the ISO Host Interface process creates a 0430 message, sets the response code to a value of 12 (host is down), and sends the 0430 message to the Authorization process. Otherwise, processing continues with step 2d.

- 2) If the RQST.ADVC-RESP-REQ field contains a value of 0 (online completions are not required), the ISO Host Interface process performs the following checks:

- a) Determines whether the DPC is marked down.
- b) Determines whether a noninterspersed store-and-forward cycle is in progress.
- c) Determines whether a send/receive station is available.

If the DPC is marked down, a noninterspersed store-and-forward cycle is in progress, or there are no available stations, the ISO Host Interface process writes the external version of the 0420 message to the SAF and takes no further action with the message until a store-and-forward cycle is started. Otherwise, processing continues with step 2d.

- d. Generates a MAC for the message, if message authentication is supported and is specified for this message type.
- e. Converts the message from ASCII to EBCDIC if the CHARACTER FORMAT field on screen 1 of the HCF is set to E for the DPC.
- f. Sends the 0420 message to the DPC.

Along with the message, the ISO Host Interface process also sends the maximum number of seconds the XPNET process can allow the message to queue to the host. This number of seconds is derived using two values from fields on HCF screen 10: QUEUE SUBTRACT and COMPLETION ACK. The value in the QUEUE SUBTRACT field is subtracted from the value in the COMPLETION ACK field and the result value is used.

- g. Stores a copy of the external message.

- h. Sets a completion acknowledgment timer.
- i. Adds one to the number of messages currently outstanding to the DPC.
- 3. The host processes the 0420 message and returns a 0430 message if text-level acknowledgments are required.

DPCs are required to return text-level acknowledgments if the value in the ACK FROM DPC field on HCF screen 1 indicates that text-level acknowledgments are required.

- 4. Upon receipt of a 0430 message, the ISO Host Interface process performs the following steps:
 - a. Locates the DPC in its internal table, identifying the necessary information for processing.
 - b. Converts the message from EBCDIC to ASCII if the CHARACTER FORMAT field on screen 1 of the HCF is set to E for the DPC.
 - c. Verifies the MAC received with the message, if message authentication is supported and is specified for the message type.
 - d. Deletes the stored external message from extended memory.
 - e. Deletes the completion acknowledgment timer.
 - f. Subtracts one from the number of messages currently outstanding to the DPC.
 - g. Drops the 0430 message.

Late Response from a DPC

If a DPC responds late to a request, the ISO Host Interface process performs the following steps for the 0210 message:

1. Checks the Response Code data element (P-39) in the external message.
If the transaction was declined, the ISO Host Interface process drops the message.
2. Checks the Processing Code data element (P-3) in the external message.
If the transaction was a PIN verification transaction, the ISO Host Interface process drops the message.
3. Checks the value of the EXTRN-RQST-TIMEOUT param in the LCONF. If this param is set to a value of 0, processing continues with step 4. If this param is set to a value of 1, the ISO Host Interface process drops the message.
4. Creates a 0420 message. In doing so, the ISO Host Interface process performs the following steps:
 - a. Sets the Original Data Elements data element (S-90) in the external message to a value of 0200.
 - b. Sets the Response Code data element (P-39) in the external message to a value of 68 (timeout).
5. Selects the station to which to send the message.

To do this, the ISO Host Interface process first checks the station from which the 0210 message was received. If it is a send/receive station and is available, the ISO Host Interface process uses that one. Otherwise, it checks for any available send/receive stations.

If a station is available, the ISO Host Interface process uses it. If no stations are available, the ISO Host Interface process writes the 0420 message to the SAF and takes no further action with the message until a store-and-forward cycle is started.
6. Generates a MAC for the message, if message authentication is supported and is specified for this message type.
7. Converts the message from ASCII to EBCDIC if the CHARACTER FORMAT field on screen 1 of the HCF is set to E for the DPC.

8. Sends the 0420 message to the selected station.

Along with the message, the ISO Host Interface process also sends the maximum number of seconds the XPNET process can allow the message to queue to the host. This number of seconds is derived using two values from fields on HCF screen 10: QUEUE SUBTRACT and COMPLETION ACK. The value in the QUEUE SUBTRACT field is subtracted from the value in the COMPLETION ACK field and the result is used.

9. Stores a copy of the external message.
10. Sets a completion acknowledgment timer.
11. Adds one to the number of messages currently outstanding to the DPC.

Failed Messages to Stations

If a message is forwarded to the XPNET process, but cannot be sent to a station, the XPNET process returns the message to the ISO Host Interface process as failed.

Upon receipt of a message failure, the ISO Host Interface process marks the station down and sets an extended network management message timer. It then performs the following steps based on message type.

Failed 0200, 0300, or 0600 Messages

The ISO Host Interface process takes the following steps when it receives a failed 0200, 0300, or 0660 message back from the XPNET process. This condition may be due to an unavailable station or the transaction becoming stale while queued by the XPNET process.

1. Locates the DPC in its internal table, identifying the necessary information for processing.
2. Retrieves the stored TSTM from extended memory.
3. Deletes the outbound request timer.
4. Updates its counters. The ISO Host Interface process performs the following steps:
 - a. Subtracts one from the number of messages currently outstanding to the DPC.
 - b. Adds one to the number of requests processed for the DPC.
 - c. Adds the amount of time taken to process this request to the total time taken for all requests to the DPC. (For performance statistics, this value is divided by the number of requests processed for the DPC.)
5. Sets the RTE.STAT field in the TSTM to a value of 12 to indicate the line was down.
6. Returns the TSTM (as a failed 0200, 0300, or 0600 message) to the BASE24-teller Authorization process.

Failed 0220, 0320, 0420, or 0620 Messages

The ISO Host Interface process takes the following steps when it receives a failed 0220, 0320, 0420, or 0620 message back from the XPNET process. This condition may be due to an unavailable station or the transaction becoming stale while queued by the XPNET process.

1. Locates the DPC in its internal table, identifying the necessary information for processing.
2. Retrieves the stored external message from extended memory.
3. Checks for a completion acknowledgment timer for the message and deletes it if found.
4. Subtracts one from the number of messages currently outstanding to the DPC.
5. Adds the message to the SAF.

Unintelligible Messages from a DPC

If the ISO Host Interface process receives a message that it cannot process, it rejects it, returning the message to the DPC that sent it.

To reject a message, the ISO Host Interface process performs the following:

- Changes the first position of the message type to a value of 9. For example, a 0200 message would be changed to a 9200 message, and a 0420 message would be changed to a 9420 message.
- Places the number of the data element that could not be processed in the status field of the external message header. For example, if the Track 2 data in the P-35 data element could not be parsed, the status field in the header would be set to the value 035.

A: Processing Information

This appendix provides information on initializing and warmbooting the ISO Host Interface process. In addition, this appendix contains information about the event messages created by the ISO Host Interface process, the configuration of the process to the XPNET system, and the text commands that the ISO Host Interface process can receive.

Initialization

Initialization occurs when the ISO Host Interface process is started. The steps that the ISO Host Interface process takes during initialization are described below and on the pages the follow. Depending on the products supported by the ISO Host Interface process, all of these steps or only a subset may be performed. The steps described under the topic “Base ISO Host Interface Process” are always performed. The steps described under product-specific headings are only performed if support for that product is bound in to the ISO Host Interface process.

Base ISO Host Interface Process

During initialization, the ISO Host Interface process performs the steps listed below. If problems occur during initialization, the ISO Host Interface process logs an event message and abends.

1. Reads the name of the Logical Network Configuration File (LCONF) into memory.

The ISO Host Interface process obtains the name of the LCONF from the startup message sent by the XPNET process. This name allows the ISO Host Interface process to access the LCONF for the assigns and params it needs for processing.

2. Opens the LCONF, reads the following assigns and params, and closes the LCONF. If the ISO Host Interface process cannot locate the required assigns and params, the process abends.

Assigns	Params
HCF (required) SAF (required) LMCF EMF KEYF TKN KEY 6	LOGICAL-NET (required) SAF-NOTIFY-INTERVAL SAF-NOTIFY-PERCENTAGE RSM-TERM-MASTER-KEY DES-TRIPLE-SINGLE MULT-LN-PIN-KEY RACAL-ANSI-METHOD PSEUDO-3DES-CBC-MAC-TYPE ENHANCED-EMV-RC-MAPPING SEND-ETX PRE-VERIFIED-PIN-PAD-CHAR CICS-ID-OFFSET

3. Opens and reads the Log Message Configuration File (LMCF) for its modified event messages. To do this, the ISO Host Interface process uses the file name specified in the LMCF assign from the LCONF.

The LMCF only contains records for modified event messages. If any records are found for the ISO Host Interface process, they are read into memory.

4. Closes the LMCF.
5. Opens the network logging facility for writing event messages.
6. Opens the Host Configuration File (HCF) using the file name specified in the HCF assign from the LCONF.
7. Opens the Key File (KEYF) or the Key 6 File (KEY6) using the file name specified in the KEYF or KEY6 assign from the LCONF.
8. Reads the HCF to determine how many DPCs the ISO Host Interface process is processing for and the maximum number of transactions that can be in progress at one time.

If no HCF records are found for the ISO Host Interface process, or if more than 20 HCF records are found for the ISO Host Interface process, the process logs an event message and abends.

If the maximum number of transactions that can be in progress at one time (the sum of the values in the MAXIMUM OUTSTANDING REQUESTS OUTBOUND and MAXIMUM OUTSTANDING REQUESTS INBOUND fields for all DPCs supported by the ISO Host Interface process) is greater than 250, the process logs an event message and abends.

9. Opens the Token File (TKN) using the file name specified in the TKN assign from the LCONF.
10. Initializes extended memory for the amount of space required to perform processing. The ISO Host Interface process allocates enough space for its internal tables, timers, and outstanding messages (each time the ISO Host Interface process sends a message, it stores a copy of the message in memory for processing if no response is returned).
11. Initializes its DPC table.

The DPC table is an internal table containing one record for each DPC the ISO Host Interface process supports. This table contains all of the information contained in the HCF for the DPC, along with other status flags and counters that it uses for processing.

At this point, the ISO Host Interface process performs the following steps:

- a. Initializes all counters in the DPC table to zeros.
 - b. Reads the HCF information into its DPC table for those DPCs it is to process.
 - c. Marks each DPC in the DPC table down.
12. Retrieves the KEYF or KEY6 information for processing and moves the KEYF or KEY6 information into its DPC table.
 13. Opens and reads the Store-and-Forward File (SAF) using the file name specified in the SAF assign from the LCONF. If a SAF does not exist, the ISO Host Interface process creates one using the file name and template specified in the LCONF.
 14. Counts the records in the SAF destined for each of its DPCs and places those counts in the DPC table.
 15. Loads its network management table.

The network management table is an internal table containing the DPC external message bit maps, message authentication bit maps, and IMS/CICS transaction code equivalents for network management messages.

To load the table, the ISO Host Interface process performs the following steps:

- a. Loads the standard default external message bit maps for network management messages (types 0800 and 0810). These bit maps dictate the structures of the external messages. The standard defaults are documented in the ***BASE24 External Message Manual***. Product-specific bit maps are loaded later.
- b. Opens the EMF.
- c. Reads the data element bit map information for the DPCs it processes. Using these EMF records, the ISO Host Interface process updates the bit maps for the network management messages.
- d. Reads the message authentication bit map information for the DPCs it processes. Using these EMF records, the ISO Host Interface process updates the message authentication bit maps for the network management messages.

- e. Reads the IMS/CICS transaction code equivalents for 0800, 0810, and 9000 messages into memory. These transaction code equivalents are set in the EMF records.

Note: Message types in the 9000 range are used to denote rejects. For example, a message type of 0200 would be changed to 9200 if it were rejected and a message type of 0420 would be changed to 9420 if it were rejected. Setting up a record in the EMF for a message type of 9000 is not to control message structure, but rather to allow for assigning a single IMS/CICS transaction code equivalent to all reject messages returned to the co-network.

- f. Closes the EMF.

- 16. Performs product-specific initialization steps for each product that is supported. Product-specific steps are performed in order by product ID. The table below identifies the product IDs for BASE24 products.

Product ID	Product Name
1	BASE24-atm
2	BASE24-pos
3	BASE24-teller
14	BASE24-telebanking

- 17. Retrieves security module information from the LCONF and opens the security modules if necessary. To do this, the ISO Host Interface process performs the following steps:
 - a. Opens the LCONF.
 - b. Retrieves the following params. These params are required to interface with the security module.

SECURE-DEV-TYP
SECURE-DEV-TIM-LMT
SECURE-DEV-FAIL-NOTIFY

If the SECURE-DEV-TYP param is not present, the ISO Host Interface process closes the LCONF and skips to step 18, processing as if a security module is not being used.

- c. Retrieves the assigns listed below that are required to interface with a security module.

SECURE-DEV1
SECURE-DEV2

If both of the above assigns are not present, the ISO Host Interface process logs an event message and sets an internal flag to prohibit the use of a security module.

- d. Opens the security modules indicated in the SECURE-DEV1 and SECURE-DEV2 assigns.

Upon opening these devices, the operating system returns a file number for each device. The ISO Host Interface process then tracks these file numbers.

If the ISO Host Interface process does not, or cannot, open at least one device, it logs event messages and abends.

- e. Retrieves the following optional params.

SECURE-DEV-CONF-CMD100
SECURE-DEV-CONF-TXT
SECURE-DEV-MONITOR-THRESHOLD

- f. Closes the LCONF.

- 18. Closes the HCF.

- 19. Generates the event message “APPLICATION INITIALIZED.”

- 20. Starts a Performance timer for each DPC.

- 21. Checks the NMM-ENABLE field taken from the HCF for each DPC.

If the NMM-ENABLE field indicates the DPC does not allow network management messages, the ISO Host Interface process marks the DPC and all of its stations up. Otherwise, the ISO Host Interface process checks the STA-TYPE field taken from the HCF for each station and performs one of the following steps:

- If the station is not a send/receive station, the ISO Host Interface process marks the station up and goes on to the next station.
- If the station is a send/receive station, the ISO Host Interface process creates and sends a logon message and sets a Network Management Message timer for the station.

- 22. If a security device is supported and message authentication is performed by the security device, the ISO Host Interface process sets a flag indicating that a MAC key management message needs to be sent. A new key message will be initiated when the DPC responds to the logon message.

- 23. If a security device is supported and PIN management is performed by the security device, the ISO Host Interface process sets a flag indicating that a PIN key management message needs to be sent. A new key message will be sent when the DPC responds to the logon message (or after the MAC keys are in place if message authentication is performed—see step 22).

24. If a security device is supported and message encryption is performed by the security device, the ISO Host Interface process sets a flag indicating that a message encryption key management message needs to be sent. A new key message will be sent when the DPC responds to the logon message (or after the MAC keys are in place if message authentication is performed—see step 22).

BASE24-atm ISO Host Interface Process

For BASE24-atm processing, the ISO Host Interface process performs the following steps during initialization:

1. Opens the LCONF, reads the following params, and closes the LCONF:

ACCT-NUM-TYP
ATM-LN-CUTOVER
HOST-SELECTED-ACCT
REVERSE-BAL-INQ

2. Derives the current BASE24-atm posting date.

To do this, the ISO Host Interface process checks the ATM-LN-CUTOVER param value it retrieved from the LCONF.

If the current time is prior to logical network cutover, the ISO Host Interface process uses the current calendar date as the posting date.

If the current time is after logical network cutover, the ISO Host Interface process adds one to the current calendar date to derive a posting date.

3. Loads its BASE24-atm table.

The BASE24-atm table is an internal table containing the DPC data element bit maps, message authentication bit maps, IMS/CICS transaction code equivalents for BASE24-atm messages, and message token requirements.

To load the table, the ISO Host Interface process performs the following steps:

- a. Loads the standard default external message bit maps and message authentication bit maps for BASE24-atm messages.

These bit maps dictate the structures of the BASE24-atm external messages. Refer to the *BASE24 External Message Manual* for the standard default bit maps.

- b. Opens the EMF.

- c. Reads the data element bit maps for the DPCs it processes and updates the data element bit maps for the BASE24-atm messages from the EMF records it finds.
- d. Reads the message authentication bit maps for the DPCs it processes and updates the message authentication bit maps for the BASE24-atm messages from the EMF records it finds.
- e. Reads the IMS/CICS transaction code equivalents set in the EMF records it finds.
- f. Reads the token group identifiers set in the EMF records it finds.
- g. Closes the EMF.
- h. Opens the TKN.
- i. Reads the TKN records for sending tokens in BASE24-atm external messages.
- j. Closes the TKN.

BASE24-pos ISO Host Interface Process

For BASE24-pos processing, the ISO Host Interface process performs the following steps during initialization:

1. Opens the LCONF, reads the following params, params, and closes the LCONF:

ACCT-NUM-TYP
ADJ-ADVICE-RC
POS-LN-CUTOVER
POS-RVSL-RTE
REVERSAL-BAL-INQ
SAF-PRIOR-TO-SEND

2. Derives the current posting date.

To do this, the ISO Host Interface process checks the POS-LN-CUTOVER param value it retrieved from the LCONF.

If the current time is prior to logical network cutover, the ISO Host Interface process uses the current calendar date as the posting date.

If the current time is after logical network cutover, the ISO Host Interface process adds one to the current calendar date to derive a posting date.

3. Loads its BASE24-pos table.

The BASE24-pos table is an internal table containing the DPC data element bit maps, message authentication bit maps, IMS/CICS transaction code equivalents for BASE24-pos messages, and message token requirements.

To load the table, the ISO Host Interface process performs the following steps:

- a. Loads the standard default external message bit maps and message authentication bit maps for BASE24-pos messages.

These bit maps dictate the structures of the BASE24-pos external messages. Refer to the ***BASE24 External Message Manual*** for the standard default bit maps.

- b. Opens the EMF.
- c. Reads the data element bit maps for the DPCs it processes and updates the data element bit maps for the BASE24-pos messages from the EMF records it finds.
- d. Reads the message authentication bit maps for the DPCs it processes and updates the message authentication bit maps for the BASE24-pos messages from the EMF records it finds.
- e. Reads the IMS/CICS transaction code equivalents set in the EMF records it finds.
- f. Reads the token group identifiers set in the EMF records it finds.
- g. Closes the EMF.
- h. Opens the TKN.
- i. Reads the TKN records for sending tokens in BASE24-pos external messages.
- j. Closes the TKN.

BASE24-teller ISO Host Interface Process

For BASE24-teller processing, the ISO Host Interface process performs the following steps during initialization:

1. Opens the LCONF, reads the TLR-LN-CUTOVER param, and closes the LCONF.
2. Derives the current BASE24-teller posting date.

To do this, the ISO Host Interface process checks the TLR-LN-CUTOVER param value it retrieved from the LCONF.

If the current time is prior to logical network cutover, the ISO Host Interface process uses the current calendar date as the posting date.

If the current time is after logical network cutover, the ISO Host Interface process adds one to the current calendar date to derive a posting date.

3. Loads its BASE24-teller table.

The BASE24-teller table is an internal table containing the DPC data element bit maps, message authentication bit maps, IMS/CICS transaction code equivalents for BASE24-teller messages, and message token requirements.

To load the table, the ISO Host Interface process performs the following steps:

- a. Loads the standard default external message bit maps and message authentication bit maps for BASE24-teller messages.

These bit maps dictate the structures of the BASE24-teller external messages. Refer to the *BASE24 External Message Manual* for the standard default bit maps.
- b. Opens the EMF.
- c. Reads the data element bit maps for the DPCs it processes and updates the data element bit maps for the BASE24-teller messages from the EMF records it finds.
- d. Reads the message authentication bit maps for the DPCs it processes and updates the message authentication bit maps for the BASE24-teller messages from the EMF records it finds.
- e. Reads the IMS/CICS transaction code equivalents set in the EMF records it finds.
- f. Reads the token group identifiers set in the EMF records it finds.
- g. Closes the EMF.
- h. Opens the TKN.
- i. Reads the TKN records for sending tokens in BASE24-teller external messages.
- j. Closes the TKN.

BASE24-telebanking ISO Host Interface Process

For BASE24-telebanking processing, the ISO Host Interface process performs the following steps during initialization:

1. Loads its BASE24-telebanking table.

The BASE24-telebanking table is an internal table containing the DPC data element bit maps, message authentication bit maps, IMS/CICS transaction code equivalents for BASE24-telebanking messages, and message token requirements.

To load the table, the ISO Host Interface process performs the following steps:

- a. Loads the standard default external message bit maps and message authentication bit maps for BASE24-telebanking messages.

These bit maps dictate the structures of the BASE24-telebanking external messages. Refer to the *BASE24 External Message Manual* for the standard default bit maps.
- b. Opens the EMF.
- c. Reads the data element bit maps for the DPCs it processes and updates the data element bit maps for the BASE24-telebanking messages from the EMF records it finds.
- d. Reads the message authentication bit maps for the DPCs it processes and updates the message authentication bit maps for the BASE24-telebanking messages from the EMF records it finds.
- e. Reads the IMS/CICS transaction code equivalents set in the EMF records it finds.
- f. Reads the token group identifiers set in the EMF records it finds.
- g. Closes the EMF.
- h. Opens the TKN.
- i. Reads the TKN records for sending tokens in BASE24-telebanking external messages.
- j. Closes the TKN.

Warmboot

Warmboot processing is similar to process initialization, except that a warmboot only includes those steps required to update the file information carried by the ISO Host Interface process in memory.

During a warmboot, the ISO Host Interface process opens the following files:

- Logical Network Configuration File (LCONF)
- Log Message Configuration File (LMCF)
- Host Configuration File (HCF)
- External Message File (EMF)
- Key File (KEYF) or Key 6 File (KEY6)

The ISO Host Interface process retains all of its current timers, processing flags, status indicators, and performance accumulators so that current processing is not affected. It does not, however, reallocate extended memory or initiate new echo-test messages to any of its DPCs. Nor does it open the SAF and recount the messages pending for each DPC.

The ISO Host Interface process does not open the Token File (TKN) or recreate the TKN extended memory table. The process must be stopped and started for this to occur.

During a warmboot, the ISO Host Interface process retrieves the names of its security modules from the LCONF and attempts to open those modules. If the security modules named cannot be opened, the ISO Host Interface process continues using those that are already open.

Following a successful warmboot, the ISO Host Interface process sends a “WARMBOOT COMPLETE” event message to the network logging facility.

Event Messages

The ISO Host Interface process generates event messages to assist network operators in performing their daily tasks. Event messages provide information regarding the internal condition of the system. They are a network operator's primary link to what is actually occurring in the system. Event messages provide operators with information ranging from system status, such as the completion of a procedure, to more serious circumstances requiring operator intervention.

The ISO Host Interface process generates event messages using a standard routing code of zero.

Event Message Documentation

ACI has developed a method that allows customers to access event message documentation electronically. Event message descriptions are contained in files on a designated subvolume. Each file contains event messages for a specific BASE24 process or module. The files are organized according to product module ID (PMID). One PMID exists for each BASE24 process or module. PMIDs identify a specific process or module in the BASE24 system (for example, BAHISO identifies the Base ISO Host Interface module and ATHISO identifies the BASE24-atm ISO Host Interface module). Providing event message documentation in files on the Tandem computer gives customers greater flexibility by allowing them to view and print event message documentation as needed at their location.

Documentation is available in file form for the event messages generated by the ISO Host Interface process. It is located on the BA_{xx}LOGM subvolume, where *xx* is the number of the current release, in a file named BAHISO. For a complete explanation of event message documentation, refer to the ***BASE24 Event Message Reference Manual***.

Log Message Configuration File

The Log Message Configuration File (LMCF) contains event message information, such as message text, routing code, and severity, that has been modified by the user. At initialization, the ISO Host Interface process builds a table in memory from the LMCF records. When an event message is generated, the ISO Host Interface process accesses the table to verify whether that event

message has been modified. If the event message has been modified, the ISO Host Interface process logs the modified information instead of the standard event message.

The LMCF is accessed using operator functions. The routing code, severity, and text of each message can be modified in the LMCF. Because LMCF records are read into memory by the processes using them, updates to the LMCF do not take effect until the processes are reinitialized. For more information on accessing the LMCF, refer to the ***BASE24 Event Message Reference Manual***.

XPNET Configuration

The ISO Host Interface process is typically configured to start automatically when the XPNET node is started. Because of the length of time required for initialization, the ISO Host Interface process should be assigned a startup logic of AUTOMATIC in the XPNET node configuration. The AUTOMATIC startup logic means that the ISO Host Interface process is started automatically by the XPNET process when the XPNET node is started. This ensures that the ISO Host Interface process is ready when it receives its first transaction.

Multiple copies of the ISO Host Interface process can be defined in the XPNET node to control or increase transaction throughput. Each ISO Host Interface process can support up to 20 DPCs.

Note: One DPC can be serviced by more than one ISO Host Interface process. ISO Host Interface processes are tied to their DPCs in the HCF.

In addition to one process configuration for each ISO Host Interface process, the XPNET node must also include station and line configurations for the stations and communications lines used to communicate with the host. For more information on configuring processes, stations, and lines to the XPNET system, refer to the *NET24-XPNET Network Control Command Reference Manual*.

BASE24 Text Commands

The ISO Host Interface process can receive a number of BASE24 text commands that cause it to perform specialized types of activities. These commands can be separated into the following categories:

- Process-generated commands – commands generated and sent by other running processes in the system. In general, these commands notify the ISO Host Interface process about events occurring elsewhere in the system.
- General use operator commands – commands entered by an operator in the daily use of the system. These commands enable the operator to control or change the way the ISO Host Interface process is operating.
- Restricted use operator commands – commands that should be entered by an operator only at the request of ACI personnel. These commands allow programmers to test functionality and determine how the ISO Host Interface process is running when changes are made to the code.

The following table lists all of the commands that the ISO Host Interface process can receive and identifies the category under which the command is described. For more information on issuing these commands to the ISO Host Interface process and on the processing that the ISO Host Interface process performs when it receives one of these commands, refer to the *BASE24 Text Command Reference Manual*.

Process-Generated Commands	General Use Operator Commands	Restricted Use Operator Commands
CUTOVER DEVICE STATE SAFEXTCOMP SAFEXTRQST	KEY CHANGE KEY REPEAT KEY VERIFY LOGOFF LOGON PACINGSTAT PERFORMANCE PERFSTAT SAFPACING STA STARTSAF STATUS STOPSAF SWL WARMBOOT	DEBUG DUMP TRACE OFF TRACE ON

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